Interaction Obstructions in Einstein–Weyl Gravity: Cubic Protection, Second-Order Metric Failure, and Krein Visibility

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Abstract

We study whether the two Lorentz-covariant free completions of scalar-free Einstein–Weyl gravity extend to the interacting theory. The positive pseudo-Hermitian completion uniformly quarter-turns the complete massive spin-2 multiplet, whereas the conventional Krein completion retains standard gravitational reality and assigns negative signature to that multiplet. We first prove a cubic-protection theorem. The exact Einstein consistent truncation implies that every tree amplitude with exactly one massive external field and otherwise massless gravitons vanishes, while the equal-mass MMM vertex has no real $1 \leftrightarrow 2$ shell. Hence the first-order positive-metric obstruction is zero.

At second order the physical process $MM \to Mh$ is open. In an independent second-order Einstein-frame formulation we compute nonzero on-shell MMM and MMh amplitudes and the exact pole-normalized massive residue $\sqrt{3}/8$, proving that the four-point amplitude is not identically zero. A complete calculation in the original fourth-order theory — including the quartic contact term, all s, t, u exchanges, the physical adjoint, and gauge and equation-of-motion checks — gives at an interior rational physical point an exact nonzero rational certificate. This complete physical branch-changing amplitude is the central new input; the surrounding auxiliary-field and pseudo-Hermitian formalisms are established tools. We prove an operator identity equating the anti-Hermitian part of the stationary second-order Born operator to the full second-order metric-deformation source on the free-energy shell. It follows that

$$\Pi_{\ker \mathrm{ad}_{h_0}} \left[(v_2^\dagger - v_2) + \frac{1}{2}[R_1, v_1 + v_1^\dagger] \right] \neq 0.$$

The canonical analytic deformation based at the Paper-IV positive real form therefore fails at second order for the complete tree-level vertex content through quartic order.

On physical BRST cohomology, canonical second quantization of the free Krein symmetry gives the particle-number grading $J_F = (-1)^{N_M}$. The gravitational obstruction survives cohomology and is not null:

$$\mathcal{O}^\sharp = \mathcal{O}, \qquad \mathrm{Tr}(\mathcal{O}^\sharp \mathcal{O}) = -8\mathcal{A}_K^2 \neq 0.$$

Unlike the perfect-square scalar theory, Einstein–Weyl gravity has no nontrivial uniform abelian charge grading capable of placing the obstruction in a one-sided null ideal. The

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result excludes both an analytic continuation of the canonical positive completion and the conventional massive-number grading as a null-sector probability mechanism. It does not exclude indefinite Krein evolution itself, nor nonlocal, non-factorizing, nonperturbative, or conformal-boundary completions. An exact scalar `REDUCED-MODE` calculation also shows that the ordinary axis-compatible independent-mass regulator does not cancel its complete final-state collinear logarithmic response. That is not a gravitational theorem and changes none of the tree-level results here; it sharpens the extra infrared/asymptotic-state input any Bateman–Turok-type gravitational completion would have to supply. A finite coherent-projector calculation cancels that scalar coefficient while preserving idempotence, but it does not derive the transport dynamically and does not transfer to the gravitational BRST carrier. An exact scalar countermodel shows that CCR preservation, projector idempotence, trace preservation, and endpoint exchange symmetry still leave three endpoint parameters. The scalar 1/48 value is compatible but not selected by those identities, sharpening rather than removing the need for an independently constructed continuum transport. The exact full off-resonant scalar composition reduces the flat three-parameter endpoint freedom to one soft normalization on its declared three-dimensional chart, but its measured cross-Gram remains $-(1/2)\,dr/r$ and hence logarithmically non-trace-class. The full normalization calculation fixes the unprojected scalar response to 1/48 per pair, but its logarithmic rows pair fixed-vacuum charges $+1$ and -1; the published one-sided projection instead makes that residue zero. The exact scalar vacuum-orbit operator makes those rows neutral and preserves 1/48, but simultaneously neutralizes the squeeze that the fixed-vacuum negative grading discarded; the charge derivation does not descend after setting the orbit operator to one. The missing full zero-mode trace therefore still decides the physical answer and supplies no gravitational or causal normalization. The same Appendix-C scalar squeeze also fails to define a vector in the ordinary massless positive Fock–Krein topology: its two-particle norm density diverges as the inverse infrared cutoff, and its pair block is not Hilbert–Schmidt. This remains a scalar `LOCAL-ALGEBRAIC`, `REDUCED-MODE` carrier obstruction. It neither transfers to the metric BV complex nor excludes an explicitly extended/non-Fock scalar representation. The full scalar exponential admits an explicit infrared-weighted vacuum carrier, but its inverse metric is unbounded, its thermodynamic representation is inequivalent. Its covariant squeeze does admit a cross-Krein core implementation and cyclic finite-rank trace, but a finite normalized orbit trace kills localized projectors and the transported vacuum projector has exponentially growing trace norm. No normal thermodynamic Born trace follows. A semifinite scalar construction nevertheless gives normalized finite-detector weights under the weak ghost hypothesis by conditioning on a finite incoming projection; its normalized corner functional is noncyclic and supplies neither a thermodynamic state nor Eq. (19). On the available zero-mode-completed logarithmic sector, an exact scalar detector pushforward is finite rank at every fixed soft cutoff and remains in that semifinite ideal after the weighted squeeze. Projector idempotence forces opposite hard and soft box traces, but their cancellation does not control the positive trace ideal: for N logarithmic cells the order-λ coefficient has squared trace norm $N/12$ and the order-λ^2 coefficient has trace norm $N/24$. Thus this sector has no uniform trace-class soft limit without an additional full-pushforward cancellation, hard matching operator, or local non-normal renormalized weight. The result neither supplies the missing scalar Eq. (19) sectors nor transfers to the gravitational BV carrier. The complete scalar signed quadratic pushforward provides that cancellation at order λ: the off-resonant symplectic phase cancels the oscillatory inverse phase, every endpoint-dangerous row vanishes, and the parent cross-Gram is zero. Its mixed-sign rows satisfy the cross-CCR Ward identity, while the unique zero-mode dressing makes the finite-mode generator neutral. Hence the unsqueezed finite-mode nonlinear factor has neutral charge support through order λ, with $Q_1 = 0$ in that charge decomposition. This is not the full Eq. (19) statement because the public map is $R = SU$. The conditional 1/48 is absent from the completed public quadratic map. This does not determine the physical scalar response because the S-matrix transition block is a distinct operator. The squeezed-vacuum and dynamical-zero-mode projector trace, continuum domains, and higher orders remain open within Eq. (19). This remains solely a scalar `LOCAL-ALGEBRAIC`, `REDUCED-MODE` result. The covariant Appendix-C squeeze conjugates the finite-core scalar detector and Born operator together, so finite-rank cyclicity preserves this zero; bare squeezed-vacuum pair occupation is not an additive Eq. (19) contribution. Trace similarity does not imply ghost evenness. For any homogeneous assignment $q(Z) = s$, the public factors obey $q_K = 1-s$ and $q_S = 2s-2 = -2q_K$. One factor has positive charge unless $s = 1$; at that value the neutral squeezed $n = 1$ projector has exact non-null ghost-odd norm $-z^2(z^2+2)$. No homogeneous regular assignment therefore realizes the full Eq. (19) package on the finite regulator. The missing thermodynamic trace domain still limits Eq. (19), while the operator distinction independently forbids reading its zero as an S-matrix coefficient. Finite paired scalar processes nevertheless have a spectator-stable non-normal cylinder functional, on which the zero coefficient survives; this is not an inclusive LSZ or gravitational construction. For a scalar orthogonal inclusive detector, higher composite amplitudes do not enter the leading transition weight, but the input kernel must still be the physical S-matrix block. The complete signed zero belongs instead to the distinct scalar Eq. (19) pushforward. The scalar five-point S-matrix response is $+1/512$ per pair and $+3/512$ in total, or $1/48$ per pair after Born normalization. The axis-compatible physical virtual response is zero, leaving $+3/512$ uncancelled. The public R_t response is also zero but is a distinct Eq. (19) object, not a physical summand. An independently normalized hard/dressed response $-3/512$ is still missing. This scalar object ledger neither completes its NLO probability nor transfers a coefficient or state construction to gravity. On a regulated positive scalar quotient, physical-shell pseudo-unitarity forces that coefficient: the real-column norm $1/16$ implies the hard survival response $-1/16$, hence absolute $-3/512$ after Born normalization. An exact finite isometric witness exists, but the statement is conditional on a continuum dressed scalar S-matrix and does not transfer to gravity. On the ordinary logarithmic scalar carrier that strong Møller limit is obstructed by fixed-norm shells escaping to the collinear endpoint. A regulator-pulled-back abstract endpoint fibre retains the leading-log isometry, but has no certified local or gravitational affiliation. The scalar six-point output carrier can nevertheless be affiliated exactly. The twenty neutral three-Ω/three-Υ assignments form ten complement pairs with $\eta = \kappa = \left(\begin{smallmatrix} 0 & I_{10} \\ I_{10} & 0 \end{smallmatrix}\right)$. The complete coefficient lies in the positive ghost-even ten-plane, its Choi matrix is strongly ghost symmetric, and every fixed-shell nine-history range embeds isometrically there with norm $9/8$. A simultaneous map of all ninety resolved histories has rank ten, so a positive channel-record ancilla is still needed. The scalar input-projector pushforward, survival block and Eq. (19) remain open, and no part of this coefficient-level result transfers to the metric BV carrier. Preparing an auxiliary state directly in that positive scalar sector yields a normalized leading probability jet without Eq. (19). The fixed-shell click effect has spectrum $0, 1/16, (2-\sqrt{3})/8, (2+\sqrt{3})/8$, and its no-click complement is positive on an exact finite interval. The source $(|\Upsilon^3\rangle + |\Omega^3\rangle)/\sqrt{2}$ has rate $\lambda^8/[2048\pi^4\kappa^4 L_x L_y^2 L_z^2]$. Its binary no-click complement is fixed by normalization; genuine forward survival also depends on unobserved output blocks. It is a scalar auxiliary-sector result, not a transported perfect-square source, metric-BV result, or all-time probability. Formal two-sidedness of R_t does, however, pull this source and its effects back to a normalized dressed perfect-square scalar projector with the same rate. Positivity forces the source to break shift charge: its projector has charge support $\{-6, 0, +6\}$ and orbit support $\{Z^{-6}, 1, Z^6\}$. This is a selected physical-scalar probability jet, not the shift-invariant $P_\chi^{(\phi)}$, general Eq. (19), or a metric-BV result. The leading scalar source is explicit on the finite covariant Gaussian core: its Z^{-3} and Z^3 Jordan-oscillator branches are individually null and cross-pair to one. Unknown $O(\lambda)$ source corrections first affect probability at λ^9, so they cannot alter the leading λ^8 rate. No ordinary massless Fock thermodynamic vector follows. The same source and positive four-frame extend to compact, disjoint and non-antipodal momentum packets supported away from zero. The scalar energy multipliers are uniformly bounded there, all degree-three creator products are closable on one weighted Gaussian image core, and normalized box approximants converge together with their fixed-strength effects. This does not compute the packet Hamiltonian strength, remove the infrared gap, prove general Eq. (19), or change any metric-BV boundary. The separate tagged scalar probability through λ^6 also extends over an exact continuous fixed-energy hard-angle family, with tree bound $W(c,T) \geq 13T/24 > 0$ and uniform compact-angle loop control. This is a fibrewise `REDUCED-MODE` scalar result, not a metric, BRST, QME or Lorentzian claim. On two orthogonal equal-normalization hard modes, the positive off-diagonal effect $E_\epsilon = P_+ + (1-\epsilon)P_-$ fixes the common leading scalar amplitude. Recorded and coherent probabilities therefore agree through λ^6; their first known difference is the order-λ^8 term $-\epsilon\|Y_4(c_1) - Y_4(c_2)\|^2/2$. The exhaustive order-eight ledger shows this is the complete coherent-minus-recorded coefficient, because the unknown leading–sixth-order cross is identical for both effects. At the complementary antisymmetric port that cross is absent. On the exact finite-volume two-angle fixture its absolute leading coefficient has a volume-uniform strictly positive lower bound. Equal-area compact packets on the invariant fixed-P two-body sphere preserve the same strict coefficient; the exact off-energy-diagonal second-Dyson kernel has the same local ultraviolet subtraction as its diagonal restriction, so a nonempty globally normalizable finite-bandwidth packet class preserves a strict coefficient. An explicit real exchange-even degree-four local quadrupole density has zero angular mean on every timelike pair fibre and selects a strictly positive absolute order-λ^8 coefficient on such a packet at leading external detector coupling. An expanding smooth compact spacetime cutoff preserves exact leading darkness for every cutoff and retains a positive order-eight coefficient for some finite, existential support radius. Its numerical bandwidth and support radius remain open. The normalized response mode also defines an exact finite-strength Julia click/no-click instrument. Under standard complement-cyclicity assumptions its nonzero exactly vacuum-dark effect cannot be a bounded local effect; a balanced contrast with nonzero vacuum baselines exists conditionally on local affiliation and domains. The public BT real structure does not yet instantiate that condition: its two-field Wightman species Gram has signature $(1,1)$. The public fundamental symmetry makes an auxiliary Hilbert Gram positive only by changing the adjoint to $\Omega^* = \Upsilon$. A Krein-selfadjoint observable is then Hilbert-selfadjoint exactly when it is ghost-even. An exact weak-ghost fixture has generalized Krein Born weight 2 but ordinary Hilbert weight 3, so a positive topology does not silently replace the public probability rule. The compact scalar quadrupole is now exactly obstructed as a regular same-chart ghost-even observable: its hidden image contains $\lambda^{-1}\log(\psi/\lambda)$ and has no continuous extension to the perturbative vacuum. A minimal cross-paired two-vacuum completion is local, ghost-even and Hilbert-selfadjoint and retains the dark response, but it is a changed doubled scalar theory not selected by the public action or R_t. A regular public auxiliary alternative is now explicit: $D_\Omega + D_\Upsilon$ is polynomial, ghost-even and positive-Hilbert symmetric on the selected packet core. A charged pointer doublet makes its ± 2 response branches total-charge neutral. The resulting positive species map is an isometry and retains the strict compact order-eight bound. This is selected public auxiliary detector physics, not the standard scalar projector, Eq. (19), a complete local net or a gravitational result. The recorded and symmetric bright-port absolute q_8 coefficients and selection by the public BT dynamics remain open, while an explicit external pointer Hamiltonian realizes the finite two-mode effect and selects its strength and phase. That apparatus is not a public-BT or gravitational interaction, and no part transfers to gravity. A local quadratic scalar vertex realizes its selected two-mode compression, but finite-derivative locality cannot isolate exactly two zero-width continuum angles. The order-20 projective Fejér filter has an antipodally even degree-38 homogeneous lift to the complete invariant fixed-P two-body sphere. Two finite solid-angle packets leave exact complement fraction below 83/125000, and the residual-retaining rotating-wave evolution has half-pulse absorption efficiency above 124917/125000. This remains fixed-P scalar apparatus physics and does not transfer to gravity. A Schwartz-switched version gives a square-integrable leading response on an open four-momentum set with total complement below $883/10^6$, while the target timelike pair sum is separated exactly from the non-timelike number-transfer cone. On a declared compact energy band an explicit off-shell successor bounds the complete undesired quadratic block below $10^{-400000}$ relative to the desired pair block at first Dyson order. This leading scalar response still supplies no unrestricted-energy theorem, gravitational interaction or full time-ordered detector.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, computational checks, claim-boundary analysis, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the corresponding human contact. The project is an experiment in whether a non-specialist human, using AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. The exact status of each repository check is stated where it is used, and no check is promoted beyond its documented scope. The manuscript remains subject to expert review.

Contents

1 Introduction

Scalar-free Einstein–Weyl gravity is the simplest four-dimensional quadratic-curvature theory whose flat-space physical spectrum consists of the ordinary massless graviton and one additional massive spin-2 multiplet. In the curvature convention used here its action is

$$S[g] = \int d^4x \sqrt{-g} \left(c_1 R + \alpha R_{\mu\nu} R^{\mu\nu} + \beta R^2 \right), \qquad \alpha = -3\beta, \tag{1}$$

with $M^2 = c_1/\alpha > 0$ in the split phase. Up to the four-dimensional Euler density, the quadratic term is Weyl-squared. The extra spin-2 branch has the opposite kinetic sign, the familiar higher-derivative ghost of quadratic gravity [1, 2]. Neither this spectrum nor its nonlinear auxiliary-field reformulation is new: quadratic gravity was put in a local second-order Einstein–scalar–massive-tensor form by Hindawi, Ovrut, and Waldram [3], and the scalar-free massive-tensor sector was developed in detail by Magnano and Sokołowski [4].

Paper IV of this series [29] did not choose between proposed interacting cures. It classified the *free* covariant completions. After symplectic reduction, all five massive polarizations form one irreducible massive spin-2 representation. Covariance therefore allows no helicity-by-helicity compromise. There are two real-form classes in the stated free, quasifree, mode-local setting:

1. a positive pseudo-Hermitian completion in which the entire massive multiplet has quarter-turned reality; and
2. a Krein completion with standard gravitational reality, positive massless signature, and uniform negative massive signature.

They determine the same free complex spectral functional but different adjoints and probability structures. The natural interacting question is therefore sharp:

> Does either covariant free completion survive the tree interaction through quartic order?

Paper V [30] supplied the deformation complex used here to ask the positive part of this question. Perturbative pseudo-Hermitian metrics, their homogeneous ambiguities, and interacting field-theory constructions have a substantial prior literature [6, 7]; we do not claim the order-by-order formalism itself as new. If ρ_0 is the free Dyson map and v_n are the transported interaction vertices, an analytic correction $\rho_\zeta = \exp[-\frac{1}{2}\sum_n \zeta^n R_n]\rho_0$ must solve

$$\begin{aligned} [h_0, R_1] &= v_1^\dagger - v_1, \\ [h_0, R_2] &= (v_2^\dagger - v_2) + \frac{1}{2}[R_1, v_1 + v_1^\dagger]. \end{aligned} \tag{2}$$

On a degenerate free-energy shell the right-hand side must have zero projection onto $\ker \mathrm{ad}_{h_0}$. That shell projection is the metric obstruction. In the perfect-square scalar, first-order protection was followed by a second-order branch-changing obstruction; at the exact massless $O(1,1)$ boundary, however, the offending component could become one-sided in continuous charge and null in the generalized Born trace.

Prior amplitudes and the precise novelty claim. The geometric premise behind the cubic selection rule is also known: Einstein metrics form an exact Bach-flat subsector. Related amplitude work proves that Ricci-curvature deformations leave all-Einstein-graviton tree amplitudes unchanged and that the corresponding pure conformal amplitudes vanish [5]. Lemma 4.1 below is the mass-eigenfield, one-massive-leg consequence of that established consistent truncation, not a new discovery of the Einstein subsector. Conformal-gravity BRST/LSZ quantization and its indefinite scattering problem have likewise been studied directly [8], and conformal or mass-deformed ghost-state amplitudes are known in adjacent conformal and supersymmetric theories [10, 11]. Those works concern the conformal/Jordan phase or different amplitude sectors, not the regular split-phase process studied here.

The closest comparison is Kuntz's PT-symmetric quadratic-gravity proposal [9]. It uses the same scalar-free quadratic action and, up to convention, the same complex rotation of the complete massive spin-2 field. It constructs the free BRST physical subspace and solves an interacting pseudo-Hermitian metric in a mode-truncated Hamiltonian in which spin and momentum labels and the exact gravitational vertex coefficients are suppressed. It does not evaluate a complete degenerate physical scattering shell. The calculation below supplies precisely that missing stress test.

To our knowledge, the narrow new input of this paper is the first complete exact calculation of the physical mass-eigenfield process $MM \to Mh$ in scalar-free Einstein–Weyl gravity, together with its independent complete massive-pole residue, and the use of that physical block to produce an explicit metric-deformation cocycle. The novelty is the verified gravitational source, not the auxiliary formulation, the general deformation equations, the elementary external-leg phase count, or indefinite-metric algebra. Propositions 2.1 and 7.1 make the amplitude-to-cocycle step an operator identity rather than a phase-based inference.

Gravity has the same perturbative-order pattern and a different mechanism:

$$\boxed{\mathfrak{o}_+^{(1)} = 0, \qquad \mathfrak{o}_+^{(2)} \neq 0.} \tag{3}$$

The cubic protection follows from the exact Einstein subsector together with equal-mass kinematics. The second-order obstruction is carried by the physically open process

$$M + M \longrightarrow M + h, \tag{4}$$

which changes massive particle number by one. The same process also separates gravity from the scalar boundary mechanism. Covariance and the free signature fix the one-particle symmetry; its natural particle-number-diagonal, cluster-multiplicative lift is $(-1)^{N_M}$. But $\mathbb{Z}_2$-odd does not imply null: the product of two odd operators is neutral and its quadratic trace is nonzero.

Results. The proof has four independent layers.

1. Section 3 proves a diagram-level phase theorem: a connected amplitude with E_M external massive legs changes by $(-i)^{E_M}$ under the uniform quarter-turn, including the compensating sign of every internal massive propagator.

2. Section 4 proves cubic protection. The Einstein consistent truncation kills Mhh, while MMM has no real physical $1 \leftrightarrow 2$ shell.

3. Sections 5–7 prove the second-order obstruction twice: first by an independent nonzero Einstein-frame factorization residue, then by a complete exact physical four-point calculation in the original fourth-order theory, joined to the deformation complex by Proposition 2.1.
4. Sections 8–11 prove conventional Krein visibility: the canonical multiplicative Fock lift is fixed in the stated class, the exact block is not null, no scalar-like abelian charge exists, and the block is not BRST-exact on physical cohomology.

Comparison with the scalar obstruction. The parallel and the difference are summarized by

Perfect-square scalar	Einstein–Weyl gravity
Källén zero and reality-factor protection	Einstein consistent truncation and equal-mass kinematics
$H + L \to L + L$	$M + M \to M + h$
$O(1, 1)$ one-sided null ideal at the exact boundary	no nontrivial uniform abelian charge grading
obstruction can become probability-invisible	obstruction remains non-null for the conventional covariant Krein form

Strength of the claims. "Metric failure" means failure of the analytic order-by-order deformation of the canonical covariant positive free metric in the operator class of (2). It is not an absolute theorem against every positive metric, and no complex interacting spectrum is proved. "Krein visibility" has the precise finite-block meaning $\mathcal{O}^\sharp\mathcal{O} \neq 0$ and $\mathrm{Tr}(\mathcal{O}^\sharp\mathcal{O}) \neq 0$ for the canonical particle-number-diagonal, cluster-multiplicative grading. It is a non-nullity statement, not a claim that an indefinite trace is a positive probability. In particular, the paper does not rule out an indefinite Krein dynamics; it rules out the naive parity/null-sector mechanism as a replacement for a positive interacting probability calculus.

Verification. The labels G13–G18 denote entries in the machine-verification table of Appendix H; they are not additional assumptions or mathematical objects. The checks use exact SymPy or rational arithmetic. The exhaustive $5^3 \times 2$ polarization scan once proposed as G16 is not used: the factorization residue, exact real-shell certificate, crossing, Ward, and gauge checks already close the proof. The scan remains optional regression hardening.

2 Free input, conventions, and the deformation class

2.1 Physical one-particle spectrum

We use signature $(+, -, -, -)$ in the main text. Around flat space the reduced physical one-particle space is

$$\mathcal{H}^{(1)}_{\text{phys}} = \mathcal{H}_{h,+2} \oplus \mathcal{H}_{h,-2} \oplus \mathcal{H}_M, \qquad \dim \mathcal{H}_M = 5. \tag{5}$$

The massless summands carry helicities ± 2 on $p^2 = 0$; the massive summand carries the spin-2 representation on $p^2 = M^2$. Paper IV derives this by explicit diffeomorphism reduction rather than by a covariant propagator count. Its result is a healthy massless sector and a uniformly ghost-signed massive sector.

2.2 Two free real forms

In the positive class the original massive mass eigenfield is ‡-anti-Hermitian. We write the positive-frame observable as

$$\widehat{M}_{\mu\nu} = iM_{\mu\nu}, \qquad M_{\mu\nu} = -i\widehat{M}_{\mu\nu}. \tag{6}$$

The same quarter-turn acts on every one of the five polarizations. The massless graviton retains standard reality. In the conventional Krein class both fields retain standard gravitational reality, while the one-particle fundamental symmetry has signature

$$J^{(1)}_{\text{free}} = I_2 \oplus (-I_5), \qquad \text{signature } (+,+;-,-,-,-,-). \tag{7}$$

The uniformity is not optional. Since $\mathcal{H}_M$ is an irreducible massive spin-2 representation, Schur's lemma makes every covariant Hermitian form on its fiber a scalar multiple of the identity. Rotating only selected helicities would violate Lorentz covariance [29].

2.3 Regulated physical Fock space and shell projection

As in Paper V, the cohomological equation is first interpreted in a periodic spatial box. We retain finitely many nonzero momentum modes, impose an ultraviolet and a finite-particle cutoff, and omit the massless zero mode. Normal ordering removes vacuum contractions, and we work with connected transition kernels. On this regulated physical Fock space h_0 has point spectrum with finite-rank eigenspaces. If P_E projects onto free energy E, then

$$\Pi_{\ker \mathrm{ad}_{h_0}} S = \sum_E P_E S P_E, \tag{8}$$

and $[h_0, R] = S$ has a solution only if this projection vanishes.

The rational point used below lies on the momentum lattice whenever $ML/(2\pi)$ is a multiple of 100. The ultraviolet cutoff may be chosen above every external and internal momentum occurring in the calculation. We take a sequence of such boxes after establishing the finite-volume obstruction. Omitting the zero mode is the infrared prescription; the exhibited momenta remain uniformly nonsoft along this sequence. Thus neither the matrix element nor its internal denominators depends on a soft limiting prescription.

2.4 Perturbative order and the stationary Born operator

Introduce a formal coupling ζ by assigning the cubic expansion of the Einstein–Weyl action order ζ and its quartic expansion order ζ^2. After the Paper-IV free Dyson map, write

$$H_+(\zeta) = h_0 + \zeta v_1 + \zeta^2 v_2 + O(\zeta^3). \tag{9}$$

The second-order metric-deformation source is

$$S_2 = (v_2^\dagger - v_2) + \frac{1}{2}[R_1, v_1 + v_1^\dagger], \qquad [h_0, R_1] = v_1^\dagger - v_1. \tag{10}$$

This is the meaning of "second order" throughout the paper: one quartic vertex or two cubic vertices at tree level.

Set $Q_E = 1 - P_E$ and define the self-adjoint reduced resolvent

$$G_E = Q_E(E - h_0)^{-1}Q_E. \tag{11}$$

The stationary connected second-order Born operator on the shell is

$$\mathcal{B}_{2,+}(E) = P_E\big(v_2 + v_1 G_E v_1\big)P_E. \tag{12}$$

At finite volume no $i0$ is required. In the continuum notation G_E is the principal-value part of $(E - h_0 + i0)^{-1}$. The possible $-i\pi\delta(E - h_0)$ contribution is absent here because the regulated physical cubic shell is empty; Lemma 4.2 below makes this statement on the complete Fock eigenspace, including spectators.

Proposition 2.1 (Born–deformation identity)**.** *Suppose $P_E v_1 P_E = 0$ and choose the canonical off-shell solution of the first equation in* (10)*,*

$$R_1^{(0)} = \sum_{E_a \neq E_b} \frac{P_{E_a}(v_1^\dagger - v_1)P_{E_b}}{E_a - E_b}. \tag{13}$$

Then the entire second-order metric source, not only its quartic part, satisfies

$$\boxed{P_E S_2 P_E = \mathcal{B}_{2,+}(E)^\dagger - \mathcal{B}_{2,+}(E).} \tag{14}$$

Consequently, a nonzero ordinary-adjoint anti-Hermitian Born matrix element on the shell is a nonzero metric-deformation cocycle.

Proof. Write $D = v_1^\dagger - v_1$ and $W = v_1 + v_1^\dagger$. The spectral formula (13) gives

$$P_E R_1^{(0)} Q_E = P_E D G_E, \qquad Q_E R_1^{(0)} P_E = -G_E D P_E. \tag{15}$$

Since $P_E v_1 P_E = 0$, insertion of Q_E between the two cubic vertices is automatic. Hence

$$\begin{aligned} \frac{1}{2} P_E[R_1^{(0)}, W]P_E &= \frac{1}{2} P_E(D G_E W + W G_E D)P_E \\ &= P_E(v_1^\dagger G_E v_1^\dagger - v_1 G_E v_1)P_E. \end{aligned} \tag{16}$$

Adding $P_E(v_2^\dagger - v_2)P_E$ gives (14), because $G_E = G_E^\dagger$. Thus the quartic contact and the two-cubic exchange contribution enter with exactly the relative combination required by the deformation equation. □

Remark 2.2 (Contact, exchange, and the logical bridge). The two terms in the deformation source have a fixed diagrammatic meaning. The shell projection of $v_2^\dagger - v_2$ is the anti-Hermitian difference of the canonical quartic vertex, including the instantaneous terms generated by the Legendre transform. Equation (16) is the anti-Hermitian difference of the complete two-cubic intermediate-state sum. Their sum, and only their sum, is the connected second-order Born block. Thus the four-point amplitude below is not being substituted for S_2 on the basis of its external-leg phase: Proposition 7.1 first identifies the complete covariant contact-plus-exchange calculation with (12), and (14) then identifies its anti-Hermitian part with $P_E S_2 P_E$. The finite-volume principal-value prescription, external normalizations, identical-particle factors, and removal of disconnected pieces are stated separately below.

2.5 Reduced amplitude, LSZ normalization, and adjoint

The perturbiner strips the universal Feynman i from the multilinear action coefficient; denote the result by $\mathcal{A}_K^{\mathrm{red}}$. At real momenta and real linear polarizations it is real. Let $|i\rangle, |f\rangle$ be unit-normalized finite-volume asymptotic states. Matching the covariant tree expansion to the stationary Born series gives

$$\langle f|\mathcal{B}_{2,K}(E)|i\rangle = \mathcal{A}_K = \eta_{\mathrm{conv}} \mathcal{N}_{fi} \mathcal{A}_K^{\mathrm{red}}, \qquad \eta_{\mathrm{conv}} \in \{\pm 1\}, \quad \mathcal{N}_{fi} > 0. \tag{17}$$

After removing the overall momentum-conservation Kronecker delta, $\mathcal{N}_{fi}$ is the product of the nonzero external pole residues, $(2\omega L^3)^{-1/2}$ factors, polarization normalizations, and Bose normalizations. At the displayed point the two incoming massive momenta are distinct, so their identical-particle factor is one; the outgoing particles have different species. The multilinear wave labels already implement the vertex symmetry factors. The real sign η_{conv} fixes the convention relating the interaction Hamiltonian to the Lagrangian Feynman rule.

Equation (17) also explains why stripping the universal Feynman i does not manufacture the obstruction: the removed factor is common to forward and reverse S-matrix elements, whereas the anti-Hermitian phase below comes from the three massive quarter-turns. Real, nonzero LSZ and normalization factors cannot change that phase or the nonvanishing result. Exact rational values are quoted for $\mathcal{A}_K^{\mathrm{red}}$; operator and trace statements use the normalized coefficient $\mathcal{A}_K$.

3 The external-massive-leg phase theorem

The quarter-turn phase must be transported through complete diagrams, not appended to an already computed external state by slogan. Internal massive lines carry the compensating kinetic sign.

Theorem 3.1 (External-massive-leg phase)**.** *Let $\mathcal{D}$ be a connected perturbative diagram written in mass-eigenfields (h, M), with E_M external massive legs and I_M internal massive lines. Under (6), its complete reduced amplitude obeys*

$$\mathcal{A}_+^{(E_M)}(\mathcal{D}) = (-i)^{E_M} \mathcal{A}_K^{(E_M)}(\mathcal{D}). \tag{18}$$

The statement includes contact terms, exchanges, and the sign of each massive inverse kernel.

Proof. Let n_v be the number of massive fields incident at vertex v. Field substitution contributes

$$\prod_v (-i)^{n_v}. \tag{19}$$

The massive quadratic form changes sign because $(-i)^2 = -1$; hence its inverse, the internal massive propagator, also contributes -1 per internal massive line. Every internal massive line is incident twice, so

$$\sum_v n_v = E_M + 2I_M. \tag{20}$$

The total factor is therefore

$$(-i)^{E_M+2I_M}(-1)^{I_M} = (-i)^{E_M}(-1)^{I_M}(-1)^{I_M} = (-i)^{E_M}. \tag{21}$$

Massless propagators are unchanged. Since the argument is diagramwise, it applies to their complete sum. Local invertible changes between regular mass-eigenfield frames can redistribute contact and exchange terms but leave the on-shell statement unchanged. □

Corollary 3.2 (Phase selection)**.** *For a real standard-frame tree amplitude, even E_M gives a real positive-frame coefficient and odd E_M gives a purely imaginary one. Only odd-E_M processes can contribute to the ordinary-adjoint anti-Hermitian shell source.*

For the four-point process (4), $E_M = 3$, and

$$\mathcal{A}_+ = i\mathcal{A}_K. \tag{22}$$

The sign is tied to the convention $M = -i\widehat{M}$; reversing the quarter-turn reverses the displayed sign but not the obstruction's nonvanishing.

4 Cubic protection

There are two odd-massive cubic structures: Mhh and MMM. The first is eliminated dynamically, the second kinematically.

4.1 Einstein consistent truncation

The scalar-free equations may be written schematically as

$$c_1 G_{\mu\nu} + \gamma_B B_{\mu\nu} = 0, \tag{23}$$

where $B_{\mu\nu}$ is the Bach tensor and γ_B depends on the normalization in (1). Every Ricci-flat Einstein metric has $G_{\mu\nu} = 0$ and is Bach-flat. More generally Einstein metrics are Bach-flat [14]. Thus the nonlinear Einstein solution space is an exact subsector of the Einstein–Weyl equations.

On the regular split branch, the higher-derivative theory admits a local second-order mass-eigenfield formulation in which the Einstein metric and the nonlinear massive spin-2 field are separate variables [4]. Denote them by (h, M) near flat space. The field map is algebraic and locally invertible on this branch, and $M = 0$ is precisely the Einstein subsector. These hypotheses are what make the following LSZ statement stronger than a basis-dependent observation.

Lemma 4.1 (One-massive-leg rule)**.** *For asymptotic mass eigenfields on the regular split branch,*

$$\mathcal{A}_{\text{tree}}(M, h, \ldots, h) = 0. \tag{24}$$

In particular $\mathcal{A}_3(Mhh) = 0$ for every physical polarization.

Proof. Write the massive-field equation in the second-order frame as

$$\mathcal{K}_M M + J_M^{(2)}[h,h] + J_M^{(3)}[h,h,h] + \cdots = 0. \tag{25}$$

For every nonlinear vacuum Einstein solution h_{E}, the pair $(h_{\text{E}}, M = 0)$ solves the full theory. Hence the terms independent of M vanish on the Einstein graviton shell. Functional differentiation and LSZ reduction say that a physical amplitude with one external M and all remaining external legs in the Einstein subsector is zero. Terms generated by a local invertible change of fields are proportional to equations of motion or external inverse propagators and vanish under LSZ, the standard equivalence-theorem mechanism [15]. □

The exact perturbiner check evaluates the physical decay point with the massive particle at rest and two back-to-back massless gravitons. All $5 \times 4 = 20$ combinations of the five massive polarizations and four graviton-helicity choices vanish exactly. This is a check of the amplitude-level content of Lemma 4.1, not its only justification.

4.2 Equal-mass cubic kinematics

The MMM vertex itself is nonzero at complex on-shell kinematics (Section 5). It nevertheless has no real physical $1 \leftrightarrow 2$ shell. If $p_3 = p_1 + p_2$ with future-directed $p_i^2 = M^2$, then

$$p_3^2 = (p_1 + p_2)^2 \geq (2M)^2, \tag{26}$$

which cannot equal M^2. Equivalently, a particle of mass M cannot decay into two particles of the same mass.

4.3 The complete regulated cubic shell

The Born–deformation identity requires a statement about the complete degenerate eigenspace, not only the two states used below to display the four-point obstruction.

Lemma 4.2 (Vanishing of the regulated physical cubic shell)**.** *On the regulated physical Fock space of Section 2, with the massless zero mode omitted,*

$$P_E v_1 P_E = P_E v_1^\dagger P_E = 0 \tag{27}$$

for every free energy E. The statement includes disconnected spectators: the connected cubic vertex cannot act nontrivially inside a degenerate many-particle shell.

Proof. A normal-ordered cubic matrix element is an active $0 \leftrightarrow 3$ or $1 \leftrightarrow 2$ process times Kronecker deltas for particles on which the vertex does not act. Positive energy forbids $0 \leftrightarrow 3$. It remains to classify the four physical cubic field contents. The Mhh amplitude is zero by Lemma 4.1; MMM is closed by equal-mass kinematics. For MMh, the channel $h \to MM$ would require $k^2 = (p_1 + p_2)^2 \geq 4M^2$, while $M \to M + h$ would require $p_M \cdot k = 0$ for a future timelike p_M and a future nonzero null k, which is impossible. The only limiting exception is $k = 0$, removed by the infrared prescription. Finally, real on-shell hhh kinematics is collinear, and the physical Einstein three-graviton amplitude vanishes there. Spectators do not alter any active momentum or energy equation, so they multiply these zeros. Taking the adjoint gives the second equality in (27). □

Theorem 4.3 (Cubic protection)**.** *For scalar-free Einstein–Weyl gravity about flat space, the Lorentz-covariant positive real form has no first-order physical on-shell metric obstruction:*

$$\boxed{\mathfrak{o}_+^{(1)} = 0.} \tag{28}$$

Proof. By Corollary 3.2, only odd-massive cubic amplitudes can enter $\Pi_{\ker \mathrm{ad}_{h_0}}(v_1^\dagger - v_1)$. The Mhh amplitude vanishes by Lemma 4.1; the MMM amplitude has no real physical shell. The remaining cubic structures carry an even number of massive legs. Equivalently, the stronger Lemma 4.2 makes the complete first-order shell block vanish. $\square$

This protection is structurally different from the Källén-factor zero of Paper V. It is a consequence of an exact nonlinear consistent truncation plus massive-particle kinematics.

5 Independent Einstein-frame factorization rail

The next odd-massive process is four-point $MM \to Mh$. Before expanding the original fourth-order action to quartic order, we prove independently that this amplitude is not identically zero.

5.1 Second-order action and convention change

This section follows the exact Einstein-frame formulation of Magnano and Sokołowski [4]. Its verification artifact uses signature $(-,+,+,+)$, only in this section and Appendix B; thus a massive pole is $P^2 = -M^2$. With $M = 1$ and in the notation of that formulation, the nonlinear massive-field Lagrangian is

$$L_M = \frac{1}{2}K(Q) + \frac{m^2}{8A}\left(-\tau^2 + \psi_{\mu\nu}\psi^{\mu\nu} + 6A\tau - 12A^2\right), \qquad \psi = g + \Phi. \tag{29}$$

Here $Q^\alpha{}_{\mu\nu} = \Gamma^\alpha{}_{\mu\nu}(\psi) - \Gamma^\alpha{}_{\mu\nu}(g)$,

$$K(Q) = g^{\mu\nu}\left(Q^\alpha{}_{\mu\nu}Q^\beta{}_{\alpha\beta} - Q^\alpha{}_{\mu\beta}Q^\beta{}_{\nu\alpha}\right), \tag{30}$$

A is the determinant ratio and τ the corresponding trace defined in [4]. The form (29) separates Einstein gravity from a minimally coupled nonlinear massive spin-2 field.

For linearly traceless external fields, the determinant and inverse- determinant contributions to the apparent $\operatorname{tr}\Phi^3$ potential cancel exactly:

$$V(\Phi) = \frac{m^2}{8}\operatorname{tr}\Phi^2 + O(\Phi^4). \tag{31}$$

The nonzero MMM amplitude below comes entirely from $K(Q)$.

5.2 Exact three-point amplitudes

Take all momenta incoming. For the MMM vertex use

$$P = (1,0,0,0), \qquad p = \left(-\frac{1}{2}, 0, 0, \frac{i\sqrt{3}}{2}\right), \qquad q = \left(-\frac{1}{2}, 0, 0, -\frac{i\sqrt{3}}{2}\right). \tag{32}$$

Then $P+p+q = 0$ and $P^2 = p^2 = q^2 = -1$. With normalized TT polarizations (M_+, M_+, M_0), direct expansion of (29) gives

$$\boxed{\mathcal{A}_3(M_+, M_+, M_0) = -\frac{\sqrt{6}}{8}.} \tag{33}$$

For MMh, take

$$-P = (-1, 0, 0, 0), \qquad r = (1, -1, -i, 0), \qquad k = (0, 1, i, 0), \tag{34}$$

so $-P + r + k = 0$, $r^2 = -1$, and $k^2 = 0$. For the polarization choice in Appendix B,

$$\boxed{\mathcal{A}_3(M_+, M, h) = -\frac{\sqrt{2}}{8}.} \tag{35}$$

The graviton Ward identity vanishes for an arbitrary symbolic vector ξ_μ under $\varepsilon^h_{\mu\nu} \mapsto k_{(\mu}\xi_{\nu)}$, and exchange of the two massive legs leaves the coefficient unchanged.

5.3 Complete massive-pole residue

Glue (33) and (35) across P. A complete orthonormal rest-frame basis contains the polarizations $+, \times, 0, xz, yz$. The exact products are

λ	$\mathcal{A}_{MMM}$	$\mathcal{A}_{MMh}$	product
$+$	$-\sqrt{6}/8$	$-\sqrt{2}/8$	$\sqrt{3}/32$
$\times$	0	0	0
0	0	$\sqrt{6}/8$	0
xz	0	0	0
yz	0	$i\sqrt{2}/4$	0

Therefore the complete polarization numerator is

$$N = \sum_{\lambda=1}^{5} \mathcal{A}_3(M, M, M_P^\lambda)\mathcal{A}_3(M_{-P}^\lambda, M, h) = \frac{\sqrt{3}}{32}. \tag{36}$$

The TT inverse kernel in the same action convention is

$$K_M(P) = \frac{P^2 + M^2}{4}. \tag{37}$$

Consequently

$$\boxed{\operatorname*{Res}_{P^2=-M^2} \mathcal{A}_4(M, M, M, h) = \frac{\sqrt{3}}{8} \neq 0.} \tag{38}$$

An overall rescaling of the action rescales the displayed residue but not its nonvanishing. A local quartic contact term is regular at this pole and cannot cancel it.

Proposition 5.1 (Factorization nonvanishing)**.** *The tree-level four-point amplitude tensor for $MM \to Mh$ is not identically zero on the relevant complexified on-shell component.*

Proof. If the rational amplitude vanished identically, every factorization residue would vanish. Equation (38) is the complete massive-polarization residue and is nonzero. □

6 The real physical shell

6.1 Open kinematics

In the center-of-mass frame, with signature $(+,-,-,-)$ restored,

$$|\mathbf{k}| = \frac{s-M^2}{2\sqrt{s}}, \qquad E_{M,\mathrm{out}} = \frac{s+M^2}{2\sqrt{s}}, \qquad s \geq 4M^2. \tag{39}$$

Thus $MM \to Mh$ is physically open on a continuum shell, unlike the cubic MMM channel.

Lemma 6.1 (Complex nonvanishing and a real component)**.** *Fix analytic physical polarization sections on an irreducible component of the complexified four-point mass shell. The nonsingular real $MM \to Mh$ region contains a Euclidean-open center-of-mass chart and is Zariski dense in the complex component containing it. Hence a rational tree amplitude that vanishes on an open real subset vanishes identically on that component and has zero factorization residues.*

Proof. Away from exceptional polarization charts, the real physical region is parametrized by $s > 4M^2$, $-1 < \cos\theta < 1$, and an azimuth. Its real points contain an open set of maximal real dimension in the relevant algebraic component. If $\mathcal{A} = P/Q$ is regular there and vanishes on a real open set, real analyticity makes P vanish there. The identity theorem, equivalently Zariski density of that real locus, makes P vanish on the component. Then all residues vanish, contrary to (38). □

The next subsection supplies a stronger direct bridge: an exact nonzero point already on the real shell. It makes Lemma 6.1 logically redundant for the obstruction theorem, while retaining it as an independent explanation of why the complex factorization rail is physically relevant.

6.2 Interior rational certificate

Set $M = 1$. With p_1, p_2 incoming and p_3, k outgoing, take

$$\begin{aligned} p_1 &= \left(\frac{5}{4}, 0, 0, \frac{3}{4}\right), & p_2 &= \left(\frac{5}{4}, 0, 0, -\frac{3}{4}\right), \\ p_3 &= \left(\frac{29}{20}, -\frac{21}{25}, 0, -\frac{63}{100}\right), & k &= \left(\frac{21}{20}, \frac{21}{25}, 0, \frac{63}{100}\right). \end{aligned} \tag{40}$$

Then

$$p_1 + p_2 = p_3 + k, \qquad p_i^2 = 1, \qquad k^2 = 0, \tag{41}$$

and

$$s = \frac{25}{4}M^2, \qquad \cos\theta = \frac{3}{5}. \tag{42}$$

This point is interior and noncollinear. The planar kinematics has a y-reflection selection rule; the verification therefore uses explicit parity-even real linear polarizations rather than interpreting symmetry zeros as dynamical cancellations.

Computational Proposition 6.2 (Exact real-shell certificate)**.** *For the real polarization tensors listed in Appendix E, the complete reduced tree amplitude at (40) is*

$$\boxed{\mathcal{A}_K^{\mathrm{red}}(MM \to Mh) = \frac{7881241032}{5584765625} \neq 0.} \tag{43}$$

It is real and nonsingular. The complete sum passes an exact Ward test for a rational gauge vector, invariance under an exact rational change of gauge representative, and equality between axial and de Donder internal gauges. Hence both $\mathcal{A}_K^{\mathrm{red}}$ and its normalized Born coefficient $\mathcal{A}_K$ are nonzero on a physical-shell neighborhood.

7 Full fourth-order calculation and the metric obstruction

The calculation of Proposition 6.2 is performed directly in the original fourth-order variables of (1), independently of the Einstein-frame factorization rail.

7.1 Perturbiner and bordered propagator

For four plane waves,

$$g_{\mu\nu} = \eta_{\mu\nu} + \sum_{a=1}^{4} \lambda_a \varepsilon_{\mu\nu}^{(a)} e^{ip_a \cdot x}, \tag{44}$$

the wave algebra retains only multilinear subsets of the labels and implements $\partial_\mu \mapsto iP_\mu$ automatically. The coefficient of $\lambda_1\lambda_2\lambda_3\lambda_4$ in the action gives the exact quartic contact. The same engine produces the quadratic 10×10 operator $K(P)$ and cubic current J_A in the symmetric-tensor basis.

No symbolic closed propagator is assumed. In each s, t, u channel one solves the exact de Donder-bordered system

$$\begin{pmatrix} K(P) & C(P)^T \\ C(P) & 0 \end{pmatrix} \begin{pmatrix} X \\ \lambda \end{pmatrix} = \begin{pmatrix} -J^{(cd)} \\ 0 \end{pmatrix}, \qquad \mathcal{A}_{\mathrm{exch}} = J^{(ab)T} X, \tag{45}$$

where

$$C_\mu(X) = P^\alpha X_{\alpha\mu} - \frac{1}{2} P_\mu X^\alpha{}_\alpha. \tag{46}$$

The exact field-equation residual and CX vanish in all three channels. Replacing C by the axial constraint $n^\mu X_{\mu\nu} = 0$ leaves the total amplitude unchanged.

7.2 Contact and exchange decomposition

At the rational point and polarization choice, the four contributions are

$$\begin{aligned} \mathcal{A}_{\mathrm{contact}} &= \frac{8364585636057}{410644531250}, \\ \mathcal{A}_s &= -\frac{836906329}{45312500}, \\ \mathcal{A}_t &= -\frac{5142800561}{111695312500}, \\ \mathcal{A}_u &= -\frac{90855828716}{205322265625}. \end{aligned} \tag{47}$$

Their exact sum is (43). The contact term alone is not gauge invariant. Replacing the external graviton by $k_{(\mu}\xi_{\nu)}$ makes the *complete* contact-plus-exchange sum vanish for an arbitrary exact rational test vector; shifting a physical representative by the same gauge tensor leaves the amplitude unchanged. Initial-massive Bose exchange is exact. The two-point coefficient against every symmetric-tensor basis direction vanishes for each external on-shell polarization, excluding equation-of-motion contamination.

Reversing all external momenta and conjugating the real polarization sections reproduces the contact and each exchange separately. The quadratic kernels are recomputed and satisfy $K(-P) = K(P)$ exactly. Thus the reverse process entering the physical adjoint is evaluated, not inferred.

7.3 From covariant graphs to the stationary Born operator

The perturbiner is a covariant Lagrangian calculation, whereas Proposition 2.1 is stated for the canonical stationary Born operator. The following standard tree-level reduction makes their relation explicit.

Proposition 7.1 (Covariant–stationary matching)**.** *For physical external states at (40), the complete contact-plus-exchange coefficient computed above obeys*

$$\langle f|\mathcal{B}_{2,K}(E)|i\rangle = \eta_{\mathrm{conv}}\mathcal{N}_{fi}\mathcal{A}_K^{\mathrm{red}} = \mathcal{A}_K \neq 0, \tag{48}$$

where $\mathcal{B}_{2,K}$ is the standard-real-form version of (12). The same identity holds for the reverse matrix element. No absorptive or prescription-dependent term occurs.

Proof. Pass to a gauge-fixed canonical formulation, equivalently to the regular second-order auxiliary-field form on the split branch. The stationary transition operator satisfies the Lippmann–Schwinger expansion

$$\mathcal{B}_K(E) = V_K + V_K(E - h_0 + i0)^{-1}\mathcal{B}_K(E), \tag{49}$$

whose order-ζ^2 coefficient is $v_{2,K} + v_{1,K}G_E v_{1,K}$. For each physical mass eigenmode, decomposing the covariant internal propagator as

$$\frac{1}{(P^0)^2 - \omega_\lambda^2 + i0} = \frac{1}{2\omega_\lambda}\left(\frac{1}{P^0 - \omega_\lambda + i0} - \frac{1}{P^0 + \omega_\lambda - i0}\right) \tag{50}$$

turns its two terms into the two old-fashioned time orderings and their $E - E_n$ denominators. Summing the complete physical polarization residues gives the intermediate-state sum in $v_{1,K}G_E v_{1,K}$, including the massive residue sign. The s, t, u graphs exhaust that sum at four points.

For a derivative or constrained theory, the canonical Legendre transform can redistribute instantaneous and seagull terms between "contact" and "exchange." Their sum is invariant: the canonical quartic matrix element plus its instantaneous pieces is precisely the four-wave contact coefficient after the complementary pieces from (50) are included. We therefore identify only the complete sum, not the individual gauge-dependent pieces, with the Born operator. This is the usual equivalence of covariant and old-fashioned tree perturbation theory [16]; local mass-eigenfield changes preserve its on-shell matrix element by the equivalence theorem [15].

At the rational point all three bordered internal systems are nonsingular, and Lemma 4.2 excludes an on-shell cubic intermediate state. Thus $i0$ in (50) reduces to the real principal-value resolvent (11). The external pole residues, box factors, polarization norms and Bose factors give exactly the real nonzero multiplier in (17). The separately evaluated reverse graphs give the ordinary-adjoint reverse matrix element with the same multiplier. □

7.4 Quarter-turn and the deformation cocycle

For the contact, three external massive fields give $(-i)^3 = i$. An exchange through an internal massless line has endpoint phase $(-i)^2(-i) = i$. An exchange through an internal massive line has endpoint phase $(-i)^3(-i)^2 = -i$, while the quarter-turned massive inverse kernel contributes -1, again giving i. Consequently every piece of (47) obeys Theorem 3.1, and

$$\langle Mh|\mathcal{B}_{2,+}(E)|MM\rangle = i\mathcal{A}_K, \qquad \langle MM|\mathcal{B}_{2,+}(E)|Mh\rangle = i\mathcal{A}_K. \tag{51}$$

Both states have exact free energy

$$E_{MM} = \frac{5}{4} + \frac{5}{4} = \frac{5}{2} = \frac{29}{20} + \frac{21}{20} = E_{Mh}. \tag{52}$$

Let $P_i = |MM\rangle\langle MM|$, $P_f = |Mh\rangle\langle Mh|$, and $\mathrm{Off}_{if}(X) = P_f X P_i + P_i X P_f$. On the ordered orthonormal basis $(|MM\rangle, |Mh\rangle)$,

$$\mathrm{Off}_{if}\, \mathcal{B}_{2,+}(E) = i\mathcal{A}_K\, \sigma_x, \tag{53}$$

and hence

$$\boxed{\mathrm{Off}_{if}(P_E S_2 P_E) = -2i\mathcal{A}_K\, \sigma_x \neq 0.} \tag{54}$$

In the reduced external-leg convention, the exact certificate is

$$(\eta_{\mathrm{conv}}\mathcal{N}_{fi})^{-1}\, \mathrm{Off}_{if}(P_E S_2 P_E) = -\frac{15762482064\, i}{5584765625}\, \sigma_x. \tag{55}$$

Equations (54) and (55) follow from the deformation-complex identity (14), rather than from an external-leg phase argument alone.

7.5 First-order metric ambiguity

The solution $R_1^{(0)}$ in (13) is defined up to a Hermitian G with $[G, h_0] = 0$. Such a change shifts the second-order source by $\frac{1}{2}[G, v_1 + v_1^\dagger]$. The complete regulated Fock-space statement in Lemma 4.2 gives

$$P_E(v_1 + v_1^\dagger)P_E = 0, \qquad P_E[G, v_1 + v_1^\dagger]P_E = 0. \tag{56}$$

Indeed, $[G, h_0] = 0$ makes G block diagonal in the P_E decomposition, so the second equality is the commutator of $P_E G P_E$ with the first zero block. Disconnected spectators and soft configurations have already been covered by the lemma's regulated Fock-space statement. The obstruction cannot be moved or canceled by choosing another analytic first-order metric representative.

Theorem 7.2 (Second-order positive-metric obstruction)**.** *The canonical analytic metric deformation based at the Paper-IV Lorentz-covariant positive real form admits no second-order solution for the complete Einstein–Weyl tree-level cubic and quartic vertex content. The obstruction is nonzero on the physical $MM \to Mh$ shell at (40) and on a nonempty open neighborhood of that point.*

Proof. Propositions 2.1 and 7.1 turn the exact four-point calculation into the nonzero source (54) in $\ker \mathrm{ad}_{h_0}$. No R_2 can have a commutator with h_0 equal to that source. Equation (56) makes the class independent of the homogeneous first-order metric freedom. The Ward, external-EOM, reverse-process, internal-gauge and LSZ arguments show that the displayed matrix element lies on the reduced physical shell. Real analyticity extends its nonvanishing from the interior point to an open neighborhood. □

8 Conventional Krein grading on physical cohomology

The positive obstruction does not by itself decide whether an indefinite completion can still be formulated. The narrower question addressed here is whether the conventional gravitational Krein grading can render the offending block null, as a continuous one-sided charge does at the perfect-square scalar boundary.

8.1 One-particle commutant

Let $J^{(1)}$ be a nondegenerate Hermitian involution on the physical one-particle cohomology (5). We require it to commute with the proper-orthochronous Poincaré representation and to agree with the free Krein real form of Paper IV.

The mass Casimir separates the massless and massive orbits, so no covariant endomorphism mixes $\mathcal{H}_h$ and $\mathcal{H}_M$. On the massive fiber, the exact spin-2 $SO(3)$ commutant is one-dimensional. There is one precision on the massless fiber: under the proper-orthochronous group, helicity $+2$ and helicity -2 are inequivalent complex representations, so covariance alone permits separate scalars. The complete commutant is therefore

$$J^{(1)} = \epsilon_+ I_{+2} \oplus \epsilon_- I_{-2} \oplus \epsilon_M I_5, \qquad \epsilon_\pm, \epsilon_M \in \{\pm 1\}. \tag{57}$$

Parity, or compatibility with the real gravitational field, exchanges the two massless helicities and imposes $\epsilon_+ = \epsilon_-$. Even without adding parity to the covariance group, agreement with the free signature (7) fixes

$$\epsilon_+ = \epsilon_- = +1, \qquad \epsilon_M = -1. \tag{58}$$

Proposition 8.1 (One-particle grading classification)**.** *Within the stated covariant class, the free real form uniquely fixes*

$$J^{(1)} = I_2 \oplus (-I_5). \tag{59}$$

No linear covariant grading can assign different signs within the massive spin-2 multiplet or mix the massive and massless shells.

8.2 Canonical second quantization and cluster multiplicativity

We do not classify every Poincaré-covariant involution on a multiparticle Hilbert space: such an operator could act on momentum-dependent multiplicity spaces. The conventional lift is the natural second quantization $\Gamma(J^{(1)})$. Explicitly, we assume that it is diagonal in asymptotic particle labels, acts on symmetrized simple tensors by the tensor product of its one-particle actions, and is multiplicative under separated clusters. Equivalently, its value on a sector is a scalar $j(n_h, n_M) \in \{\pm 1\}$ depending only on particle numbers. These assumptions imply

$$j(n_h + n'_h, n_M + n'_M) = j(n_h, n_M) j(n'_h, n'_M), \qquad j(0,0) = 1. \tag{60}$$

A homomorphism from the free commutative monoid $\mathbb{N}^2$ is fixed by its values on the two generators. Equations (58) and (60) therefore give

$$\boxed{J_F = (-1)^{N_M}.} \tag{61}$$

Assigning independent signs to unrelated multiparticle sectors would violate (60) and hence cluster factorization.

Proposition 8.2 (Canonical multiplicative Fock lift)**.** *Within the particle-number-diagonal, natural second-quantization class just specified, $J_F = (-1)^{N_M}$ is the unique lift of $J^{(1)} = I_2 \oplus (-I_5)$. No claim is made here about arbitrary momentum-dependent or non-factorizing multiparticle involutions.*

At the rational physical shell,

$$J_F |MM\rangle = +|MM\rangle, \qquad J_F |Mh\rangle = -|Mh\rangle. \tag{62}$$

The transition crosses the two Krein signatures.

9 Krein visibility of the obstruction

For an operator on the positive-frame Hilbert representatives, define the Krein adjoint

$$X^\sharp = J_F X^\dagger J_F. \tag{63}$$

On the minimal ordered basis $(|MM\rangle, |Mh\rangle)$, $J_{\min} = \mathrm{diag}(1, -1)$.

Consider first the forward block

$$X = i\mathcal{A}_K \, |Mh\rangle\langle MM|. \tag{64}$$

It is J_F-odd:

$$J_F X J_F = -X. \tag{65}$$

Its Krein adjoint and quadratic product are

$$\begin{aligned} X^\sharp &= -i\mathcal{A}_K \, J_F |MM\rangle\langle Mh| J_F = i\mathcal{A}_K \, |MM\rangle\langle Mh|, \\ X^\sharp X &= -\mathcal{A}_K^2 |MM\rangle\langle MM|. \end{aligned} \tag{66}$$

Consequently

$$\boxed{\mathrm{Tr}(X^\sharp X) = -\mathcal{A}_K^2 \neq 0.} \tag{67}$$

The sign is indefinite, but the operator is not null.

For the off-diagonal positive-metric obstruction block

$$\mathcal{O} := \mathrm{Off}_{if}(P_E S_2 P_E) = -2i\mathcal{A}_K \sigma_x, \tag{68}$$

one obtains

$$\boxed{\mathcal{O}^\sharp = \mathcal{O}, \qquad \mathcal{O}^\sharp \mathcal{O} = -4\mathcal{A}_K^2 I_2, \qquad \mathrm{Tr}(\mathcal{O}^\sharp \mathcal{O}) = -8\mathcal{A}_K^2 \neq 0.} \tag{69}$$

Proposition 9.1 (Conventional Krein non-nullity)**.** *The physical $MM \to Mh$ block and its off-diagonal metric obstruction are non-null for the canonical multiplicative fundamental symmetry* (61)*. The obstruction is Krein-self-adjoint, indefinite, and visible to the finite-block quadratic trace.*

The algebraic distinction from a one-sided continuous grading is simple. For $\mathbb{Z}_2$,

$$1 + 1 = 0 \pmod 2. \tag{70}$$

The adjoint of an odd operator is odd and their product is even. Thus oddness supplies no charge-null lemma:

$$\boxed{J_F\text{-odd} \;\not\Longrightarrow\; \text{null.}} \tag{71}$$

By contrast, at the scalar $O(1,1)$ boundary a strictly one-sided nonzero continuous charge remains nonzero after the relevant product, and an invariant graded trace can kill it [30, 19].

Remark 9.2 (What is not ruled out). Equation (69) does not say that a Krein-self-adjoint interacting Hamiltonian cannot exist. Indeed, it displays Krein-self-adjointness of the minimal obstruction block. What fails is the stronger proposal that the conventional massive-number grading makes the branch-changing component null or supplies a positive generalized Born contribution. Indefinite pseudo-unitary evolution and a positive probability calculus are different questions.

10 No scalar-like charge-null mechanism

Could a continuous internal charge make the obstruction one-sided? Poincaré covariance first restricts any linear charge on the irreducible massive fiber to one scalar q_M. The free real graviton is neutral, $q_h = 0$. The verified nonzero interaction vertices impose

$$3q_M = 0 \quad (MMM), \qquad 2q_M + q_h = 2q_M = 0 \quad (MMh). \tag{72}$$

Proposition 10.1 (Uniform abelian charge no-go)**.** *There is no nontrivial uniform abelian charge assignment to the physical massive spin-2 irrep, with neutral graviton, that is respected by the Einstein–Weyl interaction.*

Proof. Equations (72) hold in any abelian charge group. Since $q_M = 3q_M - 2q_M$, they imply $q_M = 0$. The conclusion does not rely on torsion-freeness; $\gcd(2,3) = 1$ eliminates finite torsion charges as well. □

The MMM vertex has no real cubic decay shell but is a nonzero algebraic interaction vertex, so it already breaks massive-number parity at the action level. More decisively, the exact real-shell process changes N_M from 2 to 1 and proves

$$[(-1)^{N_M}, S] \neq 0 \tag{73}$$

on the physical scattering shell. Therefore the ingredients used to prove factorization simultaneously exclude an $O(1,1)$-like charge replacement.

11 BRST cohomology

The higher-derivative massive branch is a physical reduced mode and must not be confused with Faddeev–Popov ghosts. The external states in the four-point calculation are TT massive polarizations and TT massless graviton polarizations. The exact Ward identity, invariance under changes of gauge representative, axial/de Donder equality, and external inverse-propagator zeros show that the amplitude descends to reduced physical BRST cohomology.

Let Q denote the nilpotent BRST charge and let $|i\rangle, |f\rangle$ be closed physical representatives. If an operator were BRST-exact,

$$\mathcal{O} = \{Q, Y\}, \tag{74}$$

then, under the standard decoupling assumptions,

$$\langle f|\mathcal{O}|i\rangle = \langle f|QY|i\rangle + \langle f|YQ|i\rangle = 0. \tag{75}$$

The matrix element (54) is nonzero.

Proposition 11.1 (BRST survival)**.** *The exact obstruction defines a nonzero operator on physical BRST cohomology and is not BRST-exact:*

$$[\mathcal{O}] \neq 0. \tag{76}$$

It cannot be placed in the standard BRST-null ideal.

This proposition concerns the induced operator on physical cohomology. We do not construct an extension of J_F to the complete gauge-fixed Fock space of Faddeev–Popov ghosts and contractible sectors; accordingly we do not call J_F a full-space BRST-compatible fundamental symmetry.

Appendix G records the finite-complex algebra checked by the verification suite. In a splitting into physical cohomology and a contractible doublet, the physical-to-physical block of $\{Q, Y\}$ vanishes for a completely general Y, while the physical obstruction block survives unchanged.

12 Master theorem

Theorem 12.1 (Interacting completion obstruction)**.** *Let scalar-free Einstein–Weyl gravity* (1) *be in its regular split phase about flat space. Use the covariant free completion classes of Paper IV and the analytic metric-deformation class of Paper V. Then:*

1. *the Lorentz-covariant positive pseudo-Hermitian completion is protected at first perturbative order,* $\mathfrak{o}_{+}^{(1)} = 0$*;*

2. *at second order the physical process* $MM \to Mh$ *generates the nonzero shell obstruction* (54)*;*

3. *hence the canonical Paper-IV positive base metric has no analytic second-order deformation for the complete tree-level cubic and quartic Einstein–Weyl vertex content;*

4. *on physical BRST cohomology, the free one-particle real form has the canonical particle-number-diagonal, natural and cluster-multiplicative lift* $J_F = (-1)^{N_M}$*;*

5. *for this grading the obstruction is non-null,* $\mathrm{Tr}(\mathcal{O}^\sharp \mathcal{O}) = -8\mathcal{A}_K^2 \neq 0$*, and survives physical BRST cohomology; and*

6. *the nonzero* MMM *and* MMh *vertices prohibit a nontrivial uniform abelian charge grading analogous to the scalar* $O(1,1)$ *mechanism.*

Thus, through second tree order, neither the canonical analytic metric deformation nor the conventional massive-number null mechanism supplies a positive-probability completion in the stated classes.

Proof. Item 1 is Theorem 4.3. Items 2 and 3 are Theorem 7.2, independently supported by the factorization Proposition 5.1 and the Born–deformation identity (14). Item 4 is Propositions 8.1 and 8.2. Item 5 is Propositions 9.1 and 11.1. Item 6 is Proposition 10.1. □

13 Scope and nonclaims

The hypotheses in Theorem 12.1 are part of the result. The paper rules out:

- analytic deformation through second order of the canonical Paper-IV Lorentz-covariant positive base metric by the complete tree-level cubic and quartic vertex content;
- a conserved naive massive-number ghost parity;
- conventional covariant linear one-particle fundamental symmetries that agree with the free real form but differ on selected massive helicities;
- independent particle-number-sector signs within the natural, cluster-multiplicative scalar Fock-lift class;
- a uniform abelian continuous- or torsion-charge null mechanism compatible with the verified vertices; and
- standard BRST-exact nullity of the physical obstruction.

It does *not* prove:

- that no nonperturbative or differently pointed positive completion exists;
- that the full interacting Hamiltonian has complex energies;

- that every nonlocal or non-factorizing grading fails;
- that no momentum-dependent Poincaré-covariant involution exists on multiparticle multiplicity spaces;
- that the cohomological grading J_F necessarily extends to a fundamental symmetry on the complete gauge-fixed BRST complex;
- that no singular $M \to 0$ or conformal-boundary completion can be defined;
- that a Maldacena-type boundary truncation of the solution algebra is inconsistent [24]; or
- that the result extends unchanged to quadratic gravities with a scalaron, matter, cosmological constant, or a different background.

Two prominent prescriptions lie outside the assumptions for a particularly concrete reason. In the unstable-resonance treatment of Donoghue and Menezes [12], the massive ghostlike pole does not belong to the asymptotic spectrum and only stable states enter the unitarity sum. In the fakeon construction [13], the massive mode is purely virtual and amplitudes are defined by a nonanalytic continuation. The present theorem instead puts physical M states on both sides of the scattering process and studies an analytic metric deformation. It therefore neither refutes nor subsumes either prescription.

Any surviving null-sector construction must therefore leave the conventional class. Possibilities include a grading singular at the conformal boundary, a nonlinear mixing of asymptotic particle sectors, a nonlocal observable algebra, a failure of ordinary cluster factorization, or boundary conditions that remove part of the fourth-order solution space. These are qualitatively different from a straightforward gravitational lift of the Bateman–Turok mechanism.

There is a second, logically independent reason that such a lift cannot be called straightforward. Paper V [30] proves, with dependency tag `REDUCED-MODE`, that the complete final-state collinear real normalization response does not cancel the virtual external-mass logarithm under any continuous axis-compatible parent/daughter gluing, including the physical pair threshold. This does not transfer the scalar coefficient or obstruction to gravity. It identifies an additional construction that a gravitational completion would need independently: a distributional normalization, degenerate or dressed asymptotic states, or a resummation prescription defined on the gravitational BRST carrier. The scalar certificate `REVERSE_PHYSICS_BT_ASYMPTOTIC_GENERATOR_PREFLIGHT_V1` gives an exact finite witness for the dressed alternative. After division by the scalar Born rate, projector idempotence forces dimensionless hard and three-pair blocks $-1/16$ and $+1/16$; multiplying back gives the physical $-3/512$ normalization against the scalar real response. The same certificate shows that the ordinary single-denominator cubic Fock generator vanishes in the massless collinear limit, so a Jordan/distributional asymptotic term is required. The scalar certificate `REVERSE_PHYSICS_BT_RT_ JORDAN_KERNEL_V1` derives the resonant two-annihilator part of that term from the published composite map. After a declared repair of an internal Appendix C oscillator-label inconsistency, its apparent t, t^2 growth cancels exactly on the (Ω, Υ) carrier and a formal finite-mode anti-Krein generator exists. The resulting Krein Gram has cubic splitting-endpoint poles, so its distributional continuum domain, oscillatory sectors, and the target $1/48$ remain open. Gravity would still require an independent BRST-compatible construction rather than importing either scalar coefficients or this reduced-mode endpoint obstruction. The scalar endpoint extension retains three reflection-even local constants; distinct exact plus and cutoff prescriptions disagree even on the inclusive constant test. Hence neither scalar symmetry nor the interior kernel selects the target coefficient without the omitted oscillatory and squeezed sectors, and no such ambiguity may be imported as a gravitational normalization. On the published fixed-vacuum oscillator grading, the scalar oscillatory and squeezed sectors lie in the strictly negative BT charge radical and are trace-null. The covariant vacuum-orbit calculation shows that this exclusion does not transfer to the broken-vacuum carrier. Conversely, the free cross-CCR and projection idempotence force the scalar coisometric support to be the identity to every formal order. The remaining scalar canonical identities are insufficient selectors. The exact certificate `REVERSE_PHYSICS_BT_CANONICAL_ ENDPOINT_AMBIGUITY_V1` constructs a three-parameter canonical/projector countermodel in the endpoint carrier: hard and endpoint trace shifts cancel for arbitrary parameters, and $1/48$ is compatible but not derived by those identities. This is not a claim that all parameters occur in the actual BT transport; it isolates the deferred continuum map or an independent physical endpoint condition as the missing scalar datum. It is not a range defect or phase prescription that gravity could copy. The scalar certificate `REVERSE_PHYSICS_BT_FULL_OFF_ RESONANT_PROJECTOR_V1` retains the exact parent–daughter energy deficit and restores the full radial daughter measure. The repaired carrier has no surviving t, t^2 powers and restricts exactly to the resonant kernel, but its universal soft cross-Gram is $-1/(2r^3)$. Multiplication by r^2dr leaves $-(1/2)dr/r$, whose finite part shifts by $(1/2)\log c$ under a common cutoff rescaling. Thus full phase space reduces the collinear three-jet ambiguity to one soft normalization on the declared chart without selecting it. A scalar soft-collinear asymptotic Hamiltonian or hard matching operator with cancelling regulator response is required. This is only a `LOCAL-ALGEBRAIC`, `REDUCED-MODE` classification; it is not a graviton/BRST transfer, a complete probability, or a `LORENTZIAN-CAUSAL` result. The scalar certificate `REVERSE_PHYSICS_BT_SOFT_CHARGE_ RESOLVED_FLOW_V1` fixes the numerical normalization but exposes a charge obstruction. Parent-index raising, Bose reduction, angular measure, and the outgoing $2!/3!$ projector ratio yield exactly $+1/48$ per unordered pair and $+1/16$ in total; idempotence forces the hard $-1/16$. Yet the two logarithmic scalar rows have generator charges $(+1, -1)$ and $(-1, +1)$. Projecting the published oscillator map first to its one-sided nonpositive image therefore gives zero, while obtaining the candidate neutral $1/48$ requires a background zero-mode completion carrying both compensating signs. That operator and the full order-λ pushforward are not public. This trichotomy is `LOCAL-ALGEBRAIC`, `REDUCED-MODE`; none of its scalar charge grading, coefficient, or formal finite-cutoff anti-Krein flow transfers to the metric BV complex. The exact scalar certificate `REVERSE_PHYSICS_BT_ZERO_MODE_ EQ19_TRILEMMA_V1` constructs the global shift-orbit operator $Z = e^{\lambda\phi_0}$. Equation (16) factorizes as $\Omega = Z\widehat{\Omega}$, $\Upsilon = Z^{-1}\widehat{\Upsilon}$, and the unique covariant dressing neutralizes both soft logarithmic rows while leaving their conditional coefficient at $1/48$. But it also changes the Appendix-C squeeze to neutral $Z^2(b^\dagger_\Upsilon)^2$. Fixing $Z = 1$ recovers the apparent charge -2, yet the ideal $(Z-1)$ is not invariant because $\delta(Z-1) = Z \equiv 1$ modulo that ideal. The deferred full zero-mode module, trace, and nonlinear pushforward are therefore indispensable. This is a scalar `LOCAL-ALGEBRAIC`, `REDUCED-MODE` obstruction only; it neither establishes Eq. (19) nor transfers a coefficient, state, positivity claim, or causal construction to gravity. The further scalar certificate `REVERSE_PHYSICS_BT_SQUEEZED_ VACUUM_IMPLEMENTABILITY_V1` tests the displayed Appendix-C vacuum relation in the positive topology of its ordinary carrier. Its normalized creation kernel is $1/(8|\mathbf{p}|^2)$; the first two-particle norm density is $\rho^2(\epsilon^{-1} - \Lambda^{-1})/(64\pi^2)$, and a six-mode lower bound grows linearly with box size for every uniformly equivalent fundamental symmetry. The pair block independently fails the Hilbert–Schmidt condition. Hence Krein nullity is not ordinary Fock normalizability. This exact scalar `LOCAL-ALGEBRAIC`, `REDUCED-MODE` obstruction leaves extended or inequivalent non-Fock representations open and supplies no graviton state, BRST theorem, coefficient, or `LORENTZIAN-CAUSAL` result. The full scalar exponential can be made modewise convergent by the explicit weight $\rho_\mu(p) = 4\gamma\mu^2 p^2/(p^2+\mu^2)$, $0 < \gamma < 1$, but its inverse is unbounded and the thermodynamic vacuum is inequivalent to the ordinary Fock sector. The raw scalar shear also fails the positive-Hilbert Bogoliubov relation, so generic extended positive-boson implementation theorems do not supply its cross-Krein map or cyclic Born trace. A direct scalar construction does implement the covariant squeeze on paired cores and preserves the finite-rank trace. It also proves the limiting obstruction: a finite normalized cyclic orbit trace positive on the ghost-even cone assigns zero to localized rank-one projectors, whereas the finite-rank trace has infinite identity weight, and the transported vacuum projector's trace norm grows as $\exp(V\ell)$ with

$$\ell = \frac{\mu^3}{12\pi}\left[(1+\gamma)^{3/2} + (1-\gamma)^{3/2} - 2\right] > 0.$$

Thus no normal thermodynamic Born trace follows. This remains solely a scalar carrier classification. Its dependency tags are `LOCAL-ALGEBRAIC` and `REDUCED-MODE`; it does not construct a graviton, BRST, or causal state space. A scalar semifinite construction retains localized orbit projectors without assigning finite weight to the identity. Its canonical trace satisfies $\mathrm{Tr}_\infty(E_n) = 1$ and $\mathrm{Tr}_\infty(1) = +\infty$. For a finite incoming Krein projection and a finite exhaustive output partition, the relative weights

$$p_i = \frac{\mathrm{Tr}_{\mathrm{fin}}(A_i^\dagger A_i)}{\mathrm{Tr}_{\mathrm{fin}}(P_{\mathrm{in}})}$$

are nonnegative and sum to one whenever each process A_i satisfies the BT weak ghost decomposition. The normalized corner functional is not a trace: $P = E_{00}$, $X = E_{01}$, $Y = E_{10}$ give $\omega_P(XY) = 1$ and $\omega_P(YX) = 0$. Hence this result preserves the finite-normalized-trace no-go while constructing a finite-detector conditional calculus. The exact scalar certificate `REVERSE_PHYSICS_BT_SEMIFINITE_ RELATIVE_BORN_WEIGHT_V1` carries dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`. It does not control the unbounded squeeze on the full trace ideal, construct a thermodynamic state, reproduce Eq. (19), transfer to the metric BV complex, or establish any `LORENTZIAN-CAUSAL` statement. The complete public scalar quadratic map closes over three leading inverse images for every target $b_\Upsilon(s,p)$. The off-resonant extraction phase of the opposite-sign $a_2(-s,-p)$ image exactly cancels its Appendix-C phase. The signed sum is time independent and has only

$$\delta b_\Omega = \frac{s_1 e_1 + s_2 e_2}{2e_1 e_2} b_\Omega^{(s_1)} b_\Omega^{(s_2)}, \qquad \delta b_\Upsilon = -\frac{s_2}{2e_1} b_\Omega^{(s_1)} b_\Upsilon^{(s_2)} - \frac{s_1}{2e_2} b_\Upsilon^{(s_1)} b_\Omega^{(s_2)}.$$

Its parent cross-Gram vanishes, so the resonant-only dr/r carrier and its conditional 1/48 normalization do not occur in the completed quadratic map. The mixed-sign rows satisfy the exact order-λ cross-CCR Ward identity; after the covariant Z dressing, the unsqueezed factor has $UP_0U^{-1} = P_0 + \lambda[K, P_0] + O(\lambda^2)$ with wholly neutral charge support and $Q_1 = 0$. The certificate `REVERSE_PHYSICS_BT_FULL_SIGNED_ QUADRATIC_CLOSURE_V1` has dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`. This Eq. (19) pushforward is not the scalar S-matrix transition block and establishes no metric-BV or LORENTZIAN-CAUSAL statement; its squeezed-vacuum/dynamical-zero-mode projector trace, continuum domain, and higher orders remain open. On the finite paired core, cross-Krein isometry gives $(SKS^{-1})^\dagger(SKS^{-1}) = S(K^\dagger K)S^{-1}$ while the detector is transported by the same similarity. The cyclic finite-rank trace therefore keeps the completed scalar quadratic coefficient at zero. A bare one-pair occupation of the squeezed vacuum compares a fixed b-Fock detector and is not a further Eq. (19) term. The scalar certificate REVERSE_PHYSICS_BT_SQUEEZED_ DETECTOR_SIMILARITY_V1 records this LOCAL-ALGEBRAIC, REDUCED-MODE distinction without promoting it through the non-trace-class thermodynamic limit or into the metric BV complex. For the physical scalar process, the independently computed five-point response remains $+1/512$ per pair and $+3/512$ over three pairs, equivalently $1/48$ and $1/16$ after Born normalization. The axis-compatible physical virtual daughter-ratio response is zero, leaving $+3/512$ uncancelled. The public R_t response is also zero but is not an S-matrix summand. The object-typed certificate REVERSE_PHYSICS_ BT_INCLUSIVE_NLO_ OBJECT_LEDGER_V1 therefore requires a separate hard/dressed response $-3/512$ and leaves the complete scalar NLO probability unestablished. None of this scalar ledger transfers to the metric BV complex. A scalar conditional completion theorem further shows that any regulated physical shell S-matrix with $S_{\text{phys}} = 1 + xA + x^2B + \cdots$ and $S_{\text{phys}}^\dagger S_{\text{phys}} = 1$ must obey $2\,\mathrm{Re}\,B_{hh} = -\|Ah\|^2 = -1/16$. It therefore forces the absolute hard response $-3/512$ from the measured real column, independently of phases. The exact finite witness in REVERSE_PHYSICS_BT_PHYSICAL_SHELL_ PSEUDOUNITARY_COMPLETION_V1 does not construct the continuum dressed scalar S-matrix, prove beyond-tree positivity, or supply a metric-BV carrier. The scalar logarithmic-shell limit sharpens this boundary. Disjoint endpoint shells have $\|A_nh - A_mh\|^2 = 1/8$, and their exact isometries have hard-column distance squared $2\sin^2(x/4)$, excluding an ordinary strong Møller limit. The weak limit loses the real channel and is not pseudo-unitary. The exact fixed endpoint-fibre pullback certified in REVERSE_PHYSICS_BT_LOG_SHELL_ MOLLER_LIMIT_V1 preserves the leading-log scalar probability cylinder but supplies neither a local detector affiliation nor a metric-BV construction. The scalar endpoint fibre is now affiliated with an asymptotic mass-resolution detector algebra: for a monotone profile $q_R(y) = q(y - R)$, the positive difference $d_{R,a} = q_{R+a} - q_R$ has exact trace a, and its normalized square root is translation covariant. Sharp and C^1 cubic profiles therefore give the same $+1/16$ real and $-1/16$ hard response. Certificate REVERSE_PHYSICS_BT_DETECTOR_ RESOLUTION_DILATION_V1 establishes this physical scalar leading-log resolution response only on its declared final-pair cylinder. Abel integration of the formal scalar soft Hamiltonian gives coefficient $-id/(\epsilon + id)$ and logistic norm profile $q_R(y) = (1 + e^{2(y-R)})^{-1}$ with $R = \log(\alpha/\epsilon)$. Thus inverse Abel time implements the same resolution translation. The coherent columns have fixed-ratio squared distance $\log(c)(c-1)/(c+1) > 0$; the resulting orthogonal-increment Naimark carrier is certified in REVERSE_PHYSICS_ BT_ABEL_NAIMARK_ ASYMPTOTIC_ DILATION_V1. This identifies a scalar detector automorphism, not the formal R_t lowering operator with the physical S-matrix splitting operator. Its auxiliary resolution coordinate, scalar detector trace, cancellation, and coefficient do not transfer to the metric BV complex or establish a LORENTZIAN-CAUSAL result. The scalar amplitude-level comparison has also been completed on its declared external mass-jet cylinder. The physical splitting map has full-rank raised Gram ρI_2 above unequal-mass threshold, whereas the completed public quadratic R_t map has rank-one nilpotent raised Gram $\left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$. The exact rank/Jordan obstruction in REVERSE_PHYSICS_BT_PHYSICAL_ COLLINEAR_OPERATOR_ FACTORIZATION_V1 prevents a scalar metric-compatible identification of those two operators; it neither supplies a metric-BV splitting map nor transfers the scalar 1/48, phase, carrier, or positivity statement to gravity. The scalar map also has a finite-regulator direct-integral realization, but its exact Gram $I(r) = 5/24 + r[(1/4)\log r + 1/12] + O(r^2 \log r)$ is not differentiable at the massless axis. Hence no strongly C^1 scalar Møller column on a fixed bounded-pairing carrier can implement the delta-prime derivative. The relative 1/48 scale cocycle is linearized only on the rigged affine sector $\mathrm{span}\{1, s\} \subset \mathcal{S}'(\mathbb{R}_s)$, whose generators are not L^2 vectors. Certificate `REVERSE_PHYSICS_BT_RIGGED_ RESOLUTION_JORDAN_ MOLLER_V1` does not supply a scalar endpoint state or generalized-Born functional, and none of its rigging, cocycle, or differentiability statement transfers to the metric BV complex. On the scalar resolution-local CCR net, the rank-two Gram further has a positive coherent realization. A finite interval of length a has coherent norm $a/16$, hard no-emission probability $e^{-a/16}$, and normalized Poisson count law $e^{-a/16}(a/16)^n/n!$. The compatible local Weyl implementers define a locally normal algebraic state, while the norm grows as $L/16$ and excludes a global scalar Fock implementer. The certificate `REVERSE_PHYSICS_BT_RESOLUTION_ LOCAL_COHERENT_BORN_ PROCESS_V1` constructs this scalar leading-log process only under a coherent independent- increment completion. It neither derives scalar multi-emission dynamics nor supplies a spacetime-local scalar S-matrix, and no part of its auxiliary CCR net, Poisson law, or rank-two scalar GNS factor transfers to metric BV data. The subsequent complete scalar six-point tree calculation rules out that independent-increment completion in its nested strongly ordered sector: $P_2(a) = 5a^2/512$ rather than $a^2/512$, with scalar second factorial cumulant $a^2/64$. This factor-five result is certified by `REVERSE_PHYSICS_BT_SIX_POINT_ STRONGLY_ORDERED_TREE_V1`; it requires a non-Gaussian scalar resolution state but does not select one from only two moments. The calculation uses scalar external mass jets and scalar Källén thresholds. Neither its coefficient nor its non-Poisson interpretation transfers to the metric BV carrier. The complete scalar seven-point tree now supplies the third ordered moment: $P_3(a) = 9a^3/8192$ and $\kappa_3(a) = 7a^3/2048$. It excludes the gamma–Cox completion, which predicts $15a^3/8192$, while satisfying the scalar Stieltjes moment bound. The certificate `REVERSE_PHYSICS_BT_ SEVEN_POINT_ COX_SELECTION_V1` constructs a positive locally normal two-atom Cox state matching the first three scalar moments. This state is an auxiliary resolution-local controlled Weyl mixture, not a scalar spacetime Møller operator, a complete scalar probability, or an all-order dynamical law. Its coefficient, intensity carrier, positivity construction, and threshold calculation do not transfer to the metric BV complex. The complete scalar eight-point preflight does not yet add a fourth moment. All 34300 trees give a bare projected hierarchy of ordered valuation $(0, 0, -1)$, and two fixtures with identical soft data but one hard adjacent invariant changed from 33 to 34 have leading residues differing by exactly 257/1568. Certificate `REVERSE_PHYSICS_BT_EIGHT_POINT_ HARD_PROFILE_OBSTRUCTION_V1` therefore leaves the threshold-integrated scalar fourth moment and the two-atom Cox test open until the outer Källén reduction or a hard-profile quotient is computed. The exact outer reduction retains the two full profiles and finds their leading difference to be the pure J_2 direction $771/(1568e_3u)$. Since the fixed-invariant $r \log r$ coefficient of J_2 is one, the threshold coefficient remains nonzero, $771/(1568e_3)$; at the declared $r = 5e_3$ the outer logarithms differ by 3855/1568. Certificate `REVERSE_PHYSICS_BT_EIGHT_POINT_ OUTER_THRESHOLD_PROFILE_V1` is followed by the symbolic τ_3 profile. Its exact two-fixture difference is $3(27213\tau_3^2 - 13462\tau_3 + 29485)/(156800\tau_3^2\tau_4)$. The next J_1, J_2, J_3 reduction at $a_3 = 2$ leaves $334239/313600 \neq 0$. Certificate `REVERSE_PHYSICS_BT_EIGHT_POINT_MIDDLE_THRESHOLD_ PROFILE_V1` is followed by the symbolic τ_2 profile. The two complete hard fixtures differ by $3(2090\tau_2^2 + 1146\tau_2 + 981)/(12800\tau_2^2)$, and the third J_1, J_2, J_3 reduction at $a_2 = 3$ leaves

$23229/6400 \neq 0$. Certificate REVERSE_PHYSICS_BT_EIGHT_POINT_TAU2_THRESHOLD_PROFILE_V1 leaves only the independent-mass inner threshold open. The displayed value tests profile collapse; it is not a normalized fourth moment. None of the scalar profile certificates is imported into the metric BV complex, and none supplies a gravitational fourth jump. The independent-mass test fixes only the redundant scale $a_0 = 1$ and retains $r = a_1/a_0$, $u = \tau_1/a_0$. The complete profile difference is $3\{9r^2 - 18r(u+1) + 34u^2 - 18u + 9\}/(128u^2)$. Exact invariant-cutoff residues give the two $r \log r$ coefficients $-6699/128$ and $-7149/128$, whose difference is $225/64 \neq 0$. Certificate REVERSE_PHYSICS_BT_EIGHT_POINT_INNER_THRESHOLD_OBSTRUCTION_V1 therefore obstructs the hard-independent scalar fourth jump on the declared fixtures after all four threshold functionals. A profile/channel-valued successor remains open; no normalized probability or gravitational result is inferred. This profile direction cannot be identified with the forced cross-Krein complement alone. The two fixtures share $\rho = 819/4000$, the same missing Gram and $c_1 = 14819/8000$, yet have unequal fourth responses. Certificate REVERSE_PHYSICS_BT_COMPLEMENT_HARD_PROFILE_SEPARATION_V1 requires at least one additional hard-profile base coordinate beyond ρ. It fixes an algebraic type, not a BT dynamical bundle or Eq. (19). On the resulting two-point base, the fourth effect is $K_4 = \mathrm{diag}(-6699/128, -7149/128)$. Its external eight-leg sign and every inherited normalization sign are positive, so every faithful positive profile trace is negative. Certificate
REVERSE_PHYSICS_BT_EIGHT_POINT_PROFILE_POSITIVITY_OBSTRUCTION_V1
therefore obstructs a diagonal completely positive fourth jump. It also constructs the minimal two-negative-direction Krein-skew lift over the hard base, but not a positive probability, charge-compatible BT operator, or Eq. (19). The invariant charge split further shows that this lift is not the negative-charge Q remainder. Transporting the actual target charge action through the certified missing leg C gives the null charge basis of G_{miss}, in which the vector used by the lift has two separately null charge components; its norm is entirely their cross pairing. Thus $B_+^\sharp B_+ = B_-^\sharp B_- = 0$ while $B_+^\sharp B_- + B_-^\sharp B_+ = K_4$. Certificate
REVERSE_PHYSICS_BT_EIGHT_POINT_KREIN_CHARGE_LOCALIZATION_V1
localizes the effect in the total-charge-zero sector and leaves the neutral higher-composite or dynamical trace unconstructed. The smallest neutral symmetric-bosonic carrier can now be classified. Over the two hard-profile modes its degree-two charge-zero sector has inertia $(3, 1)$ and cannot carry the rank-two negative profile block, whereas degree four has inertia $(6, 3)$. Two normalized occupation-antisymmetric degree-four vectors are total-charge zero, mutually orthogonal and have Gram $-2I_2$; the certified amplitudes therefore reconstruct K_4 in a charge-compatible Krein-skew generator. Certificate
REVERSE_PHYSICS_BT_NEUTRAL_BOSONIC_COMPOSITE_LIFT_V1
also shows why this is not yet Eq. (19): the negative neutral subspace is odd under the induced ghost parity, while the positive profile source is even. Every ghost-even forward block then has pullback $D^T(G_4\kappa)D \succeq 0$, so it cannot reproduce negative K_4. Thus the minimal block is not by itself the ghost-even neutral P term; a ghost-odd source partner or larger dynamical projector/trace remains necessary. The minimal two-dimensional odd partner does produce an exact finite projector. With partner metric $-I_2$, composite metric $-2I_2$, slope $T = \mathrm{diag}(\sqrt{6699}/16, \sqrt{7149}/16)$ and $L = (I_2, T)^T$, the graph projector $P = L(L^T\eta L)^{-1}L^T\eta$ is rank two, idempotent, Krein self-adjoint, neutral and ghost even, with finite algebraic trace $\mathrm{tr}(P^\sharp P) = 2$. Certificate
REVERSE_PHYSICS_BT_NEUTRAL_GRAPH_PROJECTOR_EXTENSION_V1
also proves that this minimal graph cannot be affiliated to the original positive even profile

source: parity forces the direct map to vanish and the graph metric is negative definite. A larger signature-balanced R_t projector or dynamical trace remains necessary. The required odd source type is nevertheless already present in the full public carrier. In the two-profile neutral degree-four J-Fock sector, which has inertia $(6, 3)$, the two occupation-antisymmetric vectors normalized by $1/2$ have Gram $-I_2$, charge zero and odd ghost parity. Since the certified complement map obeys $CS = -\rho I_2$ on the charge basis, $\mathrm{Sym}^4(C) = \rho^4 I_9$ and maps the complement vectors to $\sqrt{2}$ times these public vectors. Its inverse $I_2/\sqrt{2}$ is therefore an exact ghost-even isometry from the public odd sector to the selected composite. Certificate
`REVERSE_PHYSICS_BT_PUBLIC_FOCK_ ODD_SOURCE_AFFILIATION_V1`
replaces the abstract odd summand by a concrete public Fock subspace. It does not affiliate the original positive scalar source or establish a physical fourth probability. The graph slope belongs to the eight-point physical-response ledger, not automatically to the distinct $R_t P_\chi R_t^\dagger$ projector ledger; their identification requires an explicit intertwiner. On the covariant zero-mode Laurent algebra the charge-support part of Eq. (19) nevertheless holds to all formal orders. The exact Eq. (16) pullback sends Ω to $\lambda^{-1} Z e^{\lambda\varphi}$ and Υ to $Z^{-1} e^{-\lambda\varphi}[\Box\varphi + \lambda(\partial\varphi)^2]$, so it intertwines the source orbit derivation and target boost charge coefficient by coefficient. Free time translation preserves the identity because both free Hamiltonians are charge neutral. Perturbative two-sidedness makes the inverse pushforward equivariant. Therefore every finite shift-invariant nonzero-mode projector remains idempotent, self-adjoint and wholly charge zero, with $Q_{\text{negative}} = 0$. Certificate
`REVERSE_PHYSICS_BT_COVARIANT_FORMAL_ EQ19_CHARGE_SUPPORT_V1`
does not promote the full Eq. (19): ghost evenness, time independence, asymptotic limits and the continuum trace remain open, and the charge theorem does not descend through $Z = 1$ because $\delta_\phi(Z-1) = Z \equiv 1 \pmod{(Z-1)}$. For the unsqueezed nonlinear factor, ghost parity supplies a separate exact calculation at first nonlinear order. The complete public signed kernel has a charge-$+1$ Krein-skew species generator K_+, and covariance fixes the neutral generator to $Z^{-1}K_+$. For the standard two-species one-particle projector, $P_1 = Z^{-1}[K_+, P_0]$ is nonzero, rank four and stationary, whereas $\kappa P_1 \kappa = Z\kappa[K_+, P_0]\kappa$. Independence of the two Laurent powers proves that the neutral correction is not ghost even. Its canonical odd remainder is orthogonal to the even part but has exact finite relative norm $-7/288$, rather than zero. Certificate
`REVERSE_PHYSICS_BT_COVARIANT_GHOST_ PARITY_BRANCH_OBSTRUCTION_V1`
records this unsqueezed-factor obstruction. It is not by itself the full projector coefficient because $R = SU$ conjugates it by the Appendix-C squeeze. The minimal symmetric generator $\frac{1}{2}(Z^{-1}K_+ + Z\kappa K_+ \kappa)$ does yield an exact all-order finite ghost-even projector, but its added conjugate branch has no public R_t or source affiliation. Neither this repair nor the obstruction is a continuum probability or a gravitational/BRST result. The same-chart regular source affiliation is itself obstructed. On the off-shell commutative local-symbol algebra, zero-jet augmentation sends $O = \lambda^{-1} Z e^{\lambda\varphi}$ to the Laurent unit $\lambda^{-1}Z$, while it sends $Y = Z^{-1} e^{-\lambda\varphi}[\Box\varphi + \lambda(\partial\varphi)^2]$ to zero. The first symbol has explicit inverse $\lambda Z^{-1} e^{-\lambda\varphi}$; the second cannot be a unit. Unit preservation therefore forbids any regular unital ghost-parity automorphism exchanging them, independently of normalization. Certificate
`REVERSE_PHYSICS_BT_PERTURBATIVE_GHOST_ PARITY_UNIT_OBSTRUCTION_V1`
does not impose that augmentation on the CCR/Weyl algebra: singular, nonlocal or unbounded operator correspondences remain open. Localization at the bracketed field strength loses the zero-jet vacuum chart, while a doubled sheet changes the source theory.

The complete map has a stronger charge–squeeze exhaustion. Assign an arbitrary homogeneous real charge $q(Z) = s$. Its two public factors obey

$$q_K = 1 - s, \qquad q_S = 2s - 2 = -2q_K.$$

For $s < 1$, the full order-λ tangent has a unique positive rank-four component of charge q_K; this includes the fixed-vacuum oscillator choice $s = 0$. For $s > 1$, the free squeeze already contributes positive charges q_S and $2q_S$. Only $s = 1$ avoids positive support, when directness forces $Q_{<0} = 0$ and the full neutral projector must be ghost even.

That last condition fails exactly. On one disjoint unordered momentum pair, with basis $(|0\rangle, |\Omega\Omega\rangle, |\Upsilon\Upsilon\rangle)$, the $t = 0$ squeeze generator and transported pair vacuum are

$$Q = \begin{pmatrix} 0 & -z & 0 \\ 0 & 0 & 0 \\ z & 0 & 0 \end{pmatrix}, \qquad SPS^{-1} = \begin{pmatrix} 1 & zZ^2 & 0 \\ 0 & 0 & 0 \\ zZ^2 & z^2Z^4 & 0 \end{pmatrix}, \qquad Q^3 = 0.$$

After tensoring with the public two-species $n = 1$ projector, the canonical odd part is orthogonal to the even part but has relative norm

$$\tau_0(C^\sharp C) = -z^2(z^2 + 2),$$

which is nonzero for every real nonzero amplitude and equals $-33/256$ at the normalized finite-box fixture $z = 1/4$. Certificate
`REVERSE_PHYSICS_BT_EQ19_SPURION_ SQUEEZE_DICHOTOMY_NO_GO_V1`
therefore refutes every homogeneous regular realization of the public full map on this finite regulator. The earlier certificate `REVERSE_PHYSICS_BT_REGULAR_COVARIANT_ EQ19_NO_GO_V1` retains its exact unsqueezed rank-four tangent calculation, but that tangent is not used as the complete squeezed coefficient. Nonhomogeneous, localized, doubled, singular, non-Fock and nonperturbative completions remain open; no scalar statement transfers to the metric BV complex. The scalar history labels admit a more physical reduced-mode lift. The rooted-comb sectors have dimensions 3, 12, and 60, and their exact tree weights factor into positive per-extension rate squares 1/48, 5/64, and 27/400. The certificate `REVERSE_PHYSICS_BT_CHANNEL_ RESOLVED_BRANCHING_ INSTRUMENT_V1` constructs a finite completely positive trace-preserving branching instrument whose three first-level jumps have the physical scalar Gram $(1/48)I_2$. The higher identity-species lift and three-emission absorbing closure are not derived scalar amplitude phases or an all-order Hamiltonian. In fact, the pre-trace six-point scalar amplitude obstructs that identity lift at the second jump. In the constant/linear parent-jet basis its unique coefficients are $(2Q_{\text{inner}}, a_2/2)$, rather than $(2Q_{\text{inner}}, 2L_{\text{inner}})$, and the raised spectator-profile endomorphism has characteristic discriminant

$$-\frac{4u^2a_2^2(\tau_2 + a_2)(2\tau_2 - a_2)}{(\tau_2 - 2a_2)^2} < 0 \qquad (\tau_2 > a_2 > 0).$$

The certificate `REVERSE_PHYSICS_BT_SIX_POINT_ PARENT_JET_INTERFERENCE_V1` therefore retains the scalar 5/3072 history weight and the abstract finite instrument but refutes its amplitude affiliation above the first jump on the two-species carrier. The grading-faithful parent-jet times spectator-profile enlargement resolves that scoped obstruction. On its four-component tensor carrier the complete pullback obeys

$$A^2 = 2uvA, \qquad P = \frac{A}{2uv} = P^2 = P^\sharp.$$

The kernel of P is nondegenerate and exactly invisible to the physical collapse, while its orthogonal two-dimensional image has positive scalar raised Gram $2uvI_2$. The pointwise identity

$$(RDX)^T(3J)(RDX) = 2uv(PX)^T(J \otimes 3J)(PX)$$

reconstructs the complete scalar six-point amplitude. Together with the five-point prefix replay this affiliates the conditional second rate 5/64 on the quotient fibre. The certificate
`REVERSE_PHYSICS_BT_SIX_POINT_PROFILE_QUOTIENT_COMPLETION_V1`
is followed by the complete scalar seven-point pre-trace calculation. Its seven square-free spectator components reduce to singleton and complementary-pair profiles and factor uniquely through the same recombined parent carrier. The two parent coefficients have opposite signs throughout the nested physical threshold domain. After the seven-leg delta-prime sign, the nonzero physical quotient eigenvalue is $-2uv > 0$, and the signed arbitrary-vector identity reconstructs the complete scalar amplitude before its final trace. Consequently

$$q_2 = \frac{9/81920}{5/3072} = \frac{27}{400}$$

is amplitude-affiliated rather than merely inserted as a positive identity lift. The certificate
`REVERSE_PHYSICS_BT_SEVEN_POINT_PROFILE_QUOTIENT_AFFILIATION_V1`
checks all 2485 trees and the signed Krein quotient independently. This is still a scalar `LOCAL-ALGEBRAIC`, `REDUCED-MODE` result. Neither the scalar quotient, the growing history carrier, its three affiliated jump rates, nor its threshold sign proof transfers to metric BV data; no gravitational or Lorentzian claim is made. On that scalar quotient alone, the three jumps further close into a common finite Krein-skew coupling jet with edge weights $\sqrt{3}/12$, $\sqrt{10}/8$, and 9/20. Its exponential reproduces the scalar leading coefficients through three emissions when $x^2 = a$, but this square-root relation proves that it is not generated strongly in additive resolution on any fixed finite bounded carrier. Certificate `REVERSE_PHYSICS_BT_THREE_JUMP_KREIN_MOLLER_JET_V1` classifies the finite scalar Taylor jet and the obstruction separately. The ordinary generator remains obstructed, but a scalar quantum-stochastic continuation now exists. On the same 152-dimensional quotient, 75 rooted-comb edge operators drive a bounded Hudson–Parthasarathy cocycle [20]; its vacuum reduction is the exact branching instrument and its first three ordered-noise sectors reproduce the scalar Møller jet. The certificate
`REVERSE_PHYSICS_BT_QUANTUM_ STOCHASTIC_MOLLER_DILATION_V1`
proves minimal edge-noise multiplicity 75 for the pinned scalar GKSL map and verifies both unitary structure identities independently. None of this carrier, its auxiliary resolution noise, its rates, or its stochastic cocycle is imported into the metric BV complex. It therefore establishes no gravitational or Lorentzian Møller statement, and supplies neither a metric fourth jump nor a spacetime LSZ operator. The scalar first-emission cocycle now also has an exact Abel-regularized physical-range intertwiner: polar normalization retains both external-jet species and the full collinear direct-integral shape, while a noise-only map is rank-obstructed. The certificate
`REVERSE_PHYSICS_BT_ABEL_FOCK_ PHYSICAL_INTERTWINER_V1`
affiliates the first three scalar edge marks. The six-point positive quotient also defines an exact cumulative Källén resolution coordinate and an ordered two-noise direct-integral isometry for the twelve second-level marks. The certificate
`REVERSE_PHYSICS_BT_SIX_POINT_NESTED_ CONTINUUM_INTERTWINER_V1`

extends scalar physical continuum affiliation to marks 0 through 14. At seven points the signed quotient eigenvalue times the massless-hierarchy middle Källén measure defines a second exact cumulative coordinate with an elementary primitive and unit linear asymptote. The certificate
REVERSE_PHYSICS_BT_SEVEN_POINT_NESTED_ CONTINUUM_INTERTWINER_V1
composes its conditional column with the two-emission range and physically affiliates marks 15 through 74. All 75 marks in the available scalar three-emission hierarchy therefore have continuum columns, but no fourth jump or all-order inductive operator is constructed. Composing their direct sum with the scalar 75-channel HP unitary and vacuum inclusion gives an exact finite physical continuum column $\mathcal{M}_a = \mathcal{A}_{\leq 3} U_a I_\Omega$ with $\mathcal{M}_a^* \mathcal{M}_a = I_2$. Its nonnegative hard and three emitted-sector norms sum to one for every resolution length. In that pinned finite model the hard response $-1/16$ and real response $+1/16$ therefore cancel on the same physical column; this does not derive the full scalar NLO dynamics. The accompanying first-emission compression theorem retains the public rank-one nilpotent R_t leg only after a forced two-dimensional cross-Krein complement whose covariant determinant is $-\rho^2$ and whose raised trace is -2ρ. A positive-noise or trace-null-only completion is impossible. Certificate
REVERSE_PHYSICS_BT_FINITE_PHYSICAL_ MOLLER_COLUMN_V1
keeps that compression distinct from equality with the physical splitting map and does not derive the missing block from BT dynamics. The fixed endpoint freedom does not supply the block universally on its minimal scalar two-profile lift. Relative to the symmetric triple-plus reference, exact matching forces $c_0 = 0$, $c_1 = 7/4 + \rho/2$, and $c_2 = -7/8$; three physical rational fixtures have distinct ρ and therefore require distinct c_1. This reference-independent slope obstruction is certified by
REVERSE_PHYSICS_BT_ENDPOINT_COMPLEMENT_ MATCHING_V1.
It rules out one fixed scalar endpoint prescription, not spectator-dependent scalar dynamics, and none of its probe lift or matching coefficients transfers to the metric BV carrier. Neither the scalar polar ranges, the cumulative resolution measures, the vacuum column, nor the minimal cross-Krein complement transfers to the metric BV complex. Even in the scalar problem, the vacuum column does not determine a scattering operator on arbitrary incoming states. The exact defect-completion theorem parameterizes every same-space unitary extension as $S_W = \mathcal{M}_a I^* + W(1 - II^*)$, where W is a partial unitary from the incoming defect onto $1 - \mathcal{M}_a \mathcal{M}_a^*$. Two different finite exact extensions have the same column, and the physical continuum defect is countably infinite. The certificate REVERSE_PHYSICS_BT_PHYSICAL_MOLLER_ DEFECT_COMPLETION_V1 thus proves abstract scalar unitary extendibility and, simultaneously, exact underdetermination of the incoming-continuum dynamics. It supplies no metric defect space, gravitational asymptotic Hamiltonian, LSZ map, or causal claim. Moreover, the scalar HP cocycle cannot be transferred two-sidedly using only the vacuum-ordered maps $\mathcal{A}_k$. Its conserved auxiliary grading $Q_{\mathrm{HP}} = N_F - L$ leaves a no-spectator path sector, but closure under incoming creation and reverse annihilation requires all $k!$ temporal order chambers. The 76 certified chronological sheets therefore generate 388 sheets, of which 312 crossed sheets have no physical Källén intertwiner. The certificate REVERSE_PHYSICS_BT_CROSSED_ORDER_ CHAMBER_COMPLETION_V1 identifies this exact scalar domain deficit; none of its chamber count, auxiliary grading, or finite leakage witness transfers to the metric BV complex. The first analytic crossed block is itself obstructed on the fixed scalar sharp. On the spacelike sheet $w = -x < 0$, its quotient coefficient is $-q_\times$, where $q_\times = [2x(1+r) + (1-r)^2]/(2x^2) > 0$. The complete two-species image then has fixed-Hilbertized Gram $-3a_2 q_\times I_2$, while the six external delta-prime factors retain even sign. Hence no positive HP reversed-chamber isom-

etry exists on this one-branch carrier. Certificate REVERSE_PHYSICS_BT_CROSSED_SIX_POINT_KALLEN_OBSTRUCTION_V1 shows that an opposite branch sharp would instead give a positive bilateral Källén dilation over $L^2(\mathbb{R})$, but the scalar public map does not derive that sign. The required next object is the complete crossed scalar $3 \to 3$ detector block with both orientations. Neither this rank-two sign obstruction nor its conditional dilation transfers to the metric BV carrier. On the certified factorized scalar quotient, even the two standard momentum orientations cannot repair the sign: $p \mapsto -p$ fixes the p^2 constant/linear jet, and every positive orientation Gram tensored with the negative rank-two species block remains nonpositive. The unique real unit-sign collapse repair is $R_- = [I_2, -I_2]$, which selects the positive complementary quotient. Certificate REVERSE_PHYSICS_BT_CROSSED_DETECTOR_ORIENTATION_NO_GO_V1 identifies its source as the anti-Krein internal jet parity $\epsilon \mapsto -\epsilon$, not ordinary momentum crossing. Neither that scalar parity operation nor the scoped detector no-go transfers to the metric BV carrier. The scalar incoming Wightman dual also cannot supply that parity universally. For $W_\mu^\pm = \theta(\pm p^0)\delta(p^2 - \mu)$, momentum reflection exchanges the energy supports but commutes with $-\partial_\mu$; it is therefore the identity on the simple/dipole mass jet. Moreover the first crossed five-to-four scalar block has positive physical Gram $+\rho_\times I_2$, whereas imposing the needed $S_\epsilon = \mathrm{diag}(1, -1)$ there would reverse it to $-\rho_\times I_2$. Certificate REVERSE_PHYSICS_BT_CROSSED_WIGHTMAN_ DUAL_NO_GO_V1 thus excludes a common one-particle Wightman-reflection origin for the six-point sign. Profile-selective, higher-composite/doubled and nonfactorizing scalar mechanisms remain open. Neither the distributional identity nor this reduced scalar incompatibility transfers to the metric BV carrier or establishes a Lorentzian crossing theorem. Nor can the sign be localized by a regular same-carrier scalar parity while preserving the first crossed prefix. On the certified four-component carrier, isometries preserve the negative crossed quotient spectrum, while anti-isometries reverse both that spectrum and the nonzero prefix norm $3/2 \mapsto -3/2$. The public neutral degree-four Fock ghost parity remains a Krein isometry on its ghost-odd negative two-plane and therefore does not provide the required anti-Krein action. Certificate REVERSE_PHYSICS_BT_CROSSED_PROFILE_SELECTIVE_PARITY_OBSTRUCTION_V1 confines the surviving scalar route to singular/doubled data or a genuinely nonfactorizing crossed $3 \to 3$ pre-trace term. This carrier theorem neither transfers to the metric BV complex nor establishes a gravitational no-go. The complete finite-hierarchy leading scalar jet supplies no such term on the correlated square-free cylinder. Before the strong-order limit, the full 220-tree sum still has exactly one singleton row, one complementary-pair row, and no cubic spectator row. Its invertible outer profile matrix reconstructs unique $u(e), v(e)$ with zero residual. On the crossed positive domain exact sum-of-positive-terms identities give $u_\times < 0 < v_\times$, so the fixed-sharp scalar quotient remains rank two and negative for every finite $e > 0$. Certificate REVERSE_PHYSICS_BT_CROSSED_SIX_POINT_NONFACTORIZING_PRETRACE_NO_GO_V1 therefore excludes a hierarchy-limit artifact on that reduced scalar cylinder. It neither treats the complete non-correlated six-body phase space nor transfers its sign to the metric BV carrier; it establishes no gravitational or Lorentzian no-go. The first complete scalar calculation off that correlated boundary has the opposite local disposition. On an exact continuous planar massless $3 \to 3$ family, all 220 trees produce twenty degree-three external-mass coefficients satisfying ten complement equalities $c_S = c_{S^c}$. The full six-delta-prime scalar coefficient is consequently $2\sum_{S<S^c} c_S^2 > 0$ away from its propagator poles; the ten numerator polynomials have gcd one. This self-duality requires cancellation among all three cubic/quartic topology sectors and is not present sector-wise. Certificate REVERSE_PHYSICS_BT_SIX_POINT_PLANAR_ PHYSICAL_BORN_DENSITY_V1 shows that the reduced negative scalar quotient is not the complete local Born density. The result remains a planar scalar density, not an integrated probability, a nonplanar theorem, or anything transferred to the metric BV carrier. An exact out-of-plane successor removes the remaining coplanarity caveat on a continuous diagonal. Tilting the outgoing triple by rational stereographic parameter $u = t/2$ preserves nullness and momentum conservation while giving generic nonzero z components. The complete scalar 220-tree mass jet again has ten complement equalities and a strictly positive $2\sum c_S^2$ local kernel, with numerator gcd one and only even pole multiplicities. Certificate REVERSE_PHYSICS_BT_SIX_POINT_NONPLANAR_DIAGONAL_PHYSICAL_BORN_DENSITY_V1 shows that neither the correlated hierarchy nor planar kinematics explains the positive full-density recombination. It is still a scalar one-parameter slice, not a two-parameter phase-space theorem, integrated probability, or metric-BV result. The two rigid-rotation angles can now be separated exactly. Over $\mathbb{Q}(t,u)$, the complete scalar 220-tree degree-three mass jet obeys all ten complement identities, so its six-delta-prime kernel is $2\sum_{S<S^c} c_S(t,u)^2 \geq 0$ at every regular pair and is strictly positive on a dense open subset. Certificate REVERSE_PHYSICS_BT_SIX_POINT_TWO_ANGLE_ PHYSICAL_BORN_DENSITY_V1 does not exclude a lower-dimensional common-zero locus and does not cover the two final-state shape variables or the missing orientation angle. It is thus neither a complete five-dimensional phase-space theorem nor an integrated probability, and no part of it transfers to the metric BV carrier. The scalar local sign problem can nevertheless be completed without a five-variable brute-force expansion. If $s_A = (\sum_{i\in A} p_i)^2$ labels the ten unordered three–three channels, the full 220-tree middle jet obeys

$$c_S = c_{S^c} = \frac{1}{4}\sum_{A\neq S} s_A^{-1}.$$

In the auxiliary $O(1,1)$ quartic representation the channel incidence for a three-Ω/three-Υ assignment is $J - I$: only the channel putting all three like species on one side is forbidden. Since $\det(J - I) = -9$, the ten real coefficients cannot vanish simultaneously at a regular point and the complete six-delta-prime scalar kernel is strictly positive there. An exact two-shape/three-orientation chart has rank five, while six exact fixtures independently match the closed formula by explicit enumeration of all 220 trees. Certificate
REVERSE_PHYSICS_BT_SIX_POINT_FULL_ PHASE_SPACE_BORN_POSITIVITY_V1
is therefore a complete regular-phase-space local scalar theorem. It leaves all channel poles unregulated and supplies no integrated probability, Eq. (19), loop completion, or metric-BV transfer. Moreover the ordinary exclusive scalar integral diverges. The closed formula gives $D = (\sum_A 1/s_A)^2 + \frac{1}{8}\sum_A 1/s_A^2$, hence the universal isolated- channel behavior $9/(8s_B^2)$. An exact positive-energy rank-five chart point has one transverse zero with $\partial_t s_{11} = -1152/425$, producing a positive $(t-3/5)^{-2}$ pole. Certificate REVERSE_PHYSICS_BT_SIX_POINT_PHASE_SPACE_POLE_OBSTRUCTION_V1 therefore makes a detector or inclusive real–virtual prescription mandatory. It constructs neither one and has no metric-BV implication. The stronger scalar distributional audit shows that this exclusive pole cannot itself be made into a positive locally finite measure: its forced punctured mass is $\frac{9}{4}(\epsilon^{-1} - L^{-1})$. Linear finite-part extensions have local delta ambiguity but lose positivity, and the symmetric Feynman-modulus regulator exposes a divergent $9\pi\delta(s_B)/(8\epsilon)$ sequential-history term. The existing five-point NLO response has different order and singular support, and the certified finite Møller construction fixes only an input column, not the required two-sided BT-affiliated defect action. Certificate REVERSE_PHYSICS_BT_SIX_POINT_POSITIVE_ DISTRIBUTION_COMPLETION_NO_GO_V1 therefore redirects the scalar programme to a finite-time or wave-packet sequential-history con-

struction; it proves no finite inclusive probability and has no metric-BV or Lorentzian-causal implication. The scalar leading-shell carrier is now also explicit. The ten-channel residue map $(J-I)/4$ gives $D = \frac{9}{8}\sum_A s_A^{-2} + 2\sum_{A<B}(s_A s_B)^{-1}$. The first term contains all positive double poles as separately labelled sequential histories, while the signed interference matrix $J-I$ has inertia $(1,9)$. A universal finite-time window converts one leading pole into $9\pi T\delta(s)/(8E)$, but no BT Hamiltonian derivation or survival term is yet available. Certificate REVERSE_PHYSICS_BT_SIX_POINT_SEQUENTIAL_ HISTORY_CARRIER_V1 has no metric-BV or Lorentzian-causal implication. Separating the final species assignment from the intermediate channel gives a ninety-history incidence lift with normalized isometry $W_{\mathrm{hist}} = 4B/3$. Coherent collapse reconstructs $(J-I)/4$, while the signed $J-I$ interference is the difference of resolved and coherent positive detector weights. A normalized history effect and the exact skew rotation generated by $\begin{pmatrix} 0 & -W_{\mathrm{hist}}^T \\ W_{\mathrm{hist}} & 0 \end{pmatrix}$ produce a finite three-outcome channel-label instrument with survival. Certificate REVERSE_PHYSICS_BT_SIX_POINT_HISTORY_ INCIDENCE_ISOMETRY_V1 does not derive its angle, detector resolution or wave-packet kernel from BT dynamics and has no metric-BV or Lorentzian-causal implication. The finite-time shell kernel is now constructed at the exact channel-11 pole. There $q = (1, 9/17, -4/85, -72/85)$ is null with $E = 1$, and exact coarea gives $d\Phi_3 = 3\, da\, db\, ds\, du\, dv/[320(2\pi)^5]$. The square-integrable amplitude $\alpha_{T,E} = F_T(s/(2E))/(2E)$ has norm squared $\pi T/E$; combining it with the nine-history vector gives $Q_T = 9\pi T/(8E)$ and an isometric local survival-plus-history column. The frozen labeled phase coefficient is $27T/(81920\pi^4)$ before the common BT multiplier, incoming flux, detector normalization and generalized-Born convention. Certificate REVERSE_PHYSICS_BT_SIX_POINT_FINITE_TIME_ SHELL_COLUMN_V1 does not calibrate that multiplier, construct a global BT scattering operator, or transfer any result to the metric-BV or Lorentzian-causal programme. The public Feynman rules do fix the scalar Hamiltonian part of the multiplier: all three six-point topology classes have common magnitude $16\lambda^4$, so the density multiplier is $256\lambda^8$. The labeled fixture coefficient is consequently $27\lambda^8 T/(320\pi^4)$ per unit tangential chart volume. It has mass dimension -2, however, and the Letter does not specify the dimension-two incoming three-particle characteristic-cell normalization needed to turn it into a probability. Certificate REVERSE_PHYSICS_BT_SIX_POINT_SHELL_TREE_ NORMALIZATION_V1 therefore fixes the scalar coupling coefficient while leaving the generalized- Born detector normalization open; neither part transfers to gravity. That local detector normalization can now be closed after declaring the finite-volume incoming cell explicitly. At the exact three-beam fixture scaled by κ, eight spatial point characteristics together with the total-energy characteristic have on-shell Jacobian one. Cancelling $\delta_4(0)$ leaves

$$N_{\mathrm{in}} = \frac{5}{48\kappa^3 L_x L_y^2 L_z^2}.$$

The six incoming and six outgoing label orbits cancel the two 3! projector factors, giving the dimensionless local rate density

$$\Gamma_\Xi = \frac{9\lambda^8}{1024\pi^4\kappa^4 L_x L_y^2 L_z^2}.$$

Certificate REVERSE_PHYSICS_BT_THREE_PARTICLE_ CHARACTERISTIC_CELL_V1 establishes $q_\Xi(T) = \Gamma_\Xi T\Delta\Xi + O(1)$ for the declared symmetric local detector. It simultaneously rules out calling this a universal three-body cross section: a second idempotent coordinate cell changes the normalization by 5/3. Compact wave-packet convergence, ten-channel gluing, Eq. (19), and transfer to metric BV or Lorentzian causal gravity remain open. The time kernel itself is no longer merely kinematic. The interaction-picture Born trace gives

$$\int_0^T d\tau \int_0^T d\tau' e^{i\omega(\tau-\tau')} = |F_T(\omega)|^2 = \int_{-T}^{T} (T-|\sigma|)e^{i\omega\sigma} d\sigma,$$

after the separate external center-time volume has been cancelled by the incoming energy characteristic. Certificate `REVERSE_PHYSICS_BT_FINITE_TIME_HAMILTONIAN_ CUT_AFFILIATION_V1` therefore affiliates the $\pi T/E$ shell norm and the displayed detector rate with the BT Hamiltonian at the cut-probability level. It does not affiliate the positive survival completion: an exact J-unitary boost has conserved complementary weights $\cosh^2 r$ and $-\sinh^2 r$, proving that pseudo-unitarity alone is insufficient. Embedding the positive history rotation still requires weak ghost symmetry or Eq. (19), and none of this transfers to the metric BV carrier. That scalar output embedding is now exact. In the twenty-dimensional neutral six-leg public Fock sector, complement exchange gives $\eta = \kappa = \begin{pmatrix} 0 & I_{10} \\ I_{10} & 0 \end{pmatrix}$, whose symmetric ten-plane has positive Gram. The complete scalar coefficient obeys $c_S = c_{S^c}$ and lies wholly in this plane, while its 8×8 three-particle Choi matrix obeys $\kappa_3 A \kappa_3 = A$ and $\mathrm{tr}(A^\sharp A) = 2\sum_S c_S^2$. Each fixed-shell nine-history range embeds isometrically with norm 9/8, as certified by `REVERSE_PHYSICS_BT_SIX_POINT_ GHOST_EVEN_ HISTORY_EMBEDDING_V1`. The simultaneous map from all ninety resolved histories has rank ten and forgets an eighty-dimensional intermediate-channel record, so a detector ancilla remains necessary. The transported scalar input projector, virtual survival block and all-order Eq. (19) are still open. This `LOCAL-ALGEBRAIC`, `REDUCED-MODE` scalar result supplies no metric-BV or Lorentzian-causal conclusion. The positive plane permits a direct auxiliary-theory probability statement. At the fixed channel-$B = 1$ shell, the restriction R_+ has effect $G = R_+^T R_+$ with spectrum $0, 1/16, (2-\sqrt{3})/8, (2+\sqrt{3})/8$. For

$$\zeta = \frac{\lambda^8 T \Delta\Xi}{128\pi^4\kappa^4 L_x L_y^2 L_z^2},$$

the click/no-click pair $\{\zeta G, I_4 - \zeta G\}$ is positive and complete when $0 \le \zeta \le 16 - 8\sqrt{3}$. The normalized source $(|\Upsilon^3\rangle + |\Omega^3\rangle)/\sqrt{2}$ has click rate $\lambda^8/[2048\pi^4\kappa^4 L_x L_y^2 L_z^2]$, while pseudo-unitarity fixes the leading complementary coefficient. Certificate
`REVERSE_PHYSICS_BT_ SIX_POINT_POSITIVE_ SECTOR_PHYSICAL_ DETECTOR_EFFECT_V1`
therefore establishes a leading physical probability jet for a state prepared directly in the public scalar auxiliary theory. It is not the transported perfect-square scalar projector, an exact all-time probability, or a result on the metric BV carrier. The certified formal inverse nevertheless affiliates this selected source to the perfect-square scalar theory. With $P_u = |u_0\rangle\langle u_0|_K$, define

$$\psi_{\phi,+} = R_t^\dagger u_0, \qquad P_{\phi,+} = R_t^\dagger P_u R_t.$$

Then $\psi_{\phi,+}$ has unit Krein norm, $P_{\phi,+}^2 = P_{\phi,+} = P_{\phi,+}^\dagger$, and $R_t P_{\phi,+} R_t^\dagger = P_u$. Pulling back both effects preserves their finite generalized-Born traces, so the scalar source has the same click and no-click probability jet and the same leading rate.

This affiliation necessarily changes the preparation. The odd three-particle charge spectrum contains no zero, and every fixed-charge subspace is null under the cross-Krein metric; consequently no nonzero charge-invariant range is positive. The declared source projector has charges $\{-6, 0, +6\}$ and scalar orbit support $\{Z^{-6}, 1, Z^6\}$. Certificate

REVERSE_PHYSICS_BT_SCALAR_ DRESSED_POSITIVE_SOURCE_ AFFILIATION_V1
therefore supplies a dressed physical-scalar result while leaving the standard shift-invariant $P_\chi^{(\phi)}$, general Eq. (19), all-time scattering and the metric BV carrier open. The leading source has the explicit finite-core form

$$\psi^{(0)}_{\phi,+} = \frac{1}{\sqrt{2}} \left[Z^{-3} \prod_j \frac{a^\dagger_{1j}}{\sqrt{2E_j}} + Z^3 \prod_j \frac{a^\dagger_{2j} - 2iE_j t\, a^\dagger_{1j}}{4E_j^2 \sqrt{2E_j}} \right] |0_{\phi;t}\rangle,$$

where $|0_{\phi;t}\rangle = R_t^\dagger |0_{\rm BT}\rangle$ is the pulled squeezed vacuum. Each branch is null and their cross pairing is one by the repaired scalar CCR. Since the six-point amplitude starts at λ^4, an $O(\lambda)$ source correction first enters the probability at λ^9. Certificate
REVERSE_PHYSICS_BT_SCALAR_ DRESSED_SOURCE_FREE_ NORMAL_FORM_V1
therefore removes the leading source-symbol ambiguity while retaining the ordinary-Fock infrared obstruction and every metric-BV boundary. The finite-mode qualifier on the prepared carrier can now be removed away from zero momentum. For three normalized compact profiles on disjoint and non-antipodal supports with $0 < \varepsilon \le E \le M$, normalized scalar oscillators give

$$d^\dagger_\Upsilon(f) = Z^{-1} A_1^\dagger(2Ef), \qquad d^\dagger_\Omega(f)|0_{\phi;t}\rangle = Z\left[A_2^\dagger\!\left(\frac{f}{2E}\right) - itA_1^\dagger(f)\right] |0_{\phi;t}\rangle.$$

The full Ω operator contains a reflected annihilator; non-antipodal supports eliminate its cross-contractions in this frame. Boundedness of $2E$, $(2E)^{-1}$ and t on the support, together with the weighted Gaussian number moments, gives a common dense core and closability of every product through degree three. Complement exchange again makes the symmetric four-frame positive. Finite-volume packet approximants converge in vector norm and their fixed-ζ finite-rank effects converge in trace norm. Certificate
REVERSE_PHYSICS_BT_SCALAR_ DRESSED_SOURCE_COMPACT_ WAVEPACKET_V1
therefore constructs a compact continuum source/effect carrier. The packet-dependent Hamiltonian strength is also fixed for one finite-time, isolated-channel experiment. In the exact five-coordinate three-body chart, the labeled phase density is

$$\rho(a,b,t,u,v) = \frac{256|(1+ab)u|}{25a^2b^2(a-b)^2(1+t^2)(1+u^2)^2(1+v^2)}.$$

For compact incoming and outgoing regions crossing only channel $B = 11$, put $\delta_B = q_B^0 - |\mathbf{q}_B|$, $D_B = q_B^0 + |\mathbf{q}_B| \ge d_0 > 0$, and take a detector kernel $|\chi| \le 1$. The public two-quartic Hamiltonian then gives the Hilbert–Schmidt packet operator

$$(K_{B,T}F)(y) = \int_X \chi(y,x) \frac{F_T(\delta_B(y,x))}{D_B(y,x)} F(x)\, d\mu(x), \qquad \|K_{B,T}\|^2_{\rm HS} \le \frac{T^2 \mu(X)\mu(Y)}{d_0^2}.$$

With $A_{B,T} = 16\lambda^4 K_{B,T} \otimes R_+$, the effects $E_{\rm click} = A^*_{B,T} A_{B,T}$ and $E_{\rm no} = I - E_{\rm click}$ are positive and complete whenever

$$32(2+\sqrt{3})\lambda^8 \frac{T^2\mu(X)\mu(Y)}{d_0^2} \le 1.$$

The normalized dressed source therefore has the explicit leading probability

$$q_{\rm click} = 16\lambda^8 \|K_{B,T}F\|^2.$$

Certificate
REVERSE_PHYSICS_BT_COMPACT_WAVEPACKET_ HAMILTONIAN_PROBABILITY_V1
replaces the fitted packet strength by a BT-Hamiltonian integral. It remains a selected scalar, finite-time, single-channel result. All ten channel records can be combined without choosing between overlap strata if the detector keeps them orthogonal. A smooth square partition $\sum_B |\chi_B|^2 = 1$ gives

$$A_{\mathrm{rec}} = 16\lambda^4 \bigoplus_{B=0}^{9} (K_{B,T} \otimes R_B), \qquad \|A_{\mathrm{rec}}\|^2 \le 144\lambda^8 \frac{T^2\mu(X)\mu(Y)}{d^2}.$$

Here every R_B has squared Hilbert–Schmidt norm $9/16$. One exceptional channel is the hard split $q = P$, $q^2 = 256/25$, and is dark for the chosen source; the other nine are mixed exchange channels and contribute

$$q_{\mathrm{click}} = 16\lambda^8 \sum_{B=1}^{9} \|K_{B,T} F\|^2.$$

Thus $A^*_{\mathrm{rec}} A_{\mathrm{rec}}$ and its complement form a positive normalized instrument when the displayed bound is at most one. Certificate
REVERSE_PHYSICS_BT_TEN_CHANNEL_RECORDED_ COMPACT_WAVEPACKET_INSTRUMENT_V1
proves positivity even where shell neighborhoods overlap, because the output records are orthogonal. The record can now also be erased coherently without discarding its cross terms. The exact residue-interference Gram is

$$H_{AB} = \mathrm{tr}(R_A^T R_B) = \frac{\delta_{AB} + 8}{16}, \qquad \mathrm{spec}\, H = \left\{ \frac{81}{16}, \left(\frac{1}{16}\right)^{\times 9} \right\}.$$

Thus

$$A_{\mathrm{coh}} = 16\lambda^4 \sum_{B=0}^{9} K_{B,T} \otimes R_B, \qquad \|A_{\mathrm{coh}}\|^2 \le 1296\lambda^8 \frac{T^2\mu(X)\mu(Y)}{d^2},$$

and $A^*_{\mathrm{coh}} A_{\mathrm{coh}}$ and its complement form a positive normalized instrument when the last bound is at most one. On the selected source,

$$q_{\mathrm{click}} = 16\lambda^8 \left\| \sum_{B=1}^{9} K_{B,T} F \right\|^2,$$

which retains every coherent channel cross term. Certificate
REVERSE_PHYSICS_BT_COHERENT_COMPACT_ WAVEPACKET_DETECTOR_DILATION_V1
also constructs the exact Julia unitary with no-click Kraus $(I - A^*_{\mathrm{coh}} A_{\mathrm{coh}})^{1/2}$. Expanding any actual pseudo-unitary BT completion on the same positive source/output sector would force its order-λ^8 source survival coefficient to have Hermitian part $-A_4^* A_4/2$ only if this selected output block is exhaustive. In general, writing the complete positive transition column as $C_4 = (A_4, B_{\mathrm{out}})$ gives

$$\mathrm{Herm}(M_8)_{\mathrm{source}} = -\frac{1}{2}(A_4^* A_4 + B^*_{\mathrm{out}} B_{\mathrm{out}}).$$

This is the $C_4^* C_4$ cut component; other perturbative products and cut-free order-λ^8 terms remain separate. True survival loses both terms and outside leakage restores the second inside the detector's no-click event, leaving exactly $I - A_4^* A_4$. Certificate

REVERSE_PHYSICS_BT_COMPACT_DETECTOR_ SURVIVAL_LEAKAGE_FACTORIZATION_V1 proves with two exact completions that the compact click data determine the binary detector probability but not the BT forward coefficient. The public calculation supplies neither B_{out} nor a common-domain evolution, so the Julia complement is the minimal zero-leakage detector completion, not a dynamical affiliation. The zero-gap limit, general Eq. (19), all-time scattering, and every metric-BV or Lorentzian claim remain open. The connected order-λ^4 output type is nevertheless complete. Graph counting gives $d_\lambda = E + 2L - 2$, so the possible connected types are $(6,0), (4,1), (2,2), (0,3)$. The timelike three-particle source excludes all but the $3 \to 3$ tree: the vacuum and two-leg graphs do not attach it, and a single massless output cannot carry $P^2 = 256/25$. Complement parity further forces every output of the positive even source into the same positive four-plane. Thus the only connected outside block is three-body momentum outside the detector. On a common compact regular acceptance the actual unpartitioned public tree column is

$$A_{\text{full},C} = 16\lambda^4 \sum_{B=0}^{9} K_{B,T} \otimes R_B, \qquad \|A_{\text{full},C}\|^2 \le 12960\lambda^8 \frac{T^2 \mu(X)\mu(Y)}{d^2}.$$

Certificate
REVERSE_PHYSICS_BT_COMPLETE_CONNECTED_ ORDER4_PACKET_COLUMN_V1 therefore replaces the channel-partitioned detector amplitude by the actual unit-weight ten-channel tree on the compact acceptance. Its adjoint square and complement are positive on the displayed contraction domain. In fact the compact momentum cutoff can be removed at fixed finite time. With $P = (M, \mathbf{0})$, $M = 16/5$, every exchange channel is $q = P - p - k$ for one incoming and one outgoing spectator of energies $E, K \le M/2$. Its only zero has $E = K = M/2$ and antipodal directions. The four-dimensional transverse Jacobian there has determinant $-2M^2 = -512/25$. The exact recursive measure

$$d\Phi_3(P) = \frac{E\, dE\, d\Omega\, d\Omega_*}{512\pi^5}$$

then gives

$$\int \frac{d\Phi_3(x) d\Phi_3(y)}{D_B^2} = \frac{1}{3200\pi^6}$$

for each exchange channel and $1/(6400\pi^6)$ for the hard channel. Hence

$$\|A_{\text{full}}\|^2 \le \frac{1539}{400\pi^6} \lambda^8 T^2.$$

Certificate
REVERSE_PHYSICS_BT_GLOBAL_CONNECTED_FINITE_ TIME_PACKET_COLUMN_V1 proves that the apparent $q_B = 0$ singularity is globally square-integrable, closing the connected leading momentum domain. Disconnected spectators, the matching forward coefficient, all-time scattering, and Eq. (19) remain open. For a fully rearranged detector the disconnected terms can instead be removed by support. At the exact rational incoming/outgoing centers, the minimum Euclidean squares of every all-incoming subset momentum sum of sizes one, two, and three are

$$2, \qquad 32/625, \qquad 17794/10625.$$

Small compact neighborhoods therefore avoid every component momentum delta. There are 202 disconnected set partitions of six labels in ten size profiles, and every profile has a component with at most three legs. Distributional and external-mass derivatives preserve support, so all disconnected terms through order λ^4 vanish on this detector. Consequently

$$P_Y(U_T - I)P_X = \lambda^4 P_Y A_{\text{full}} P_X + O(\lambda^5),$$

while $P_Y P_X = 0$ removes identity and forward contributions from the leading click coefficient. For the dressed scalar packet,

$$q^{(8)}_{Y\leftarrow X} = 16\lambda^8 \left\| \sum_{B=1}^{9} P_Y K_{B,T} P_X F \right\|^2 \leq \frac{81}{200\pi^6}\lambda^8 T^2.$$

Certificate
`REVERSE_PHYSICS_BT_FULLY_REARRANGED_ PHYSICAL_PACKET_PROBABILITY_V1`
therefore establishes the complete leading finite-time physical transition for a nonempty selected detector class, not merely its connected part. It does not establish overlap sectors, higher orders, an all-time S operator, or Eq. (19). The first overlap stratum is nevertheless separately tractable. An exact hard nonforward rational configuration has one tagged identity spectator and no competing component momentum delta. Exhaustive enumeration of all 203 set partitions leaves only the connected six-leg partition and the tagged two-plus-four partition on support. The latter is the unique order-λ^2 transition, while the connected six-leg tree starts at order λ^4. The active four-point mass-jet vector $r_4 = 2\sum_{i<j} e_{ij}$ lies in the positive complement-pairing eigenspace and has norm $r_4^\sharp r_4 = 24$. Hence a normalized spectator packet gives

$$q_{\text{tag}} = \frac{3\lambda^4 \Delta\Omega}{32\pi^2 s\,\text{Area}} + O(\lambda^5),$$

with rational coefficient 75/2048 at $s = 64/25$. Certificate
`REVERSE_PHYSICS_BT_TAGGED_SPECTATOR_ PHYSICAL_PACKET_PROBABILITY_V1`
therefore closes this selected spectator-overlap stratum at leading order. It does not construct one detector coherently crossing the spectator and fully rearranged strata, the forward/collinear sectors, higher orders, all-time scattering, or Eq. (19). An exhaustive support theorem shows that these two calculations cover every local hard nonforward stratum. Of the 202 disconnected six-label partitions, 162 contain an impossible singleton. The fifteen $2+4$ cases give nine spectator cylinders and six impossible same-side pairs; all ten $3+3$ cases require either the nonzero total momentum or a collinear same-side null pair; and the fifteen perfect matchings give six removed forward permutations plus nine same-side-impossible matchings. Pairwise intersections of the nine spectator cylinders either identify same-side momenta or force the third spectator equality and hence a forward permutation. Certificate
`REVERSE_PHYSICS_BT_HARD_NONFORWARD_PHYSICAL_ STRATIFIED_ATLAS_V1`
therefore leaves exactly the fully rearranged order-λ^8 probability and the nine labeled spectator order-λ^4 probabilities as the local leading atlas. It does not yet compute one unresolved finite-resolution record: there the spectator and connected amplitudes first interfere at probability order λ^6. That scalar cross kernel is now computed on a common positive external-jet carrier. The tagged four-point vector embeds isometrically with norm 24 in six of the ten positive

six-leg coordinates; its incidence pairing with the connected column has weights five and six. Four weight-six channels become exactly resonant at the tagged fixture, giving

$$I^{(6)}_{\text{tree}} = 16\sqrt{2}\lambda^6 W(T), \qquad W(T) > 0 \ (T > 0), \qquad W(T)/T \longrightarrow 12,$$

with $W(T) = 12T + 125\sin(16T/5)/256 + 125\sin(8T/5)/128 + 125\sin(2(\sqrt{17}-3)T/5)/8$ and the exact lower bound $W(T) \geq [221 - 50\sqrt{17}]T/8$. Certificate
`REVERSE_PHYSICS_BT_TAGGED_CONNECTED_ FINITE_TIME_INTERFERENCE_V1`
therefore rules out scalar tree-sector decoupling on this stratum. It does not supply the common distributional packet normalization, active one-loop, source or survival terms needed for the complete order-λ^6 probability. In particular this scalar result supplies no metric BV–BRST transfer, gravitational detector, Eq. (19), or Lorentzian causal theorem. On a declared finite-volume spectator mode the missing relative normalization is also exact. The positive ghost-even spectator has norm $N_s = 2E_sV = 12\kappa V/5$. Its identity contraction cancels the two external normalizers in the disconnected tree, whereas the connected tree retains $1/N_s$. Hence

$$q^{(6)}_{\text{cross}} = \frac{125\sqrt{2}\lambda^6 W_\kappa(T)\Delta\Omega}{12288\pi^2\kappa^3 \text{Area}\, V}, \qquad W_\kappa(T) = \kappa^{-2}w(\kappa T).$$

Certificate
`REVERSE_PHYSICS_BT_TAGGED_CONNECTED_ FINITE_VOLUME_NORMALIZATION_V1`
proves fixed-time V^{-1} suppression but also a nonuniform long-time limit: $q^{(6)}_{\text{cross}}/q^{(4)}_{\text{tag}}$ grows as $(10\sqrt{2}/3)\lambda^2 T/(\kappa^2 V)$. This finite-box scalar coefficient is not the compact-packet or complete order-λ^6 probability and supplies no gravitational or Lorentzian transfer. The compact-packet replacement is nevertheless exact as a functional. For normalized positive spectator packets f, g in $d\nu = d^3p/(2E_p)$, write $c_{gf} = \langle g, f\rangle_\nu$ and $C_{gf} = \langle g, \mathcal{W}_{\kappa,T} f\rangle_\nu$. On the same active cell,

$$\frac{q^{(6)}_{\text{cross}}[g,f]}{q^{(4)}_{\text{tag}}} = \frac{2\sqrt{2}\lambda^2}{3}\Re(\overline{c_{gf}}C_{gf})\,.$$

Certificate
`REVERSE_PHYSICS_BT_TAGGED_CONNECTED_ COMPACT_PACKET_INTERFERENCE_V1`
shows that a fixed compact packet contains $O(V)$ coherent box modes: their double sum compensates the single-mode V^{-1} factor and converges to a finite, generally nonzero packet integral. Continuity of the finite-time kernel and positivity at the exact fixture give a nonempty positive packet family for every fixed $T > 0$. This reduced scalar tree functional is not the complete order-λ^6 probability, and none of these statements transfers to gravity. The possible preceding odd order is, however, selected exactly. Total Fock parity $\Pi_F = (-1)^N$ is distinct from BT ghost parity and $SO^+(1,1)$ charge. From $S_\lambda[\phi] = S_{-\lambda}[-\phi]$ and the corresponding covariance of the BT composites, the order-λ^2 tagged output is Fock-odd while its complete order-λ^3 correction is Fock-even. The cross-Krein metric pairs only equal total particle parity, so

$$q^{(5)}_{\text{tag}} = 2\Re\langle y_2, y_3\rangle_K = 0.$$

Certificate
`REVERSE_PHYSICS_BT_TAGGED_PACKET_ LAMBDA5_PARITY_SELECTION_V1`
therefore moves the first unresolved correction to order λ^6, where the tree cross, active loop, second-order source/detector and survival terms remain unassembled. This scalar selection rule

has no gravitational or Lorentzian implication. The exact object ledger reduces the remaining scalar calculation further. On fixed odd three-particle BT projectors, parity kills the complete order-λ^3 block. Exhaustive tagged support leaves at amplitude order four the connected six-point tree, spectator identity times the active four-point loop, and spectator self-energy times the active tree. Thus

$$q_{\text{tag}}^{(6)} = 2\Re\langle T_2, C_{4,\text{tree}}\rangle_K + 2\Re\langle T_2, I_s \otimes L_{4,\text{active loop}}\rangle_K + 2\Re\langle T_2, S_{2,s} \otimes A_{2,\text{active tree}}\rangle_K.$$

Certificate
`REVERSE_PHYSICS_BT_TAGGED_PACKET_ LAMBDA6_OBJECT_LEDGER_V1`
shows that nonforward support removes pure survival, not spectator dressing times active scattering, and that the selected scalar source/effect dressing cancels inside the common two-sided R_t similarity; counting it separately would double count. The compact tree cross is known, so the active four-point one-loop and spectator two-point packet kernels are the two missing scalar coefficients. The existing hard logarithm is only an asymptotic scale check on the former, not either kernel, and nothing here transfers to the gravitational BRST carrier. There is a valid selected auxiliary scheme in which the spectator term vanishes. Certificate
`REVERSE_PHYSICS_BT_TAGGED_PACKET_NORMAL_ ORDERED_SPECTATOR_REDUCTION_V1`
proves that the quartic BT interaction has only a momentum-independent one-vertex tadpole in the order-λ^2 two-point sector. Normal ordering plus massless unit-residue conditions gives $S_{2,s} = 0$ on the finite-time packet carrier. The active four-point loop is then the sole missing scalar $q^{(6)}$ coefficient. This is a declared auxiliary renormalization scheme, not a scheme-independent theorem or a gravitational result. In that same auxiliary scheme the finite covariant active loop is computable. Certificate
`REVERSE_PHYSICS_BT_AUXILIARY_ACTIVE_ ONE_LOOP_MSBAR_V1`
derives the exact six-species bubble tensor and the real MSbar density

$$\frac{d\sigma^{(6)}_{\text{active},\overline{\text{MS}}}}{d\Omega} = \frac{5\lambda^6}{256\pi^4 s}(L_s + L_t + L_u + 6).$$

The logarithmic part independently matches the scalar Callan–Symanzik row. This covariant calculation alone does not give equality with the finite-duration auxiliary Dyson packet kernel. No coefficient or carrier statement transfers to the gravitational BRST complex. On the selected energy-diagonal hard carrier that scalar equality is now proved. Certificate
`REVERSE_PHYSICS_BT_FINITE_TIME_ACTIVE_ LOOP_AFFILIATION_V1`
derives the ordered second-Dyson dispersive kernel and its exact Fejer convolution. With $C(z) = \sin(z)/z - \mathrm{Ci}(z)$, each channel bubble is

$$B_T(P) = B_{\overline{\text{MS}}}(P) - C(T|P^0 - |\mathbf{P}||) - C(T|P^0 + |\mathbf{P}||).$$

This supplies the finite-time transient, a compact-packet bound, and the covariant large-time boundary. Nothing in this affiliation constructs a metric loop, a gravitational QME, residual transfer, or Lorentzian causality. Combining it with the connected-tree functional, Certificate
`REVERSE_PHYSICS_BT_COMPLETE_TAGGED_Q6_ PHYSICAL_PROBABILITY_V1`
gives

$$q_{\text{tag}}[f;T] = q_4\{1 + \lambda^2 R_6[f;T,\mu]\} + O(\lambda^8), \qquad R_6 = \frac{2\sqrt{2}}{3}\Re C_{ff}(T) + \frac{5B_*(T,\mu)}{24\pi^2},$$

where $q_4 = 75\lambda^4\Delta\Omega/(2048\pi^2\kappa^2\,\mathrm{Area})$ and B_* is the finite-time three-channel bubble above. The normal-ordered spectator coefficient vanishes and parity removes the odd orders, so no declared $q^{(6)}$ object remains missing. This is a selected auxiliary scalar probability theorem; its packet- and scheme-dependent sign supplies no gravitational coefficient and no shortcut through the metric BV–BRST and quantum-master-equation gates. The nine spectator-label copies of this selected scalar experiment can be assembled into one recorded coefficient instrument. Exact enumeration of $S_3^{\mathrm{in}} \times S_3^{\mathrm{out}}$ gives a transitive orbit of the nine Δ_{ia} cylinders and an order-four stabilizer. The six weight-five and four weight-six tree-cross channels are recomputed for every tag. With one additional fully rearranged bulk record, the ten characteristic effects are orthogonal and complete. For $\rho = I_{10}/10$, their cylinder union has weight 9/10; the bulk has no term through λ^6. Certificate
`REVERSE_PHYSICS_BT_NINE_CYLINDER_RECORDED_ Q6_INSTRUMENT_V1`
therefore gives

$$q_{\mathrm{rec}}[f;T] = \frac{9}{10} q_4\{1 + \lambda^2 R_6[f;T,\mu]\} + O(\lambda^8)$$

on the selected permutation orbit. This remains an explicitly recorded, randomized auxiliary scalar experiment. It is not a coherent unresolved cross-stratum detector, a full generic-kinematics theorem, a metric probability, or evidence for a gravitational QME or Lorentzian causal construction. The scalar coefficient does extend along an exact continuous fixed-energy angle family. For $c = \cos\theta \in (-1, 1)$ the active invariants are

$$s = \frac{64\kappa^2}{25}, \qquad t = -\frac{32(1-c)\kappa^2}{25}, \qquad u = -\frac{32(1+c)\kappa^2}{25}.$$

Four of the ten connected channels remain exactly resonant. In $\kappa = 1$ units their complete tree bracket is

$$\begin{aligned} W(c,T) = 12T &+ \frac{125}{256}\sin\frac{16T}{5} + \frac{125}{128}\sin\frac{8T}{5} \\ &+ \frac{10\sin(a_t T)}{-t} + \frac{10\sin(a_u T)}{-u}, \end{aligned}$$

where $a_t = 2(\sqrt{17-8c} - 3)/5$ and $a_u = 2(\sqrt{17+8c} - 3)/5$. Exact denominator factorizations give

$$W(c,T) \geq \frac{13}{24} T > 0 \qquad (-1 < c < 1,\ T > 0).$$

The finite-time loop contains the corresponding six light-cone transients, with exchange gaps $4\kappa\sqrt{2(1 \mp c)}/5$, and is uniformly bounded on each compact $|c| \leq c_\star < 1$. Certificate
`REVERSE_PHYSICS_BT_CONTINUOUS_ANGLE_Q6_ FAMILY_V1`
therefore proves that the selected scalar q_6 interference is not a ninety-degree artifact and supplies the fibrewise coefficient $R_6(c; f, T, \mu)$. Its dependency tags are

`LOCAL-ALGEBRAIC`, `EUCLIDEAN-SPECTRAL`,
`REDUCED-MODE`.

It does not construct an off-diagonal two-angle detector, either forward endpoint, a scalar all-order scattering operator, a metric BV–BRST transfer, a gravitational coefficient or Lorentzian causality. An operational two-angle detector can be supplied separately. With $P_+ = \frac{1}{2}\left(\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right)$,

$P_- = I - P_+$ and $E_\epsilon = P_+ + (1-\epsilon)P_-$, the click effect has nonzero off-diagonal entries, eigenvalues $1, 1-\epsilon$, and positive complement ϵP_-. Because the fixed-s leading scalar output is proportional to $(1,1)$, E_ϵ fixes it and hence leaves both its norm and its cross with the complete order-λ^4 output unchanged. Certificate
`REVERSE_PHYSICS_BT_TWO_ANGLE_COHERENT_ Q6_DETECTOR_V1`
therefore gives

$$q_\epsilon(c_1, c_2) = 2q_4 \left\{ 1 + \lambda^2 \frac{R_6(c_1) + R_6(c_2)}{2} \right\} + O(\lambda^8)$$

for every $0 < \epsilon \leq 1$. At $\epsilon = 1$ the click is the pure symmetric angle mode. The first known detector-sensitive term is $-\epsilon \|Y_4(c_1) - Y_4(c_2)\|^2/2$ at order λ^8. Exact total-Fock parity leaves only the X_4 norm and X_2–X_6 cross in the complete order-eight ledger. Since $E_\epsilon X_2 = X_2$, self-adjointness gives

$$\langle X_2, E_\epsilon X_6 \rangle = \langle X_2, X_6 \rangle$$

for arbitrary complete X_6. Certificate
`REVERSE_PHYSICS_BT_TWO_ANGLE_Q8_COHERENCE_ DIFFERENCE_V1`
therefore proves that the entire relative coefficient is

$$q_8[E_\epsilon] - q_8[I_2] = -\frac{\epsilon}{2} \|X_4(c_1) - X_4(c_2)\|^2 \leq 0.$$

Neither the recorded nor the symmetric bright-port absolute q_8 coefficient is computed: both the complete X_4 Gram data and the common X_2–X_6 interference remain open. The complementary effect P_- is different because $P_- X_2 = 0$. Its probability begins at absolute order eight,

$$q_-(\lambda) = \lambda^8 Q_{8,-} + O(\lambda^{10}), \qquad Q_{8,-} = \frac{1}{2} \|X_4(c_2) - X_4(c_1)\|^2,$$

and the complete q_6 contrast implies

$$\frac{Q_{8,-}}{\bar{q}_4} \geq \frac{(\Delta R_6)^2}{8}, \qquad \bar{q}_4 = q_4/\lambda^4.$$

For the common-lattice modes $c_1 = 0, c_2 = 3/5$ at $\kappa T = 1$, exact alternating-series intervals give $\Delta W > 1/16$ and $\Delta B > 1/220$. Since

$$\Delta R_6 = \frac{2\sqrt{2}\,\Delta W}{3N_s} + \frac{5\Delta B}{24\pi^2} > \frac{49}{511104} \qquad (N_s > 0),$$

one obtains the strict volume-uniform coefficient bound

$$\boxed{\frac{Q_{8,-}}{\bar{q}_4} > \frac{2401}{2089818390528} > \frac{1}{10^9}.}$$

Certificate
`REVERSE_PHYSICS_BT_DARK_PORT_ABSOLUTE_Q8_ LOWER_BOUND_V1`
certifies this finite-volume auxiliary-scalar coefficient. By itself it does not construct continuum packets, compute either still-open bright/recorded coefficient, or transfer the result to the metric theory. This is an operational `LOCAL-ALGEBRAIC`, `REDUCED-MODE` scalar effect, not a detector selected by the public closed-system BT dynamics and not a gravitational result.

The zero-width modes can be thickened within the invariant fixed-P sphere. In coordinates with $d\Omega = dx\, d\varphi$, choose equal azimuth half-width $\delta = 1/10000$ around $\varphi = \pi/2$ and $\varphi = \arccos(3/5)$. Exact interval arithmetic proves the uniform equatorial margins

$$\Delta W > \frac{1}{20}, \qquad \Delta B > \frac{1}{225}.$$

Joint continuity of the hard finite-time tree and loop kernels supplies a positive latitude thickness together with compact incoming and spectator packet neighborhoods. On those neighborhoods the tree packet-functional contrast is positive and $\Delta B_{\rm pkt} > 1/230$. The two equal-area output indicators are normalized, orthogonal and have the same leading amplitude. Consequently

$$\boxed{\frac{Q_{8,-}}{\bar{q}_{4,\rm pkt}} > \frac{2401}{2284119687168} > \frac{1}{10^9}.}$$

Certificate
`REVERSE_PHYSICS_BT_COMPACT_SPHERE_DARK_PORT_ ABSOLUTE_Q8_V1`
proves this compact fixed-P auxiliary-scalar result without assigning a fabricated numerical continuity radius. It supplies neither finite invariant-mass or total-momentum bandwidth nor a metric-theory coefficient. For finite bandwidth, put $x = \Omega T$ and $y = \delta T$. The exact ordered second-Dyson interference kernel is

$$k(x, y) = \frac{1 - \cos((y - x)/2)\,\mathrm{sinc}(y/2)/\,\mathrm{sinc}(x/2)}{y - x}, \qquad k(0, y) = \frac{1}{y} - \frac{\sin y}{y^2}.$$

On $|x| \le 1/2$, $|y| \ge 1$, it obeys

$$|k(x, y) - k(0, y)| \le \frac{32}{5} \frac{|x|}{y^2}.$$

The mismatch-dependent remainder is therefore ultraviolet integrable on compact hard support, while the common $1/y$ divergence is removed by the already fixed local $\overline{\rm MS}$ counterterm. Joint external-momentum continuity supplies a nonempty compact direct-integral neighborhood with

$$\boxed{\frac{Q_{8,-}^{\rm band}}{\bar{q}_{4,\rm band}} > \frac{2401}{9136478748672} > \frac{1}{10^{10}}.}$$

Certificate
`REVERSE_PHYSICS_BT_FINITE_BANDWIDTH_DARK_PORT_ Q8_V1`
proves this globally normalizable finite-bandwidth auxiliary-scalar packet class. It supersedes only the preceding fixed-fibre certificate's sharp-bandwidth boundary. Its bandwidth radius is existential, not numerical, and it supplies neither a local apparatus for the fibrewise projector nor a metric-theory coefficient.

A local dark apparatus exists for a quadrupole direction. With $P = k_1 + k_2$, $r = (k_1 - k_2)/2$, $P \cdot r = 0$, a calibrated spacelike axis a, and target pair mass M_0, take

$$F_2(P, r) = \frac{6}{M_0^4} \left[P^2 (a \cdot r)^2 - \frac{P^2 a^2 - (a \cdot P)^2}{3} r^2 \right].$$

It is a real exchange-even polynomial of degree four and hence the pair symbol of a Hermitian finite-derivative local quadratic scalar density. The invariant fibre identity

$$\langle r_\mu r_\nu \rangle = -\frac{P^2}{12}\left(g_{\mu\nu} - \frac{P_\mu P_\nu}{P^2}\right)$$

gives $\langle F_2 \rangle = 0$ for every timelike P. Thus the local density annihilates the leading angle-independent amplitude exactly throughout the packet bandwidth. At the central centre-of-mass point it reduces to $P_2(c) = (3c^2 - 1)/2$. Exact tree and loop moment bounds give

$$J_{\mathrm{tree}} > \frac{1}{100}, \qquad J_{\mathrm{loop}} > \frac{252416}{73828125} > \frac{1}{400}, \qquad J_R > \frac{1}{19200}.$$

Both contributions have the same positive sign. The exact normalized spherical Cauchy factor is 5/16, and continuity retains half the central relative moment on a nonempty compact four-momentum neighborhood. Hence

$$\boxed{\frac{Q_{8,\mathrm{local}}}{\bar{q}_4} > \frac{1}{4718592000} > \frac{1}{5000000000}.}$$

Certificate
`REVERSE_PHYSICS_BT_LOCAL_QUADRUPOLE_DARK_ DETECTOR_Q8_V1`
proves this `LOCAL-ALGEBRAIC`, `EUCLIDEAN-SPECTRAL`, `REDUCED-MODE` auxiliary-scalar coefficient. With smooth compact Fourier support inside the continuity neighborhood, selecting pointer excitation together with the active-field vacuum and unchanged spectator yields

$$p_{\mathrm{selected}} = g_{\mathrm{det}}^2 \lambda^8 Q_{8,\mathrm{local}} + O(g_{\mathrm{det}}^2 \lambda^{10}) + O(g_{\mathrm{det}}^4).$$

The switching is Schwartz rather than compactly supported in spacetime. The apparatus is external to BT dynamics, the detector-coupling remainder is uncontrolled, and no metric BV–BRST, Eq. (19), or Lorentzian-causal result follows.

The noncompact switching is not essential to the coefficient. Choose a real $\chi \in C_c^\infty(\mathbb{R}^4)$ equal to one on the unit ball and supported in the radius-two ball, and set

$$h_R(x) = \chi(x/R) h_0(x).$$

Then $h_R \to h_0$ in Schwartz topology by Leibniz scaling and arbitrary Schwartz tail gain. The imported global Hilbert–Schmidt tree bound and local integrability of the loop's logarithmic endpoints make the selected finite-order response a tempered distribution in h. Consequently some finite R_0 retains more than half the nonzero order-λ^4 amplitude. For every finite R, however,

$$A_2(h_R) = \int dP\, \widehat{h}_R(P) C_2(P) \int_{S_P^2} F_2(P, r)\, d\Omega_P = 0,$$

because the inner quadrupole mean vanishes on each timelike fibre. The noncompact Fourier tails forced by compact spacetime support therefore do not restore a lower order. It follows that

$$\boxed{\frac{Q_{8,\mathrm{compact}}}{\bar{q}_4} > \frac{1}{18874368000} > \frac{1}{20000000000}.}$$

Certificate
`REVERSE_PHYSICS_BT_COMPACT_SPACETIME_ QUADRUPOLE_DARK_DETECTOR_Q8_V1`
proves this compactly switched auxiliary-scalar coefficient with an existential rather than numerical support radius. It is a local insertion at leading detector order, not a construction of renormalized Lorentzian time-ordered products, causal perturbative AQFT, Eq. (19), or metric gravity.

The compact response also has an exact operational normalization. Let u_2 be its normalized nonzero two-particle response mode and set

$$K_{\text{click}} = \frac{1}{2}|0\rangle\langle u_2|, qquad K_{\text{no}} = I + \left(\frac{\sqrt{3}}{2} - 1\right)|u_2\rangle\langle u_2|.$$

Their effects are $|u_2\rangle\langle u_2|/4$ and its complement, and the corresponding Julia matrix

$$\begin{pmatrix} \sqrt{3}/2 & 0 & -1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & \sqrt{3}/2 \end{pmatrix}$$

is exactly unitary. Hence the finite-strength click probability is

$$p_{\text{click}}(\Psi) = \frac{1}{4}|\langle u_2, \Psi\rangle|^2.$$

It annihilates the complete fibrewise angle-independent X_2 subspace and has nonzero overlap with X_4, so its first BT term is a strictly positive order-λ^8 coefficient without a detector-coupling remainder.

This exact instrument is not the full local Hamiltonian exponential: $\int_{-1}^{1} P_2(c)^3 dc = 4/35$, so three quadrupole insertions contain a scalar component. Nor is it a local-algebra instrument: the click map is a normalized $P_0 D_h P_2$ compression and the no-click map contains the global rank-one response projector. Certificate
`REVERSE_PHYSICS_BT_EXACT_QUADRUPOLE_JULIA_ INSTRUMENT_V1`
therefore classifies an exact operational completion, not public BT dynamics, all-order BT probability, causal local AQFT, Eq. (19), or gravity.

The locality boundary is a conditional no-go under standard local-net hypotheses. Let $\mathcal{M} = \mathcal{A}(O)$ commute with the spacelike-complement algebra $\mathcal{N} = \mathcal{A}(O')$, and suppose $\mathcal{N}\Omega$ is dense. Then $A\Omega = 0$ for $A \in \mathcal{M}$ implies $AB\Omega = BA\Omega = 0$ on the dense set $\mathcal{N}\Omega$, hence $A = 0$. Thus Ω is separating. If a positive effect $E \in \mathcal{M}$ has zero vacuum probability, then $\|E^{1/2}\Omega\|^2 = \langle\Omega, E\Omega\rangle = 0$, so $E = 0$. The nonzero effect $P_u/4$ is exactly vacuum-dark and therefore is not local under these hypotheses. The same applies to both Kraus operators and to every normal local tensor-pointer dilation inducing this effect by a slice map. Certificate
`REVERSE_PHYSICS_BT_REEH_SCHLIEDER_LOCAL_ DETECTOR_OBSTRUCTION_V1`
proves this `LOCAL-ALGEBRAIC` statement. It does not construct a positive Haag–Kastler net or Reeh–Schlieder vacuum for public BT dynamics [21].

This does not prohibit a balanced local contrast: $E_\pm = (I \pm B)/2$ for a self-adjoint local contraction B. If $\langle\Omega, B\Omega\rangle = 0$, both outcomes have vacuum probability 1/2 but their difference has zero vacuum mean. This contrast exists exactly under a sharp additional hypothesis. If the compact quadrupole density has a self-adjoint realization D affiliated with $\mathcal{M}$ and $\Omega, X_2, X_4 \in \text{Dom}(D)$, its zero vacuum and X_2 matrix elements and nonzero X_4 matrix element pass to the limit of the bounded spectral truncations $D_n = D1_{[-n,n]}(D)$. Their five real response coordinates span a closed subspace of $\mathbb{R}^5$, so that nonzero limit is an exact finite combination of at most five truncation vectors. The corresponding finite combination of D_n, normalized in operator norm, is a bounded self-adjoint local B with exact X_2 cancellation and strict X_4 vacuum-to-pair response. Operationally this is read by the phase-reversal contrast

$$\frac{1}{4}\{\langle\Omega+X,B(\Omega+X)\rangle-\langle\Omega-X,B(\Omega-X)\rangle\}=\Re\langle\Omega,BX\rangle,$$

which is exactly zero for X_2 and strictly nonzero for a calibrated phase of X_4. It is not a one-shot dark-port probability. Thus the remaining local gate is the positive BT local net, self-adjoint affiliation and domain theorem, not bounded functional calculus itself. Eq. (19) and gravity remain open.

The public BT carrier supplies an exact additional real-structure test. On a nonzero positive-frequency packet its two-field Wightman matrix factors through

$$G=\begin{pmatrix}0&1\\1&0\end{pmatrix},$$

whose $(1,1)$ and $(1,-1)$ directions have norms 2 and -2. Hence it cannot be a positive-Hilbert vacuum two-point matrix while Ω and Υ remain individually Hermitian. With the public fundamental symmetry $\kappa = G$, one instead has $G\kappa = I$ and

$$A^*=\kappa A^\sharp\kappa,\qquad \Omega^*=\Upsilon,\qquad T^*=T,\qquad X^*=-X.$$

For every Krein-selfadjoint A, its ghost-even part is Hilbert-selfadjoint and its ghost-odd part is Hilbert-anti-selfadjoint; thus

$$A^*=A\quad\Longleftrightarrow\quad\kappa A\kappa=A,\qquad (iA_{\mathrm{odd}})^*=iA_{\mathrm{odd}}.$$

This is not merely an adjoint convention. The exact weak-ghost fixture $B = I$, $Q = E_{21}$ obeys $Q^\sharp = Q$, $Q^2 = 0$ and

$$\operatorname{tr}(Q^\sharp Q)=\operatorname{tr}(B^\sharp Q)=0,\qquad \operatorname{tr}[(B+Q)^\sharp(B+Q)]=2,$$

whereas the positive-Hilbert adjoint gives

$$\operatorname{tr}(Q^*Q)=1,\qquad \operatorname{tr}[(B+Q)^*(B+Q)]=3.$$

Certificate
`REVERSE_PHYSICS_BT_POSITIVE_LOCAL_REAL_ STRUCTURE_DICHOTOMY_V1`
therefore requires the compact quadrupole to have certified internal ghost-evenness, or a controlled i-times-odd projection, before the positive local contrast is instantiated. Likewise, the negative-charge remainder in Eq. (19) needs a local dynamics-compatible quotient or conditional expectation: Krein nullity is not positive-Hilbert nullity. This is solely a `LOCAL-ALGEBRAIC` scalar boundary and supplies no metric-BV, gravitational, QME or Lorentzian claim.

The scalar quadrupole itself decides this parity test negatively on the regular perturbative chart. If $D[\phi] = B(\phi, \phi)$ and $g = \lambda^{-1}\log(\psi/\lambda)$, then hidden exchange gives

$$D[g-\phi]=D[\phi]-2B(\phi,g)+D[g].$$

Both parity projections contain the logarithmic mirror term. Along $\phi = 0$, $\psi_t = \lambda t e^{\lambda f}$, every finite ψ_t jet tends to zero, but the selected nonzero-momentum pair coefficient of $D[g]$ is the coefficient of $D[f]$ and is independent of t. The rational quadrupole fixture $P = (1, 0, 0, 0)$, $r = (0, 1/2, 0, 0)$, $a = (0, 1, 0, 0)$ gives coefficient one, whereas $f = 0$ gives zero at the same limiting vacuum jet. Thus no single-valued regular same-chart extension exists.

On two exchanged vacuum sheets a constructive but changed completion is available. With

$$G = \kappa = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad D_{\text{dbl}} = \text{diag}(D_A, D_B),$$

one has $G\kappa = I$ and, for identically transported mirror dynamics, $\kappa D_{\text{dbl}} \kappa = D_{\text{dbl}}$ and $D^{\sharp}_{\text{dbl}} = D^{*}_{\text{dbl}} = D_{\text{dbl}}$. Normalized symmetric source and detector sheets retain the single-sheet amplitude, exact leading darkness, and the inherited strict compact q_8 bound. Certificate
`REVERSE_PHYSICS_BT_QUADRUPOLE_MIRROR_SHEET_ DICHOTOMY_V1`
proves the regular-chart obstruction and the doubled repair. Since the public scalar action and R_t do not select the second vacuum sheet, this is not a public-BT or gravitational result and does not prove Eq. (19).

There is, however, a regular one-sheet detector directly in the public auxiliary variables. With the same STF pair symbol, set

$$D_{\text{aux}} = D_{\Omega} + D_{\Upsilon}, \qquad D_X = :X(x_1) F_2 X(x_2): \big|_{x_1 = x_2}.$$

The complete real symmetric ghost-even quadratic family is $C(a, b) = \left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$. Boost neutrality forces $a = 0$, but the remaining cross bilinear is exactly blind to the pure positive source $u_0 = (|\Upsilon^3\rangle + |\Omega^3\rangle)/\sqrt{2}$. The responding choice $C = I$ has charge branches ± 2. Pointer click states e_-, e_+ with charges ∓ 2 give the neutral interaction

$$V = h \left(|e_-\rangle\langle g| \otimes D_{\Omega} + |e_+\rangle\langle g| \otimes D_{\Upsilon}\right) + \text{Krein adjoint}.$$

Its selected pair map preserves charge, intertwines ghost parity and maps u_0 isometrically to the positive even state $(|e_-\Upsilon\rangle + |e_+\Omega\rangle)/\sqrt{2}$. The STF identity removes the order-λ^2 response fibrewise. The pure connected-tree quadrupole moment is one fifth of the certified positive moment and the loop moment remains positive, hence

$$\frac{Q_{8,\text{aux,compact}}}{\bar{q}_4} > \frac{1}{18874368000}.$$

Certificate
`REVERSE_PHYSICS_BT_AUXILIARY_POLYNOMIAL_ QUADRUPOLE_POSITIVE_DETECTOR_V1`
checks the exact species, charge, parity, adjoint and response identities. This is a `LOCAL-ALGEBRAIC/ EUCLIDEAN-SPECTRAL/ REDUCED-MODE` public auxiliary result. Full self-adjoint local affiliation does not follow from the finite selected-core matrix alone.

It does follow from a closed-column construction on the positive free carrier. Since $\Omega^* = \Upsilon$, the auxiliary pair is a complex scalar. For compact h, locality and the adjoint relations on the common Weyl-invariant exponential-polynomial domain [22, 23] give

$$D_{\Upsilon}(\bar{h}) \subset D_{\Omega}(h)^*, \qquad D_{\Omega}(\bar{h}) \subset D_{\Upsilon}(h)^*.$$

Hence $K_h\psi = (D_\Omega(h)\psi, D_\Upsilon(h)\psi)$ is closable, without assuming that either branch is essentially self-adjoint. For its closure,

$$V_h = \begin{pmatrix} 0 & K_h^* \\ K_h & 0 \end{pmatrix}$$

is self-adjoint: its explicit resolvent at $z = \pm i$ has diagonal denominators $K_h^* K_h - z^2$ and $K_h K_h^* - z^2$, so both deficiency ranges are full. Free-field locality affiliates V_h with the three-pointer matrix amplification of the dual local net. The charged pointer makes the block neutral and ghost even. Bounded functional calculus then gives

$$U_g = e^{-igV_h}, \qquad E_{\text{click}} = \sin^2(g|K_h|), \qquad E_{\text{no}} = \cos^2(g|K_h|)$$

as exact positive complementary local effects for every real g.

A phase-reversed coherent ground/click input, followed by pointer-only readout, has half-contrast tangent

$$C'_\theta(0) = \operatorname{Im}\Big(e^{-i\theta}\langle v, K_h u\rangle\Big), \qquad \max_\theta |C'_\theta(0)| = |\langle v, K_h u\rangle|.$$

Thus the certified positive compact q_8 response is an operational local unitary tangent and needs no final field-vacuum projection. Certificate
`REVERSE_PHYSICS_BT_AUXILIARY_POINTER_LOCAL_ UNITARY_V1`
checks the closability, block resolvent, affiliation, symmetries and exact contrast. The local unitary is a bounded functional-calculus operation in the free auxiliary carrier, not interacting BT time evolution. Equality of the positive Hilbert and public generalized Krein Born rules, higher λ control, Eq. (19), metric BV–BRST transfer, QME restoration and every Lorentzian claim remain open.

The Born-rule comparison is nevertheless exact for this selected process. For $\alpha(A) = \kappa_{\text{tot}} A \kappa_{\text{tot}}$ and $A = A_+ + A_-$, the relation $A^* = \kappa_{\text{tot}} A^\sharp \kappa_{\text{tot}}$ implies

$$\operatorname{Tr}(A^\sharp A) = \|A_+\|_2^2 - \|A_-\|_2^2.$$

The canonical expectation $E_\kappa(A) = A_+$ changes this public weight by $\|A_-\|_2^2$ and therefore preserves it if and only if A is already fixed. It cannot remove a general weak-ghost remainder probability-preservingly.

For the pointer theorem, V_h, U_g, P_g and P_{click} are all fixed by total ghost parity. Consequently

$$A_g = P_{\text{click}} U_g P_g, qquad A_g^\sharp = A_g^*, qquad \boxed{A_g^\sharp A_g = A_g^* A_g = \sin^2(g|K_h|).}$$

The public generalized-Born and positive-Hilbert prescriptions thus give the same selected local click effect and the same strictly positive compact q_8 tangent. Certificate
`REVERSE_PHYSICS_BT_KAPPA_FIXED_BORN_ DESCENT_V1`
checks the expectation obstruction and operator-level selected descent. It does not prove arbitrary-process equivalence, Eq. (19), interacting BT dynamics, metric BV–BRST transfer or a Lorentzian theorem.

On the same two-mode carrier, the bounded interaction

$$H_{\text{int}} = gP_-(\phi) \otimes \sigma_y$$

with a degenerate pointer qubit gives, after duration τ,

$$K_{\rm click} = P_+(\phi) + \cos(g\tau)P_-(\phi), \qquad K_{\rm no} = \sin(g\tau)P_-(\phi).$$

Thus its measured effect is the one above with $\epsilon = \sin^2(g\tau)$, while the conjugation $P_-(\phi) = S_\phi P_-(0)S_\phi^\dagger$ selects the relative angle phase. Certificate
REVERSE_PHYSICS_BT_TWO_ANGLE_FINITE_APPARATUS_ HAMILTONIAN_V1
checks the exact four-dimensional unitary, both Kraus maps, phase calibration and the mismatch response. It constructs external detector physics coupled to the scalar BT output modes; it is not contained in the closed-system BT Hamiltonian, is not spacetime local, and supplies no gravitational detector. On the two rational modes, a microscopic scalar vertex can nevertheless be made local. The vacuum-to-pair weights of $D_+ =:\phi^2:$ are $(1, 1)$, while

$$D_- =:\phi^2: -\frac{625}{72\kappa^2} :(\partial_y\phi)^2:$$

has weights $(1, -1)$. Two real detector quadratures consequently synthesize the arbitrary phased weights $(1, -e^{-i\phi})/\sqrt{2}$, and the compression to $|g, +_\phi\rangle, |g, -_\phi\rangle, |e, 0_{\rm field}\rangle$ is the exact pair-absorption Hamiltonian with the same two effects as the finite apparatus. This is only a selected compression. At fixed total momentum a compact smearing factor is common to the continuous angular family, and every finite-derivative local quadratic kernel is a finite Laurent polynomial in $e^{i\theta}$. Such a nonzero polynomial cannot vanish on the open complement of two angles. Certificate
REVERSE_PHYSICS_BT_TWO_ANGLE_LOCAL_DETECTOR_ COMPRESSION_V1
therefore obstructs exact continuum two-angle locality and forbids identifying the exponential of the compression with the compression of full evolution. Finite angular packets therefore require a quantitative leakage bound. The construction is anisotropic scalar apparatus physics and supplies no metric, BV–BRST, QME, graviton-state or Lorentzian-causal result. Such a bound is explicit on the complete invariant angular shell at the selected fixed total momentum. In the centre-of-momentum frame the massless pair is parametrized by $n \in S^2$, and the invariant measure is a common factor times $d\Omega$. The earlier equal-lab-energy family is the equator $n_x = 0$, whose angular measure is $d\varphi$, not dc. For $\zeta = e^{2i\varphi}$, the normalized difference of two order-20 projective Fejér kernels has exact values $+1, -1$ at the target pair lines. On the equatorial bins

$$B_1^\varphi = [\pi/4, 3\pi/8], \qquad B_0^\varphi = [3\pi/8, 5\pi/8]$$

its exact invariant leakage is below 331/500000. Its homogeneous lift

$$F(n) = (1 - n_x^2)^{19} p(\varphi)$$

is an antipodally even degree-38 polynomial and hence the symbol of one finite local quadratic density with at most 38 derivatives on the entire sphere. Adding the common band $|n_x| \leq 1/2$ gives exact latitude leakage below 3×10^{-6} and total complement fraction

$$\eta_{S^2} = 0.000663615867055246985\ldots < \frac{83}{125000} < 10^{-3}.$$

The inequality is certified in $\mathbb{Q}(\pi, \sqrt{2})$ by strict rational intervals, not by the displayed decimal. If B is the normalized selected bright packet and R the complement, the complete coupled direction is $v = \sqrt{1-\eta_{S^2}}B + \sqrt{\eta_{S^2}}R$. Exponentiating $H/G = |e\rangle\langle v| + |v\rangle\langle e|$ with R retained gives

$$E_{\mathrm{absorb}} = (1-\eta_{S^2})\sin^2(G\tau)P_B.$$

Certificate
`REVERSE_PHYSICS_BT_FIXED_P_TWO_SPHERE_PACKET_ DETECTOR_V1`
therefore supplies a sub-per-mille invariant full-solid-angle packet detector at fixed P and a half-pulse efficiency above 124917/125000, without substituting the exponential of a compression. It does not control invariant mass or total-momentum thickness, number-scattering or counter-rotating sectors, nor transfer the earlier equal-weight relative-q_8 formula to these unequal-norm packets, and has no metric or Lorentzian-causal implication. The fixed-P scalar response also has a four-momentum-thick leading successor. In the target centre-of-momentum apparatus frame, smear the same degree-38 pointwise local density with a complex Gaussian modulation whose squared Fourier envelope is

$$|\widehat{h}(P)|^2 = e^{-\|P-P_0\|_E^2/\sigma^2}, \qquad \frac{\sigma}{M_0} = \frac{1}{50000}, \qquad M_0 = \frac{8\kappa}{5}.$$

On the five-sigma core, the rotationless centre-of-momentum boost changes the angular filter by a rigorously bounded amount. The Fejér coefficient ℓ^1 bounds control its degree-38 gradient. Outside the core, the degree-76 squared polynomial is dominated by a four-dimensional Gaussian with a rationally enclosed radial tail. Combining both estimates gives

$$\eta_{\mathrm{complete}} < 0.00088259945714573480 7\ldots < \frac{883}{10^6}.$$

The resulting pair-annihilation response is a nonzero L^2 vector on an open set of total momenta and defines a positive normalized leading packet effect.

The switching also separates quadratic frequency sectors. For future null momenta, every number-scattering transfer satisfies

$$(k_{\mathrm{in}} - k_{\mathrm{out}})^2 \le 0.$$

Its squared apparatus-frame distance from the timelike centre P_0 is at least $M_0^2/2$, while the wrong-sign future-pair argument in the past causal cone is at least M_0^2 away. The corresponding pointwise squared Fourier envelopes are bounded by $e^{-1{,}250{,}000{,}000}$ and $e^{-2{,}500{,}000{,}000}$.
Certificate
`REVERSE_PHYSICS_BT_SCHWARTZ_FOUR_MOMENTUM_ PACKET_RESPONSE_V1`
proves the thick leading scalar response and the cone-distance statement. The Gaussian is not compactly supported, the pointwise suppression is not an operator-norm estimate on the unrestricted number sector, and the complete time-ordered evolution remains open. None of these scalar bounds transfers to the metric BV–BRST or gravitational carrier. An operator-bounded scalar successor makes the off-shell density explicit:

$$\mathcal{D}_F(x) =: F\left(\frac{i\partial_1 - i\partial_2}{M_0}\right)\phi(x_1)\phi(x_2) : \Big|_{x_1=x_2=x}.$$

Its pair and number symbols are respectively $F((k_1-k_2)/M_0)$ and $F((k+k')/M_0)$, with switching transfers k_1+k_2 and $k-k'$. On

$$K = \{k : M_0/4 \le |k| \le 3M_0/4\}, \qquad \mu(K) = \frac{\pi M_0^2}{2},$$

the compressed number kernel is Hilbert–Schmidt and the exact comparisons are

$$\frac{\|N_K\|^2}{\|w\|^2} < 2^{153-1,250,000,000} < 10^{-1,000,000}, \qquad \frac{\|w_-\|^2}{\|w\|^2} < 2^{47-2,499,999,924} < 10^{-1,000,000}.$$

After both adjoints are included,

$$\frac{\|E_{\text{undesired}}\|}{\|A_{\text{pair}}\|} < 10^{-400,000}$$

at first Dyson order on the declared energy-windowed domain. Certificate
`REVERSE_PHYSICS_BT_COMPACT_ENERGY_QUADRATIC_ SECTOR_BOUND_V1`
proves this scalar operator statement. It does not control scattering from K to its complement, unrestricted energies, higher time ordering, recorded or bright-port absolute q_8, Eq. (19), or any metric BV–BRST or Lorentzian-causal carrier. There is nevertheless a genuine scalar hard-scattering baseline. The certificate
`REVERSE_PHYSICS_BT_UV_ HARD_SCATTERING_LAW_V1`
proves at leading logarithmic order that every fixed nonforward scalar acceptance falls as $24\pi^2/[25s\log^2(s/\Lambda^2)]$ per unit solid angle. This uses the scalar Born normalization, scalar beta function, and scalar projected hard log. None of its coefficient, positivity statement, or Callan–Symanzik closure transfers to the graviton/BRST carrier; the gravity paper records it only as the kind of physical endpoint an independent gravitational construction would have to reach.

14 Discussion

14.1 The same order pattern, for different reasons

Paper V and the present gravity calculation share the sequence

$$\text{first-order protection} \longrightarrow \text{second-order scattering obstruction.} \tag{77}$$

The agreement in order should not obscure the difference in structure. The perfect-square scalar is protected by its special cubic kinematic factor and reality assignments. Gravity is protected because the Einstein solution space is an exact nonlinear subsector and the remaining odd cubic vertex is kinematically closed. In the scalar, an enhanced $O(1,1)$ charge appears exactly at the perfect-square boundary and can sort an obstruction into a one-sided null ideal. In gravity, the MMM and MMh interactions remove every uniform abelian charge before the four-point obstruction is considered.

14.2 Relation to the PT-symmetric quadratic-gravity proposal

Kuntz [9] starts from the same scalar-free Einstein–Weyl vertex content used here. His auxiliary-field complex deformation makes the massless and massive free spin-2 sectors Hermitian and positive and places factors of i on interactions with an odd number of massive fields. At this level his rotation and (6) are the same construction up to the sign chosen for the quarter-turn. His transverse physical asymptotic space also retains both massless helicities and all five massive polarizations, so the distinction is not that the M states have been removed.

The difference is in the interacting test. The displayed metric calculation in Ref. [9] replaces the full tensor field theory by a mode Hamiltonian with spin and momentum dependence suppressed and with schematic interaction coefficients. It demonstrates a positive perturbative metric for that truncation and offers it as evidence for the gravitational theory, but it does not include the exact energy-degenerate $MM \to Mh$ block. Proposition 2.1 shows why that omission is decisive: a nonzero anti-Hermitian component inside a degenerate free-energy block is not divided by a free-energy difference and cannot be removed by an analytic R_2.

Thus our result is a direct full-vertex stress test of that proposal, not a general rejection of PT-symmetric gravity. Any extension of the truncated construction to the complete theory must evade (54) by leaving at least one of the present hypotheses: for example through a nonanalytic metric, a different free base point, different asymptotic states, or a prescription that excludes the massive mode from the physical scattering spectrum.

14.3 Real forms become dynamical

At the free level a real form selects which complex solutions are called real and which adjoint defines observables. The choice can appear kinematic. The phase theorem makes its interacting content explicit: every physical process with odd external massive number crosses the two reality assignments and becomes anti-Hermitian in the positive frame. The decisive diagnostic is therefore

$$\boxed{\text{Which physical on-shell processes change the real-form sector?}} \tag{78}$$

For Einstein–Weyl gravity the first answer is $MM \to Mh$.

14.4 Why the factorization and physical rails are both useful

The Einstein-frame residue (38) is compact, independent, and immune to cancellation by local contact terms. It proves a genuine interaction before the large quartic calculation is attempted. The original-variable physical point then supplies what factorization alone does not: an exact real-shell value, the full gauge cancellation, the physical adjoint, and the deformation-complex matrix element. Agreement between the two rails is substantially stronger than either calculation in isolation.

14.5 What Krein visibility means

The negative traces in (67) and (69) are not advertised as negative probabilities. They show that the conventional quadratic Krein form sees the block: it is neither zero nor killed by a grading selection rule. A viable interacting probability construction would need extra structure beyond the free fundamental symmetry. The scalar boundary supplies a continuous one-sided charge at tree level, but the exact scalar loop calculation shows that this algebraic structure alone does not select an ordinary axis-compatible infrared prescription. The finite scalar projector transport shows that coherent cross-multiplicity normalization is algebraically compatible after Born-rate normalization, while the ordinary cubic Fock generator has zero collinear Gram; it does not by itself show that the required Jordan term exists. The exact scalar calculation derives its two-annihilator coefficient and proves secular cancellation, but stops at a cubic endpoint-pole obstruction before a continuum projector or probability exists. The verified Einstein–Weyl vertices do not even supply the corresponding uniform charge, and no gravitational infrared completion is constructed here.

The unresolved gravitational problem is whether conformal-boundary conditions, changed observable algebras, or genuinely nonlocal gravitational structures can evade the hypotheses of Theorem 12.1 without destroying covariance and factorization. Any proposal that retains massless asymptotic gravitons must additionally declare how its real and virtual collinear sectors are glued and prove regulator independence on the BRST quotient; the scalar certificate motivates this requirement but is not evidence that gravity passes or fails it.

References

[1] K. S. Stelle, *Renormalization of higher-derivative quantum gravity*, Phys. Rev. D **16** (1977) 953, doi:10.1103/physrevd.16.953.

[2] A. Salvio, *Quadratic gravity*, Front. Phys. **6** (2018) 77, arXiv:1804.09944.

[3] A. Hindawi, B. A. Ovrut and D. Waldram, *Consistent spin-two coupling and quadratic gravitation*, Phys. Rev. D **53** (1996) 5583–5596, arXiv:hep-th/9509142.

[4] G. Magnano and L. M. Sokołowski, *Nonlinear massive spin-two field generated by higher derivative gravity*, Ann. Phys. **306** (2003) 1–35, arXiv:gr-qc/0209022.

[5] P. Donà, S. Giaccari, L. Modesto, L. Rachwał and Y. Zhu, *Scattering amplitudes in super-renormalizable gravity*, JHEP **08** (2015) 038, arXiv:1506.04589.

[6] A. Mostafazadeh, *Metric operator in pseudo-Hermitian quantum mechanics and the imaginary cubic potential*, J. Phys. A **39** (2006) 10171–10188, arXiv:quant-ph/0508195.

[7] C. M. Bender, D. C. Brody and H. F. Jones, *Extension of PT-symmetric quantum mechanics to quantum field theory with cubic interaction*, Phys. Rev. D **70** (2004) 025001, arXiv:hep-th/0402183.

[8] J. Kubo and J. Kuntz, *Analysis of unitarity in conformal quantum gravity*, Class. Quantum Grav. **39** (2022) 175010, arXiv:2202.08298.

[9] J. Kuntz, *Unitarity through PT symmetry in quantum quadratic gravity*, arXiv:2410.08278.

[10] T. Adamo, S. Nakach and A. A. Tseytlin, *Scattering of conformal higher spin fields*, JHEP **07** (2018) 016, arXiv:1805.00394.

[11] H. Johansson, G. Mogull and F. Teng, *Unraveling conformal gravity amplitudes*, JHEP **09** (2018) 080, arXiv:1806.05124.

[12] J. F. Donoghue and G. Menezes, *Unitarity, stability and loops of unstable ghosts*, Phys. Rev. D **100** (2019) 105006, arXiv:1908.02416.

[13] D. Anselmi, *Fakeons and Lee–Wick models*, JHEP **02** (2018) 141, arXiv:1801.00915.

[14] H.-S. Liu, H. Lü, C. N. Pope and J. F. Vázquez-Poritz, *Not conformally-Einstein metrics in conformal gravity*, arXiv:1303.5781.

[15] S. Kamefuchi, L. O'Raifeartaigh and A. Salam, *Change of variables and equivalence theorems in quantum field theories*, Nucl. Phys. **28** (1961) 529–549, doi:10.1016/0029-5582(61)90056-6.

[16] S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*, Cambridge University Press, 1995.

[17] C. M. Bender and P. D. Mannheim, *No-ghost theorem for the fourth-order derivative Pais–Uhlenbeck oscillator model*, Phys. Rev. Lett. **100** (2008) 110402, doi:10.1103/physrevlett.100.110402.

[18] C. M. Bender and P. D. Mannheim, *Exactly solvable PT-symmetric Hamiltonian having no Hermitian counterpart*, Phys. Rev. D **78** (2008) 025022, arXiv:0804.4190.

[19] S. Bateman and N. Turok, *Escape from Ostrogradsky via hidden ghost parity*, arXiv:2607.00096.

[20] R. L. Hudson and K. R. Parthasarathy, *Quantum Itô's formula and stochastic evolutions*, Commun. Math. Phys. **93** (1984) 301–323, doi:10.1007/BF01258530.

[21] H. Reeh and S. Schlieder, *Bemerkungen zur unitäräquivalenz von lorentzinvarianten Feldern*, Nuovo Cim. **22** (1961) 1051–1068, doi:10.1007/BF02787889.

[22] R. Brunetti, K. Fredenhagen and M. Köhler, *The microlocal spectrum condition and Wick polynomials of free fields on curved spacetimes*, Commun. Math. Phys. **180** (1996) 633–652, arXiv:gr-qc/9510056.

[23] K. Sanders, *Essential self-adjointness of Wick squares in quasi-free Hadamard representations on curved spacetimes*, J. Math. Phys. **53** (2012) 042502, arXiv:1010.3978.

[24] J. Maldacena, *Einstein gravity from conformal gravity*, arXiv:1105.5632.

[25] T. Ya. Azizov and I. S. Iokhvidov, *Linear Operators in Spaces with an Indefinite Metric*, Wiley, 1989.

[26] GPT-5.6.sol and A. A. Palm, *Canonical Positive Symplectic Diagonalization of the Pais–Uhlenbeck Oscillator*, Paper I of the pure-Weyl gravity programme, 2026; file `paper/01-symplectic-diagonalization.pdf` in the repository `https://github.com/area9innovation/weyl-gravity`.

[27] GPT-5.6.sol and A. A. Palm, *The Pais–Uhlenbeck Metric as a Minimum-Distortion Principle, and the Representation Problem for the Fourth-Order Field*, Paper II of the pure-Weyl gravity programme, 2026; file `paper/02-variational-fock.pdf` in the repository `https://github.com/area9innovation/weyl-gravity`.

[28] GPT-5.6.sol and A. A. Palm, *The Universal Vacuum of the Fourth-Order Scalar Field: Metric Orbits, Fock Sectors, and the Krein Boundary*, Paper III of the pure-Weyl gravity programme, 2026; file `paper/03-fourth-order-vacuum.pdf` in the repository `https://github.com/area9innovation/weyl-gravity`.

[29] GPT-5.6.sol and A. A. Palm, *Gauge Reduction and the Completion Problem in Fourth-Order Gravity: PU Pairing, Covariant Real Forms, and the Conformal Jordan Boundary*, Paper IV of the pure-Weyl gravity programme, 2026; file `paper/04-fourth-order-gravity.pdf` in the repository https://github.com/area9innovation/weyl-gravity.

[30] GPT-5.6.sol and A. A. Palm, *Interaction Obstructions, Resonant PT Breaking, and Doubled Jordan Symmetry in Fourth-Order Theories*, Paper V of the pure-Weyl gravity programme, 2026; file `paper/05-interaction-obstructions.pdf` in the repository https://github.com/area9innovation/weyl-gravity.

A Einstein consistent truncation and LSZ

This appendix spells out the hypotheses behind Lemma 4.1. For the scalar-free combination in (1), the field equation is the Einstein tensor plus a multiple of the Bach tensor. In four dimensions the Bach tensor may be written

$$B_{\mu\nu} = \left(\nabla^\rho \nabla^\sigma + \frac{1}{2} R^{\rho\sigma}\right) C_{\mu\rho\nu\sigma} \tag{79}$$

up to a convention-dependent overall sign. It vanishes on every Einstein metric [14]. In the flat-background problem the relevant exact subsector is Ricci-flat vacuum Einstein gravity.

On the regular branch of the Legendre transformation from Ricci-tensor gravity to its second-order form, introduce the Einstein metric and a symmetric massive field $\Phi_{\mu\nu}$. The transformation is algebraic in the auxiliary variables and locally invertible wherever its determinant does not vanish [4]. Linearization about the common flat vacuum diagonalizes the asymptotic equations into

$$\mathcal{K}_h h = 0, \qquad \mathcal{K}_M M = 0, \tag{80}$$

with $p^2 = 0$ and $p^2 = M^2$, respectively. Thus h and M used in the main text are LSZ mass eigenfields, not arbitrary components of the original metric perturbation.

Let $E_M[h, M] = 0$ be the exact massive-field equation in this frame. The consistent truncation states

$$E_M[h_E, 0] = 0 \tag{81}$$

for every nonlinear Einstein solution h_E. Expanding about flat space,

$$E_M[h, M] = \mathcal{K}_M M + \sum_{n\geq 2} J_M^{(n)}[h^n] + \sum_{r\geq 1, s\geq 1} J_{r,s}[M^r h^s]. \tag{82}$$

Equation (81) does not require each off-shell functional $J_M^{(n)}$ to vanish. It requires its value on a full perturbative Einstein solution to cancel, including nonlinear corrections to that solution. LSZ amputation removes precisely the external inverse- propagator and field-redefinition terms, leaving

$$\mathcal{A}(M, h_1, \ldots, h_n) = 0 \tag{83}$$

for physical on-shell Einstein gravitons. This is the standard consistent-truncation selection rule.

The perturbiner supplies an explicit cubic check. In signature $(+,-,-,-)$, take all momenta incoming,

$$p_M = (1,0,0,0), \qquad p_2 = \left(-\frac{1}{2},0,0,-\frac{1}{2}\right), \qquad p_3 = \left(-\frac{1}{2},0,0,\frac{1}{2}\right). \tag{84}$$

They obey $p_M^2 = 1$, $p_2^2 = p_3^2 = 0$ and sum to zero. The exact action coefficient vanishes for all five massive TT polarizations and the four products of the two graviton helicities. Replacing either graviton polarization by $p_{(\mu}\xi_{\nu)}$ also gives zero.

The selection rule is tree-level and concerns external mass eigenfields. It does not state that the original fourth-order metric action lacks terms apparently linear in a chosen massive component, nor does it exclude loop effects, anomalies, or a singular failure of the mass-eigenfield map at the conformal boundary.

B Einstein-frame cubic expansion

We record enough of the independent G14 calculation to make the exact numbers reproducible. This appendix uses signature $(-,+,+,+)$ and $M = 1$. It matches `symbolic/verify_gravity_factorization.py`.

For a symmetric matrix Φ, write

$$X = g^{-1}\Phi, \qquad t_n = \operatorname{tr}(X^n), \qquad A = \sqrt{\det(I+X)}. \tag{85}$$

Through cubic order,

$$\begin{aligned} A &= 1 + \frac{1}{2}t_1 + \frac{1}{8}t_1^2 - \frac{1}{4}t_2 + \frac{1}{48}t_1^3 - \frac{1}{8}t_1 t_2 + \frac{1}{6}t_3 + O(\Phi^4), \\ A^{-1} &= 1 - (A-1) + (A-1)^2 - (A-1)^3 + O(\Phi^4). \end{aligned} \tag{86}$$

In the potential of (29), $\tau = 4 + t_1$ and the relevant quadratic contraction is $4 + 2t_1 + t_2$. Substitution gives, when $t_1 = 0$,

$$A^{-1}\left(-\tau^2 + \psi_{\mu\nu}\psi^{\mu\nu} + 6A\tau - 12A^2\right) = t_2 + O(\Phi^4), \tag{87}$$

with zero t_3 coefficient. This is the potential cancellation (31).

At first order in one plane wave (p, E),

$$Q^{(1)\alpha}{}_{\mu\nu}(E,p) = \frac{i}{2}\eta^{\alpha\beta}\left(p_\mu E_{\beta\nu} + p_\nu E_{\beta\mu} - p_\beta E_{\mu\nu}\right). \tag{88}$$

The quadratic correction $Q^{(2)}$ comes from the inverse-ψ factor. The cubic MMM coefficient is the sum of all contractions $K(Q^{(1)}, Q^{(2)}) + K(Q^{(2)}, Q^{(1)})$, with the factor $1/2$ in (29). Exact substitution of (32) and normalized TT tensors gives (33).

For MMh, the script expands g^{-1}, ψ^{-1}, the connection of g, and Q through the orders required for the coefficient of three independent wave labels. At (34), take

$$v_r = (1,-1,0,0), \qquad e_h = (1,0,0,1), \qquad E_r = v_r^\flat \otimes v_r^\flat, \qquad H = e_h^\flat \otimes e_h^\flat. \tag{89}$$

Together with the internal + polarization at rest these tensors are transverse and traceless, and the coefficient is (35). For

$$H^{\text{gauge}}_{\mu\nu} = k_\mu \xi_\nu + k_\nu \xi_\mu \tag{90}$$

with four independent symbolic components of ξ, the coefficient vanishes identically.

Finally, the massive TT two-point coefficient for waves (p, E) and $(-p, E)$ with $E \cdot E = 1$ is

$$L_M^{(2)} = \frac{p^2 + 1}{4}. \tag{91}$$

This fixes the factor of four between the polarization numerator $\sqrt{3}/32$ and pole residue $\sqrt{3}/8$; it is not inferred from a unit-residue completeness convention.

C Polarization basis and factorization sum

At the rest momentum $P = (1, 0, 0, 0)$ in the Einstein-frame signature, let e_x, e_y, e_z be unit spatial vectors. A normalized massive basis is

$$\begin{aligned}
E_+ &= \frac{1}{\sqrt{2}}(e_x e_x - e_y e_y), & E_\times &= \frac{1}{\sqrt{2}}(e_x e_y + e_y e_x), \\
E_0 &= \frac{1}{\sqrt{6}}(e_x e_x + e_y e_y - 2 e_z e_z), & E_{xz} &= \frac{1}{\sqrt{2}}(e_x e_z + e_z e_x), \\
& & E_{yz} &= \frac{1}{\sqrt{2}}(e_y e_z + e_z e_y).
\end{aligned} \tag{92}$$

Each is transverse, traceless, and orthonormal under the induced tensor inner product. For a massive momentum $(E, 0, 0, p_z)$, replace e_z by the normalized longitudinal transverse vector; the same formulas give the transported helicity basis.

The factorization numerator is basis-independent. In a nonorthonormal basis T_a with Gram matrix $G_{ab} = T_a \cdot T_b$, it is

$$N = V_a (G^{-1})^{ab} U_b, \tag{93}$$

where $V_a = \mathcal{A}_3(M, M, T_a)$ and $U_b = \mathcal{A}_3(T_b, M, h)$. The orthonormal basis above reduces this to the sum displayed after (36). All five intermediate states are included; the result is not one selected nonzero channel.

The original fourth-order perturbiner supplies a separate complex factorization check at a rational-i point. It constructs an exact five-dimensional TT basis at the internal massive momentum, contracts the two cubic vectors with its Gram inverse, and obtains a nonzero reduced numerator. The Einstein-frame calculation then fixes a compact normalized representative and the explicit inverse-kernel slope.

D Physical-shell geometry and the rational point

For incoming equal masses in the center-of-mass frame,

$$p_1 = (E, 0, 0, p), \qquad p_2 = (E, 0, 0, -p), \qquad E = \frac{\sqrt{s}}{2}, \quad p = \frac{1}{2}\sqrt{s - 4M^2}. \tag{94}$$

Choose the outgoing graviton at scattering angle θ,

$$k = (\kappa, \kappa \sin\theta, 0, \kappa \cos\theta), \qquad \kappa = \frac{s - M^2}{2\sqrt{s}}, \tag{95}$$

and $p_3 = p_1 + p_2 - k$. Then $p_3^2 = M^2$ and $p_3^0 = (s + M^2)/(2\sqrt{s})$. The domain

$$s > 4M^2, \qquad -1 < \cos\theta < 1 \tag{96}$$

is a real open chart away from forward/backward and threshold degeneracies.

At $s = 25M^2/4$ one has $E = 5M/4$ and $\kappa = 21M/20$. Taking $\cos\theta = 3/5$ and $\sin\theta = 4/5$ gives exactly (40). All energy, momentum, and mass-shell identities are rational. Because all spatial y components vanish, reflection $y \mapsto -y$ is a symmetry. An odd number of y-odd polarization tensors forces an amplitude zero. The certificate uses the first nonzero all-even combination.

The exact threshold regression uses two incoming massive particles at rest and an outgoing graviton of energy $3M/4$. In the separate polarization convention of that test,

$$\mathcal{A}_{\text{threshold}} = -\frac{509784}{390625}. \tag{97}$$

Threshold is not used for the open-set claim; it guards the implementation against drift.

E Full four-point engine and deformation certificate

E.1 Wave algebra

A wave-algebra element is a dictionary from subsets of external labels to sparse tensor components. Multiplication discards products in which a wave label repeats, so multilinear truncation is automatic. A derivative acting on subset S multiplies by

$$iP_{S\mu} = i \sum_{a \in S} p_{a\mu}. \tag{98}$$

From the metric, inverse metric, determinant, Christoffel symbols, and Ricci tensor, the engine extracts

$$\sqrt{-g}R, \qquad \sqrt{-g}R_{\mu\nu}R^{\mu\nu}, \qquad \sqrt{-g}R^2 \tag{99}$$

at the desired wave subset. Before any cubic or quartic claim it checks that the two-point coefficient vanishes on $p^2 = 0$ and $p^2 = M^2$, is nonzero off shell, and obeys cubic Ward identities.

E.2 Exact polarization tensors

For a massive momentum p in the (t, x, z) plane, choose two independent transverse in-plane vectors v_1, v_2. With

$$\Pi_{\mu\nu} = \eta_{\mu\nu} - \frac{p_\mu p_\nu}{M^2}, \tag{100}$$

the parity-even TT tensor used in the certificate is

$$E_{12,\mu\nu} = v_{1\mu}v_{2\nu} + v_{2\mu}v_{1\nu} - \frac{2}{3}(v_1 \cdot v_2)\Pi_{\mu\nu}. \tag{101}$$

The three massive external legs use the corresponding E_{12} at $p_1, p_2, -p_3$. For the graviton momentum direction proportional to $(5, 4, 0, 3)$, use transverse vectors

$$e_1 = (0, -3, 0, 4), \qquad e_2 = (0, 0, 1, 0), \tag{102}$$

and the real plus tensor

$$\varepsilon_{+} = \frac{1}{25}e_1e_1 - e_2e_2. \tag{103}$$

These are the convention-fixed polarizations entering (43).

E.3 Gauge and adjoint checks

For each exchange, the exact bordered solution obeys

$$CX = 0, \qquad KX + C^T\lambda + J = 0. \tag{104}$$

The total amplitude with graviton gauge polarization is zero, while the contact contribution alone is

$$-\frac{596335802837691}{361367187500000} \neq 0. \tag{105}$$

This cancellation is a sensitive check of relative signs and combinatorics. De Donder and axial constraints give identical physical totals.

Under reversal $p_a \mapsto -p_a$, all contacts and cubic currents are recomputed. The action contains two- and four-derivative terms, and the independently recomputed quadratic matrices obey $K(-P) = K(P)$. Contact, s, t, and u pieces equal their forward values separately. Since the external linear tensors are real, this is the physical reverse matrix element used in $\mathcal{B}_{2,+}^{\dagger}$.

E.4 Pole regression

An exact shell-preserving complex shift

$$p_1(z) = p_1 + z\eta, \qquad k(z) = k - z\eta, \qquad \eta = (3, 0, 4i, 5), \tag{106}$$

has $\eta^2 = \eta \cdot p_1 = \eta \cdot k = 0$. In the relevant channel

$$P(z)^2 = \frac{25}{4} + 15z \tag{107}$$

hits the massive pole at $z_* = -7/20$. The bordered kernel there has exactly the five TT zero modes. If T collects them and G_5 is their Gram matrix,

$$W = T^T K'(z_*)T = -\frac{15}{2}G_5 \tag{108}$$

entrywise, including 24 nontrivial identities. The residue

$$-j_1^T W^{-1} j_2 = -\frac{4551434384\, i}{328515625} \neq 0 \tag{109}$$

equals the independently normalized cubic contraction divided by the kinetic slope. A direct near-pole numerical probe of the actual bordered exchange agrees to relative error below 10^{-4}. This regression is not used to set the convention of the compact Einstein-frame residue; it validates the original-variable propagator machinery.

F Covariant grading and Fock-lift equations

The exact grading check solves the commutant rather than assuming Schur's lemma numerically. At a standard massive momentum, represent the spin-2 fiber by symmetric traceless 3×3 tensors. The three $SO(3)$ generators act as

$$J_i(T) = L_i T + T L_i^T. \tag{110}$$

Solving $[X, J_i] = 0$ for a general 5×5 matrix gives $X = a_M I_5$. On the two complex massless helicity fibers, the little- group rotation generator is $\mathrm{diag}(2, -2)$, whose commutant is diagonal. Solving simultaneously with the mass Casimir on the full seven-dimensional fiber gives

$$X = \mathrm{diag}(a_+, a_-, a_M I_5). \tag{111}$$

Hermiticity and $X^2 = I$ restrict the three scalars to signs. The helicity-exchange real structure sets $a_+ = a_-$; the free signature fixes $(a_+, a_-, a_M) = (1, 1, -1)$.

For the Fock lift, cluster factorization is the monoid equation (60). Iterating it gives

$$j(n_h, n_M) = j(1, 0)^{n_h} j(0, 1)^{n_M} = (-1)^{n_M}. \tag{112}$$

The exact script verifies multiplicativity for all partitions in a finite table and the analytic monoid argument establishes it for all particle numbers.

G BRST block and non-nullity

The finite verification sets the common real external normalization $\eta_{\mathrm{conv}} \mathcal{N}_{fi}$ to one and therefore uses $\mathcal{A}_K^{\mathrm{red}}$ in place of $\mathcal{A}_K$. Restoring the nonzero multiplier rescales every quadratic trace by its square and cannot change nullity.

The minimal non-null calculation uses

$$J_{\min} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 0 \\ i\mathcal{A}_K & 0 \end{pmatrix}. \tag{113}$$

Then

$$X^\sharp = J_{\min} X^\dagger J_{\min} = \begin{pmatrix} 0 & i\mathcal{A}_K \\ 0 & 0 \end{pmatrix}, \qquad X^\sharp X = \begin{pmatrix} -\mathcal{A}_K^2 & 0 \\ 0 & 0 \end{pmatrix}. \tag{114}$$

For $\mathcal{O} = -2i\mathcal{A}_K \sigma_x$,

$$J_{\min} \mathcal{O}^\dagger J_{\min} = \mathcal{O}, \qquad \mathcal{O}^2 = -4\mathcal{A}_K^2 I. \tag{115}$$

To encode the cohomology argument, take a canonical finite splitting into two physical representatives and one contractible doublet $u \mapsto v \mapsto 0$. In the ordered basis (i, f, u, v),

$$Q = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad Q^2 = 0. \tag{116}$$

For a completely general 4×4 matrix Y, direct multiplication shows

$$\big(QY + YQ\big)_{\mathrm{phys,phys}} = 0. \tag{117}$$

The upper-left physical block of the exact obstruction is nonzero after embedding, so it cannot be of this exact form. This finite matrix is not a replacement for gravitational BRST quantization; it records the universal algebraic implication used after the Ward and reduced-helicity checks have established that the amplitude descends to cohomology.

H Machine-verification index

All scripts named below are in `symbolic/` and print `ALL PASS`. Checks use SymPy exact arithmetic except for the explicitly labelled near-pole regression.

Archive. The engine, polarization data and manuscript sources are archived in a GitHub repository:

`https://github.com/area9innovation/weyl-gravity`.

The historical pre-extraction verification snapshot was

`aeb9acaa944f53cbf29fba37476bfe888c6a1ddd`

in the former monorepo; current paths are relative to the standalone repository root.

Environment. The recorded versions are Python 3.14.4 and SymPy 1.14.0. The pinned dependency and the SHA-256 manifest are, respectively,

`symbolic/paper6-requirements.txt`
`symbolic/paper6-verification.sha256`.

Momentum, signature, polarization, action and reduced-amplitude conventions are recorded in Appendices B–E; they are explicit script inputs rather than hidden generated defaults.

Check	Claim	Artifact/type
G13a–b	two-point on-shell validation and cubic Ward identities	`gravity_perturbiner.py`, exact
G13	$Mhh = 0$ for 5×4 physical polarization choices	`verify_gravity_cubic.py`, exact
G14a–c	nonzero original-variable MMM, MMh, and reduced massive-pole factorization	`verify_gravity_cubic.py`, exact
G14d–i	potential cancellation, compact $-\sqrt{6}/8$ and $-\sqrt{2}/8$ amplitudes, symbolic Ward identity, five-state sum, inverse kernel	`verify_gravity_factorization.py`, exact
G15a–f	full physical contact+s, t, u, Ward, Bose, gauge, and threshold checks	`verify_gravity_g15.py`, exact
G15g	shifted-pole kernel, Schur slope, residue; near-pole probe	same script, exact plus numerical regression
G16	crossing needed for the adjoint is included in G17; exhaustive 250-polarization scan is deferred	not a theorem dependency
G17a–d	exposed contact/exchanges, reverse process, EOM/Ward/gauge, termwise quarter-turn	`verify_gravity_obstruction.py`, exact
G17e–g	exact shell obstruction, first-order metric-ambiguity independence, and Born–deformation identity	same script, exact
G18a–b	one-particle commutant and cluster-factorizing Fock lift	`verify_gravity_krein.py`, exact
G18c–e	non-null quadratic traces, charge no-go, and BRST block	same script, exact

The analytic inputs not reducible to finite symbolic identities are identified in the main text: the induced-representation interpretation of the free commutant (Paper IV), the consistent-truncation/LSZ argument, covariant–stationary tree matching, real-analytic continuation on the physical shell, and the standard BRST decoupling statement. The symbolic checks test every convention-sensitive coefficient consumed by those arguments.

Reproduction commands. From the repository root for this project:

```
python3 symbolic/verify_gravity_paper6.py
```

The expected terminal line is `PAPER VI GRAVITY SUITE: ALL PASS`. The optional `--quick` flag skips the slower shifted-pole regression; the theorem-critical obstruction script independently recomputes the exact physical certificate. Individual script outputs provide the canonical coefficient-level verification evidence mapped in the table above.